

ADVENTURE · AI PLAY STRATEGY

Adventure AI Play Strategy
A Post-Hoc Analysis

Why does the AI recommend that move?

The AI sometimes rolls when a move card would draw a card right away, and plays a small multiplier while holding a large one. These choices have reasons that can be computed. This report breaks down the value VELA and LYRA give to each outcome of an action, then tests the reasons with card values, deck contents, dice distributions and paired play-outs. It also lists the positions where the AI's choice differs from the exact value.

1 Reading the game and the AI — rules, base rates, how the AI decides, how to read a case

2 Choice patterns across full game records

3 Strategy cases — eleven strategies, 29 cases

4 Endgame — card order checked against exact values

5 Where the AI goes wrong — eight cases by type, and why

6 How reliable the recommendations are, and the cost of overriding them

7 High-scoring games — the score distribution of a million games and their resources

AIs analyzed
VELA v4.3 · v4.2 · LYRA v2

Evidence
Rule math · value breakdowns · play-outs · exact values

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SUMMARY

Decision principles at a glance

The numbers come from the cases and the full game records. The principles interpret how the two AIs calculate; they are not rules that hold in every position.

1 **Early on, keep the cards and roll**

With many rolls left, the AI rolls even when a move card would draw a card now. In these positions, changing only the rolls used to 92 (8 left) makes the AI play the card at once in 92.9% of cases. Of the cards it kept, 99.8% were later used to draw. It plays early anyway when the drawn card can draw again right away, or when the card it spends is very cheap (Part 3, Section 1).

2 **Late on, turn cards into distance**

With few rolls left, the AI will play several cards in a row to draw just one. At 1,686 it used four cards to cover 106 tiles. Changing the rolls used to 20 (80 left) in the same positions turns 72.3% of these chains into a roll (Section 2).

3 **With a full hand, play a card to clear a slot**

With a full hand a roll cannot draw. Removing one other card to leave four in hand turns most of these choices into a roll. The card played is usually a cheap one, and whether it rides a jump or leads to the next draw matters too. The AI still rolls with a full hand when a card in hand can draw right after the roll (Section 3).

4 **Before a stop tile, save move cards and jump it with a multiplier**

Even when a roll can only move 1–2 tiles, the AI does not spend a move card to go a few tiles further. With a multiplier in hand, it jumps the stop tile even when the chance of drawing is low (Sections 4 and 5).

5 **Play the card that lands on a jump or starts a chain**

The AI plays a small card that rides a jump without drawing, a negative card that keeps the better cards or opens a chain, and a stage card from whose landing tile a card in hand can draw right away (Sections 6, 7 and 9).

6 **Small multipliers first; the deck settles close calls**

A small multiplier is worth little in hand, and its shorter move makes it easier for a card in hand to draw from where it lands. Changing only the deck flipped 2.4% of choices, but with 2–3 cards left in the deck the decision follows the deck (Sections 10 and 11).

7 **In the endgame, order is score**

Before the last roll, free cards go in the order that lines up jumps and draws one after another. With two rolls left, the order of two multipliers also depends on where they land. In endgame positions, the share of choices matching the optimal action was 99.7% for VELA v4.3 Normal, 98.8% for VELA v4.2 and 97.4% for LYRA v2. Part 5 sorts the mistakes by type.

How to read a recommendation

① Where the action really lands (after jumps and stop tiles)
② What it spends (a card, a roll, a bonus roll) and what it gains
③ Whether a card left in hand can draw again from the landing tile
④ The quality of the cards left in the deck
⑤ The rolls used (and left)

When two actions are within 1 point, treat them as equally good. Taking the second-best action instead of the recommendation at a random decision costs 15.1–17.6 points on average, so following every recommendation scores higher on average even though the AI is sometimes wrong (Section 6.4). The value tables and outcome breakdowns in Part 3 come from VELA v4.2 (= v4.3 Fast); in all 29 cases v4.3 Normal and High and LYRA v2 chose the same action.

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PART 1

Reading the game and the AI

The rules, base probabilities, the AI's method and the value of a single card, as needed to follow the recommendations.

1.1 Actions and resources

On each turn the player either rolls the dice or plays a card from the hand. Cards come in three types.

Type	Effect	Cost
 Move card	Moves a fixed number of tiles (+1 to +12, −1 to −3). Jumps and draws on the landing tile apply as usual. Does not stop at stop tiles.	One card. Uses no roll (a free action).
 Multiplier card	Moves the sum of two dice times the multiplier (×2, ×3, ×5, ×7, ×8, ×10). Passes over stop tiles.	One card and one roll (the bonus roll, if there is one).
 Stage card	Moves to the first tile of the next stage. The distance depends on the current tile (about 20 tiles on average).	One card. A free action.
Roll	Moves the sum of two dice. Stops at the first stop tile ahead.	One roll (or the bonus roll).

A game starts with 100 rolls, and the number of the final tile is the score. A double on a paid roll (a plain roll or a multiplier card) makes the next roll free; this report calls it the **bonus roll**. A double on the bonus roll does not earn another one. The game ends when no rolls and no bonus roll are left, even if cards remain in hand.

The game screen shows how many rolls have been used. This report also describes positions by **rolls used**, adding the rolls left (100 minus rolls used) in brackets where helpful. Only the endgame parts (Parts 4 and 5), where the timing of the last roll matters, count the rolls left.

The hand holds at most five cards. With five cards (a **full hand**), landing on a draw tile gives nothing. A played card leaves the hand first, so a move card played from a full hand can still draw on its landing tile. Cards come at random from the **deck** (the cards not yet drawn). The deck starts with 30 cards and is **refilled** to 30 as soon as its last card is drawn.

Board tiles

Draw tile

Landing here draws one card from the deck (nothing with a full hand). 401 of the 2,898 tiles.

Jump tile

Landing here sends the token along a ladder or pipe, forward (up to +34) or back (up to −20). 238 tiles; the destination may be a draw tile.

Stop tile (boss tile)

A plain roll cannot pass it and stops there. 279 tiles; 232 of them form four runs of 58.

1.2 Base probabilities

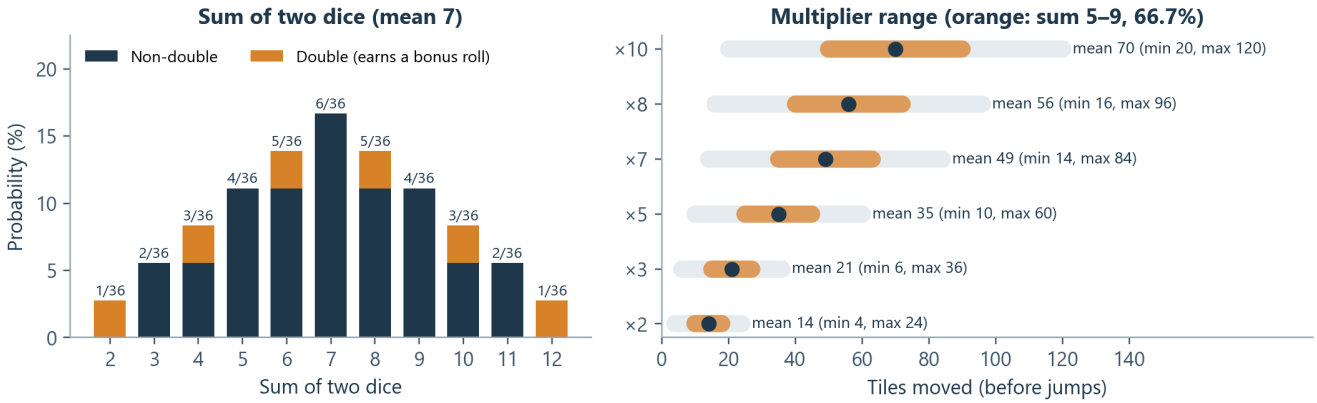


Figure 1: Sum of two dice (left) and the range of each multiplier (right). A multiplier scales the sum, so it stretches the same distribution.

Averaged over the whole board, a roll moves 7.01 tiles and lands on a draw tile with probability 13.1%. Move cards land on a draw tile with similar probability whatever their number (13.7% to 15.0%). A stage card's destination (the first tile of a stage) is never a draw tile. A roll is cut short by a stop tile 19.2% of the time.

Read the probabilities in the cases against this figure of about 14%. The AI's choice depends on how much the current tile shifts that probability and on whether the hand left after landing leads to the next draw. In the game records, 94.2% (VELA v4.2) and 94.4% (LYRA v2) of the move cards played led to a draw.

1.3 How the AI chooses

Both AIs work the same way. For each action they expand every possible outcome by the rules (dice results, cards drawn), estimate the final score reachable from each outcome, and take the probability-weighted average. That average is the **AI value** of the action (an estimate of the expected final score), and the AI recommends the action with the highest value.

AI	How it values an outcome
VELA v4.2	A large precomputed value table (by tile, rolls left and hand) with a correction for the remaining deck. It looks two steps ahead: the action, then the best reply (depth 2). The production service fixes it at depth 2.
VELA v4.3	The same value table, with a choice of search strength. Fast is identical to v4.2. Normal, High and Max look three steps ahead at every action; when the top two actions are within 1.5 points, they extend the actions within 1.5 points of the best up to six steps. The three differ only in their computation cap (2,400, 47,400 and 383,400 evaluated states). Normal is the service default.
LYRA v2	A neural network (a model trained on many games to predict how good a position is) values each outcome directly. It looks one step ahead.

Over 20,000 games, the search strengths averaged 1,877.2 (Fast), 1,878.6 (Normal), 1,879.3 (High) and 1,879.8 points (Max). Searches of three or more steps do not split cleanly into per-outcome values, so the value tables and outcome breakdowns in the cases show the two-step calculation of VELA v4.2 (= v4.3 Fast). The evidence tables list "VELA v4.2, depth 1·2 (production)·3". Depths 1 and 3 are analysis runs with the same value table and a different search depth. The service recommends at depth 2; when the choice changes with depth, it shows how many steps ahead the judgment has to look. Each case also lists how v4.3 Normal, High and Max evaluated the same position. In all 29 cases in Part 3, all four strengths chose the same action as v4.2.

Splitting an action's value by outcome (an **outcome breakdown**) shows which outcomes drove the choice. In every case, recombining the outcome values with their probabilities reproduced the AI value to within 0.001 points.

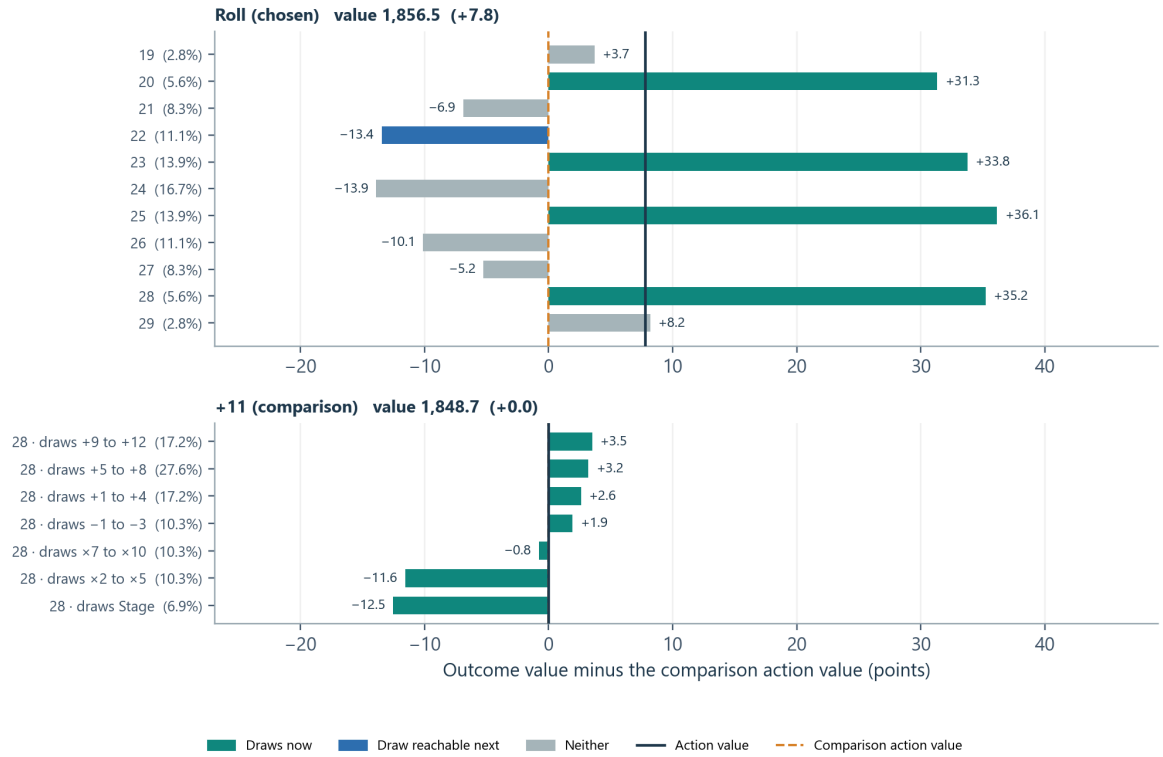


Figure 2: How to read an outcome breakdown (Case 1-1 in Part 3). Each bar is the AI's value for one outcome minus the comparison action's value (dashed orange line). The solid dark line is the action's own value. Color shows whether the outcome draws a card now (green), leaves a hand that can draw on the next action (blue), or neither (gray). When many different cards can be drawn, they are grouped by kind.

The explanations rest on four kinds of evidence, each with its own level of certainty.

Evidence	Meaning and limits
Rule math	Moves, jumps, draws and dice probabilities computed from the rules. Exact.
AI value	Action values and outcome breakdowns. They show what the AI based its choice on, but they are estimates, not exact expected scores. Against exact endgame values, the value of the chosen action was off by 0.74 points on average for VELA v4.2, 0.26 for v4.3 Normal and 0.73 for LYRA v2 (Part 5).
Full-game comparison	Games played to the end with the same dice sequence; the gap in final score is measured. Used for the Part 3 cases (both actions played on to the end) and for the override experiment in Section 6.4, with a correction that removes much of the dice luck (a control variate: the luck of each step, measured with the AI's search values, is subtracted) and a 95% interval (the range likely to contain the true mean gap).
Exact value	The expected final score computed over every remaining possibility. Only feasible in the endgame, with 0–2 rolls left.

1.4 What one card is worth

Playing a card means comparing what the card is worth in hand with what playing it gains now. This report defines a card's **holding value** as the difference in AI value between the same position with and without that card. The figures below come from 22,838 cards in 7,000 positions that VELA v4.2 actually reached.

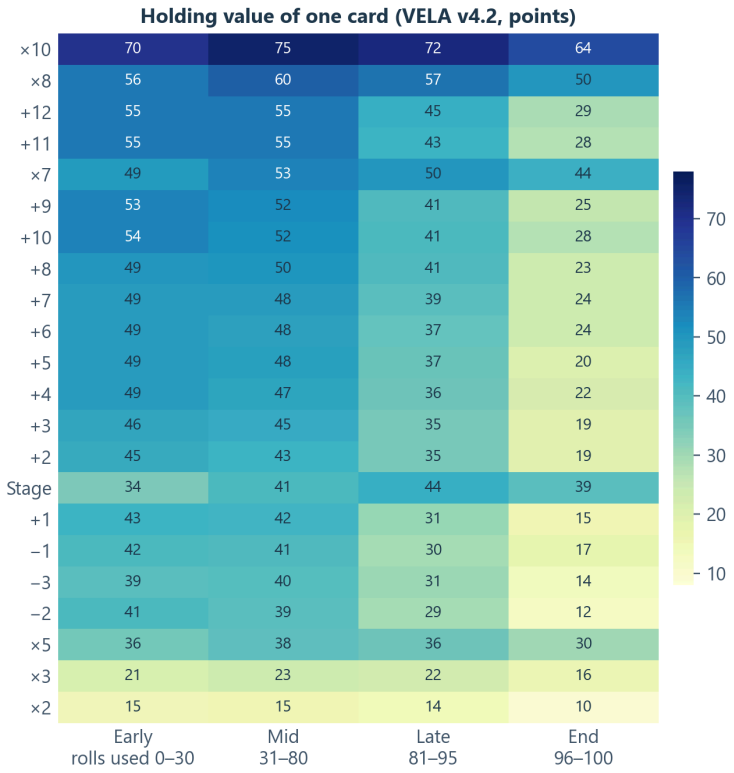


Figure 3: Average holding value by card type and game phase. Phases are defined by the number of rolls used.

The chart shows four things.

1. The same move card changes value a lot over the game. A +10 drops from 54.0 points early to 27.7 at the end. With few rolls left there is less time to wait for a good spot to play it, and fewer chances to use the card it draws.
2. Multipliers vary widely. The ×10 is the most valuable at 72.5 points and the ×2 and ×3 the least at 14.9 and 22.4. A ×2 moves only 7 tiles more than a roll on average but costs a card, so the net gain is small.
3. The −1 to −3 cards are worth 37.5 to 38.4 points, about the same as a +1 or +2. Where moving back reaches a jump or a draw tile, a backward card does the same job as a forward one.
4. The stage card rises to 44.2 points late in the game. Stage jumps get longer toward the end of the board (16.6 tiles on average before tile 700, 26.9 after tile 2,500), and a stage card skips a run of stop tiles without using a roll.

Holding value also depends on the tile and on the other cards in hand. The same card is worth much more where it reaches a jump or a draw tile from the current position. And the more cards are held with it, the less one card is worth. In the position of case 6-2 in Part 3, the −3 alone was worth 49.7 points with no other card in hand, 46.6 with one, 42.5 with two and 37.8 with three (average over 60 randomly filled hands). Moving the position by 30 tiles changed this by about a point. A full hand blocks the next draw, so it lowers the value further. After an action that spends several cards, each remaining card is therefore worth more; after an action that keeps the hand, each is worth less. This explains the Part 3 cases where removing a card unrelated to the chain still flips the choice (Sections 3 and 6). A card play can be read as the following trade.

The trade in a card play

Cost: the holding value of the card played (plus one roll for a multiplier)
Gain: distance, the value of the card drawn if the landing tile is a draw tile, and the chance for a card left in hand to draw again from the landing tile

The deck plays a smaller part in this trade than one might expect. Cards are drawn without replacement and the deck refills when it runs out, so drawing a weak card now leaves a better deck behind. In 600 positions the AI actually reached, replacing only the deck with a random deck of the same size changed the AI's choice 2.4% of the time: 0% for decisions with a value gap above 10 points, and 21.0% for close calls under 1 point. The deck mostly settles close calls; it changes the decision itself when only 2–3 cards are left (Part 3, Section 11).

1.5 How to read a case

Each case in Part 3 explains the AI's choice with four kinds of evidence.

Evidence	How to read it
Outcome breakdown	The AI value split by dice result and card drawn (Section 1.3). It shows which outcomes pushed the action's value up.
Split after four rolls	Starting from the chosen action and from the comparison action, VELA v4.2 plays on with the same dice sequence until four rolls have been used (1,000 pairs). The gap in VELA v4.2 value at that point is split into a **position part** (the value with the hand emptied: tile, rolls left and deck) and a **hand part** (the rest). A large position part means "it got further ahead"; a large hand part means "it kept better cards".
Counterfactuals	The position re-evaluated with one key factor changed: the rolls used (20, 70 or 92), one card removed from the hand, or a different deck. If the choice flips, that factor decides it.
Full-game comparison	Both actions played on to the end of the game with the same dice; the final score gap with a control-variate correction and a 95% interval. 2,048 pairs with VELA v4.2 playing on and 512 with v4.3 Normal; when the interval included 0, more pairs were added.

Played on to the end, all 29 Part 3 cases favored the AI's choice within the 95% interval when VELA v4.2 played on (gaps of 1.4–48.7 points). With v4.3 Normal playing on, 28 did; the remaining one (6-2) was a close call not distinguishable from 0.

Each case's "value gap with only the rolls used changed" chart re-evaluates the position with VELA v4.2, changing only the rolls used in steps of four. Two things in these charts should be read differently.

- **Large changes at the end (after about 90 rolls used) come from the game itself.** With only a few rolls left there is no chance to use the cards in hand, so the value of holding a card drops sharply (Section 1.4). Actions that keep cards and roll lose ground at the end (1-1, 1-2, 4-1, 4-2, 11-1), and actions that turn cards into distance now gain (1-3, Sections 2, 6 and 9, 11-2). Computing 4-1 exactly with two rolls left (98 used), the roll is 7.8 points below the +5, almost exactly the chart's end value (−7.7).
- **Wiggles of 1–2 points in the middle are within the value table's approximation.** Evaluating the same positions with LYRA v2 gives larger wiggles in different places: the two AIs estimate value differently, so the small bumps do not line up. Read only the overall trend in the middle.

PART 2

Choice patterns across full game records

How often each AI chose each kind of action in comparable situations, from 10,000 games of VELA v4.2 and 10,000 games of LYRA v2.

2.1 Other actions even when a move card can draw

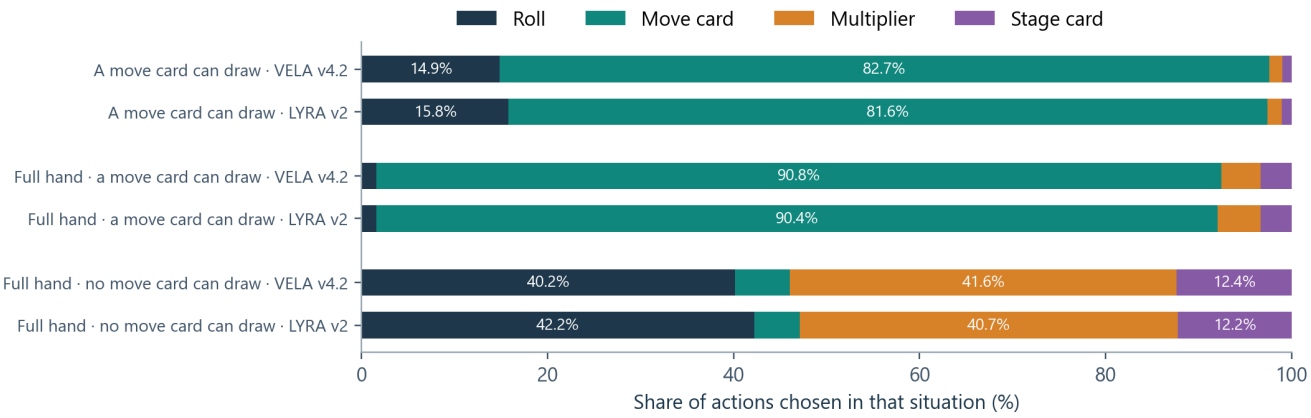


Figure 4: Share of actions by situation. "A move card can draw" means at least one move card in hand lands on a draw tile after jumps.

When a move card could draw (VELA v4.2 522,731 times, LYRA v2 532,490 times), the AIs played a move card 82.7% and 81.6% of the time. In the remaining 17–18% they rolled or played a multiplier or a stage card. With a full hand and no move card that could draw, they mostly rolled (40.2% and 42.2%) or played a multiplier (41.6% and 40.7%). Rather than playing cards to make room, they often kept the hand and changed position.

The two AIs agree to within about one percentage point. LYRA v2, however, started its training from VELA's choices, so it may have inherited similar tendencies. The two data sets also come from different games, so this is not a position-by-position comparison.

2.2 Actions by game phase

Rolls used	Roll	Move card	Multiplier	Stage card
0–30	67.9% · 68.3%	23.4% · 23.0%	6.7% · 6.7%	2.0% · 2.0%
31–80	61.4% · 61.2%	28.3% · 28.5%	7.7% · 7.7%	2.6% · 2.6%
81–95	57.2% · 57.0%	31.5% · 31.6%	8.5% · 8.6%	2.8% · 2.8%
96–100	34.4% · 34.5%	46.0% · 46.3%	14.0% · 13.6%	5.7% · 5.5%

Each cell shows VELA v4.2 · LYRA v2. The denominator is decisions with at least one card in hand. These are shares among the positions actually reached in each phase, not the same positions compared across phases.

Early in the game about 68% of decisions are rolls; in the endgame (96–100 rolls used), move cards rise to 46%. This matches the drop in endgame holding values seen in Section 1.4.

2.3 Choosing among multipliers

In positions where the AI held two or more multipliers and played one (51,845 and 51,427 times), it played the smallest one 54.4% and 54.3% of the time. It chose the multiplier with the highest chance to draw on this roll (ties included) 89.8% and 90.0% of the time, and the one with the longest average move 45.6% and 45.7% of the time. Part 3, Section 10 works through this choice in detail.

PART 3

Strategy cases

Eleven strategies behind AI choices that may puzzle a human player, each examined in two or more real positions where that strategy shows.

Every case comes from positions VELA v4.2 reached in 2,000 of its own games, selected by three criteria. Conditions on the nearby tiles keep other strategies out as far as possible (for example, a strategy unrelated to stop tiles and jumps requires no stop tile or jump within the next 12 tiles). Changing the strategy's key factor alone must flip the choice. And VELA v4.2, v4.3 (Normal and High) and LYRA v2 must all choose the same action. Positions with a bonus roll pending were excluded.

Each case follows the same order: the board and the value of every action, the outcome breakdown (Section 1.3), the value split after four rolls played with the same dice and the evaluation with only the rolls used changed, and the evidence table. The last two rows of the evidence table give the score gap when both actions are played on to the end of the game with the same dice (Section 1.5).

1. Early on, roll first even when a card can draw

In short

Even when a move card would draw a card now, the AI keeps that card and rolls while many rolls are left. In 1,157 such positions, changing only the rolls used to 92 (8 left) makes the AI play the card at once in 92.9% of cases, and changing them to 70 (30 left) does so in 20.4%. It does play early when the drawn card can draw again right away (1-2) or when the card it spends is very cheap (1-3).

The saved card is not wasted. Following 1,200 move cards saved this way to the end of the game, 100% were played later and 99.8% of those drew a card. The median wait was 5 rolls. A move card can buy one draw now or a few rolls later, so with many rolls left the AI moves by dice first and plays the card on a later draw tile. Each roll in between also lands on a draw tile with about a 14% chance.

Case 1-1 · VELA v4.2 record

Rolling when the +11 would draw now

Position **17** · Rolls used **3** (97 left)
Deck (29): −3 · −2 · −1 · +1 · +2 · +3 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2





Figure 5: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; “draw” = draw tile). Orange dashed: the comparison action.

The hand is a single +11, which would draw at 28. VELA v4.2 chose to roll, by 7.8 points. A roll itself lands on a draw tile with probability 38.9%.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,856.5**	**+7.0**	**38.9%**	**50.0%**
+11 · comparison	1,848.7	+11.0	100.0%	100.0%

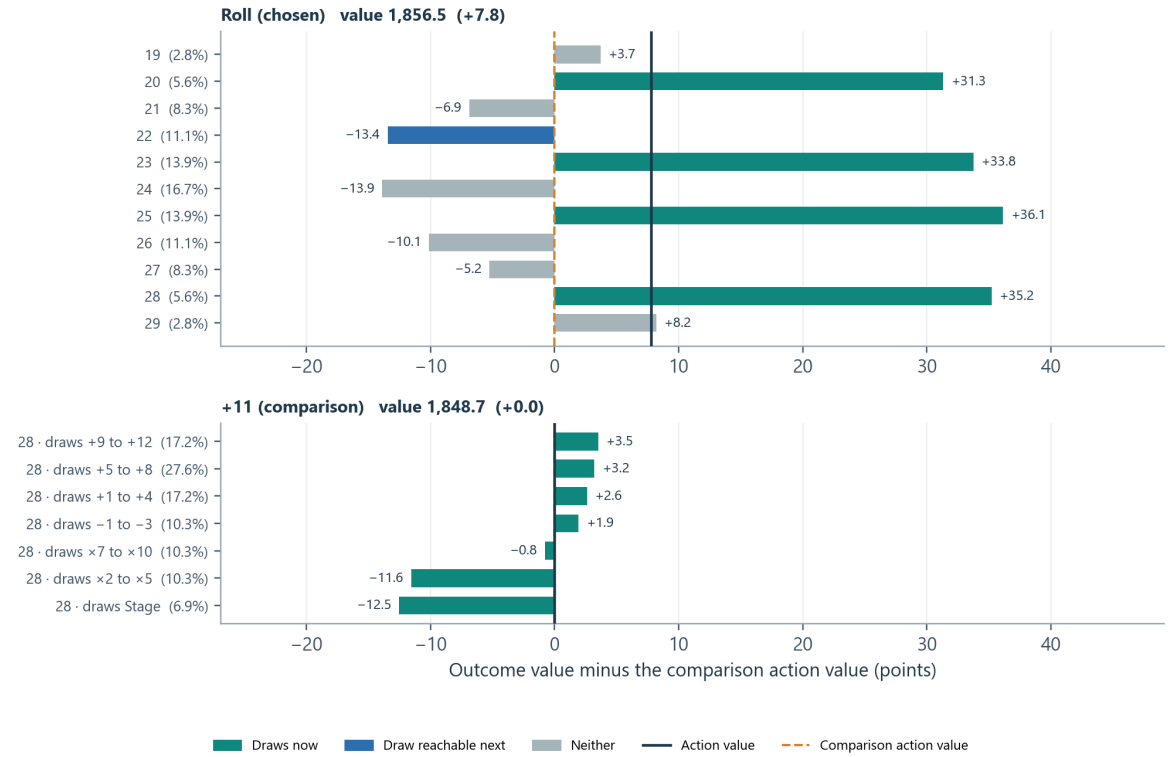


Figure 6: Outcome breakdown (reference: the value of the +11). Roll outcomes that land on a draw tile (green, 38.9%) average +34.5 points; the rest (61.1%) average −9.2. The +11 depends on the card drawn, and none of its outcomes beats the roll's value (7.8 to the right of the reference).

The roll's value splits two ways. The 38.9% of outcomes that land on a draw tile draw a card while still holding the +11, so they are worth +34.5 points more than the +11 on average; the rest average −9.2. Playing the +11 makes the draw certain, but it trades the +11 in hand for a random card from the deck.

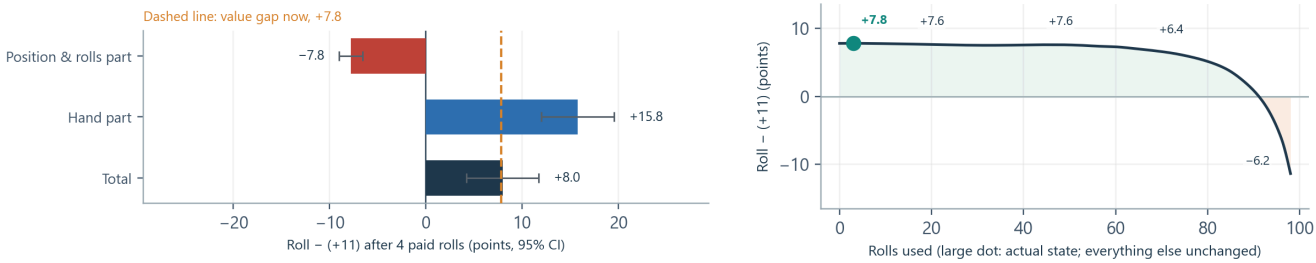


Figure 7: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+11)) when only the number of rolls used changes.

After four rolls the gap comes from the hand part (+15.8 points). By then the roll side had played 0.6 cards on average and the +11 side 1.6. The +11 side drew slightly more cards (2.6 against 1.8), but the roll side still holds the original +11. The position part goes the other way: the +11 side is 7.8 points ahead, because the move to 28 leaves it about 13 tiles further on. One card is worth more (about 50 points) than 13 tiles, so the total favors the roll. With the rolls used changed to 92 (8 left) the choice flips to the +11: with few rolls left there are fewer later chances to play it, and the certain card now wins.

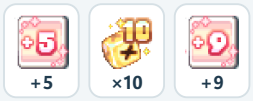
Evidence	Result (in brackets: value of Roll − (+11))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+11): 9.2)
Change only the rolls used	20 used: Roll (+7.6) · 70 used: Roll (+6.6) · 92 used: +11 (−0.8)
Replace the deck with a full 30	Roll (+7.6)
Gap after 4 rolls used	position part −7.8 (±1.2) · hand part +15.8 (±3.8) · total +8.0 (±3.7)
Progress over 4 rolls (mean)	Roll: 42.6 tiles, 1.8 cards drawn, 0.6 played · +11: 56.0 tiles, 2.6 drawn, 1.6 played
Full game (then VELA v4.2)	+10.0 [+7.8, +12.1], 2,048 pairs
Full game (then v4.3 Normal)	+7.2 [+3.1, +11.4], 512 pairs

Case 1-2 · VELA v4.2 record

Roll over the +9 draw, unless the deck backs it

Position **509** · Rolls used **32** (68 left)

Deck (11): −1 · +3 · +8×2 · +9 · +11 · +12 · ×2 · ×7 · ×8 · Stage



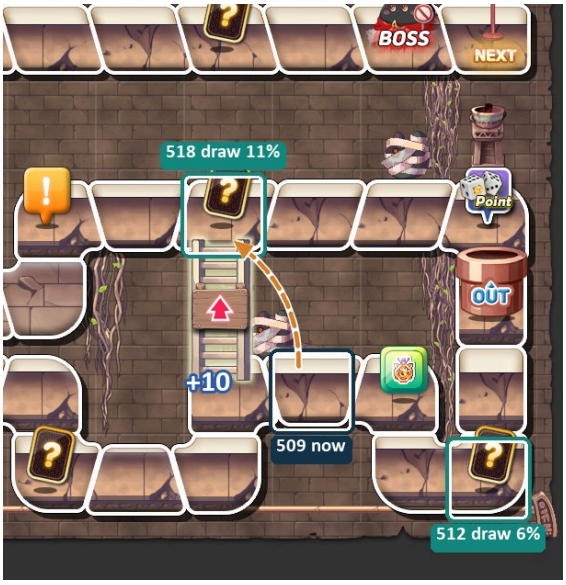


Figure 8: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the comparison action.

Of the +5, ×10 and +9 in hand, the +9 would draw at 518. The AI rolled (by 7.7 points). A roll draws with probability 16.7%.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,856.4**	**+7.0**	**16.7%**	**33.3%**
+9 · comparison	1,848.7	+9.0	100.0%	100.0%
×10	1,841.7	+71.5	19.4%	22.2%
+5	1,809.9	+5.0	0.0%	0.0%

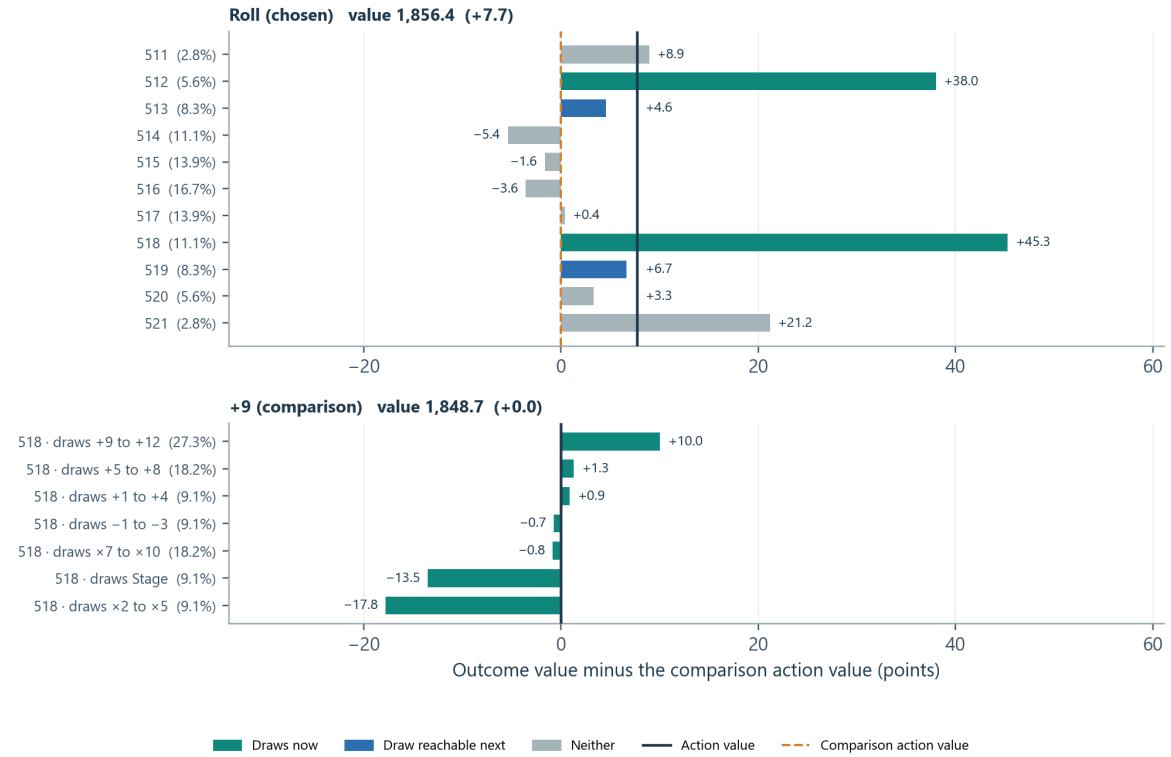


Figure 9: Outcome breakdown (reference: the value of the +9). 58.3% of roll outcomes are worth more than the +9. Of the cards the +9 can draw, only 18.2% beat the roll's value.

This position is sensitive to the deck. Replacing the deck with move cards that draw again from 518 (+10 · +10) makes the AI play the +9 at once; replacing it with the same number of move cards that cannot draw from 518 (+8 · +12) makes it roll (evidence table). The two decks have the same average card value. The only change is whether the card drawn at 518 can draw again right away. Removing another card from the hand (the +5 or the ×10), by contrast, does not change the choice.

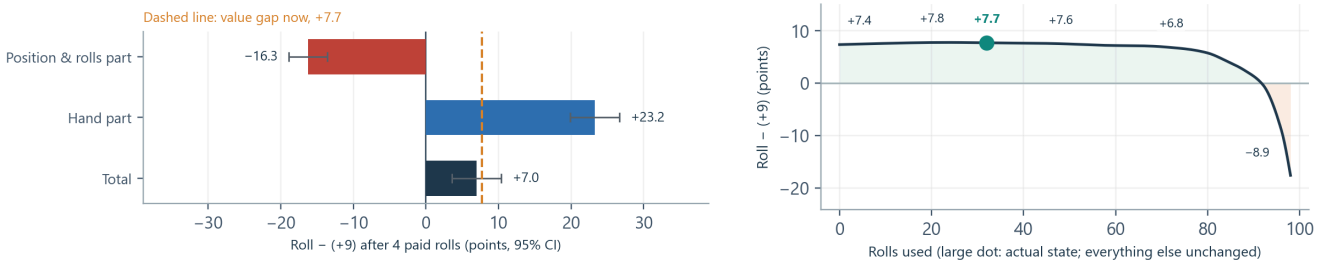


Figure 10: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+9)) when only the number of rolls used changes.

After four rolls the picture matches 1-1. The roll side still holds the +9 and leads by 23.2 points on hand; the +9 side is about 19 tiles further on and leads by 16.3 points on position. The same check was run on all 1,156 positions of this strategy. With the deck replaced by move cards that draw again from the same landing tile, the AI plays the card at once in 74.2% of them; with move cards that cannot, in 37.6%. Playing at once gained +3.1 points on average (higher in 89.4% of positions). "Save cards early" describes what the AI does when the card drawn now cannot draw again right away.

Evidence	Result (in brackets: value of Roll − (+9))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+9): 7.8)
Change only the rolls used	20 used: Roll (+7.8) · 70 used: Roll (+7.0) · 92 used: +9 (−0.3)
Remove one other card	no change (without +5: Roll · without ×10: Roll)
Replace the deck with a full 30	Roll (+6.4)
Gap after 4 rolls used	position part −16.3 (±2.7) · hand part +23.2 (±3.4) · total +7.0 (±3.4)
Progress over 4 rolls (mean)	Roll: 50.1 tiles, 1.4 cards drawn, 1.0 played · +9: 69.4 tiles, 2.4 drawn, 2.1 played
Full game (then VELA v4.2)	+7.6 [+5.9, +9.4], 2,048 pairs
Full game (then v4.3 Normal)	+8.2 [+4.9, +11.4], 512 pairs
Change only the deck	backing deck: +9 (−6.7) · non-backing deck: Roll (+5.4)

Case 1-3 · VELA v4.2 record

Contrast: a cheap card is played even early

Position **98** · Rolls used **5** (95 left)
Deck (22): −3 · +2 · +3 · +4 · +5×2 · +6×2 · +7×2 · +9 · +10×2 · +11 · +12 · ×2 · ×5 · ×7 · ×8 · ×10 · Stage×2



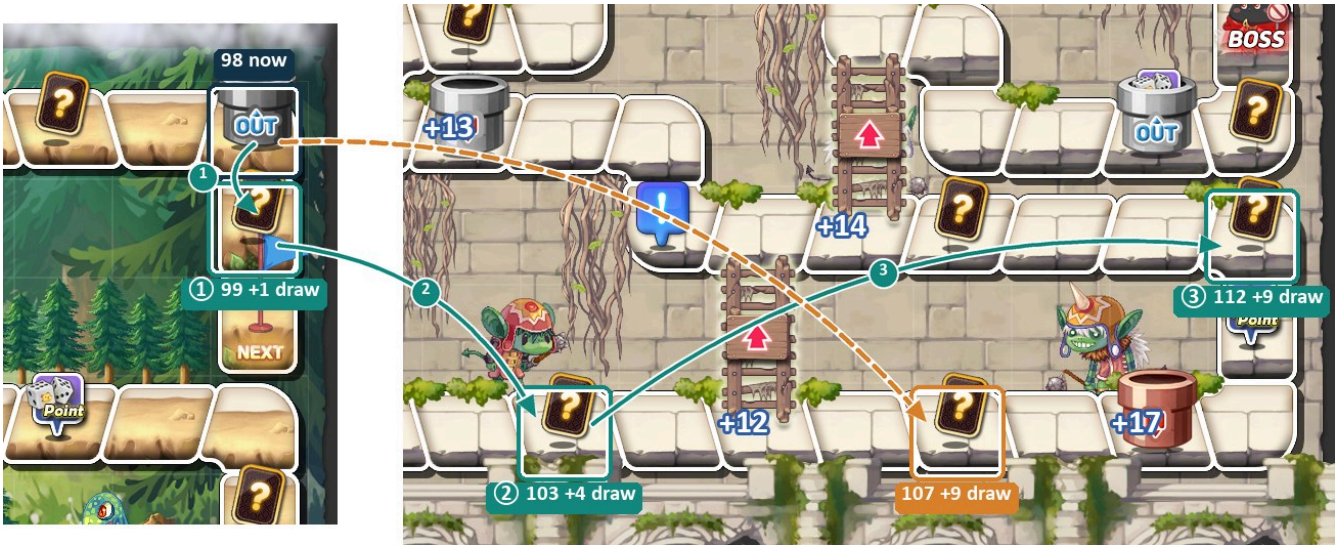


Figure 11: Board. Green arrows: the chosen action and what follows; numbers give the order. Orange dashed: the comparison action. Arrows that cross between boards are moves into another stage.

With 95 rolls left this is early in the game, yet the AI plays the +1 at once and draws at 99. What differs from 1-1 and 1-2 is the card spent. The +1 moves only one tile (holding value 38.0 points), so the trade is one cheap card for one card from the deck. The comparison action is the +9, which draws at 107.

Action	AI value	Mean move	Draws now	Draw possible by next action
+1 · chosen	**2,012.6**	**+1.0**	**100.0%**	**100.0%**
+9 · comparison	2,006.9	+9.0	100.0%	100.0%
Roll	2,005.0	+9.9	22.2%	75.0%
−1	1,974.9	−1.0	0.0%	0.0%
+4	1,971.6	+4.0	0.0%	100.0%

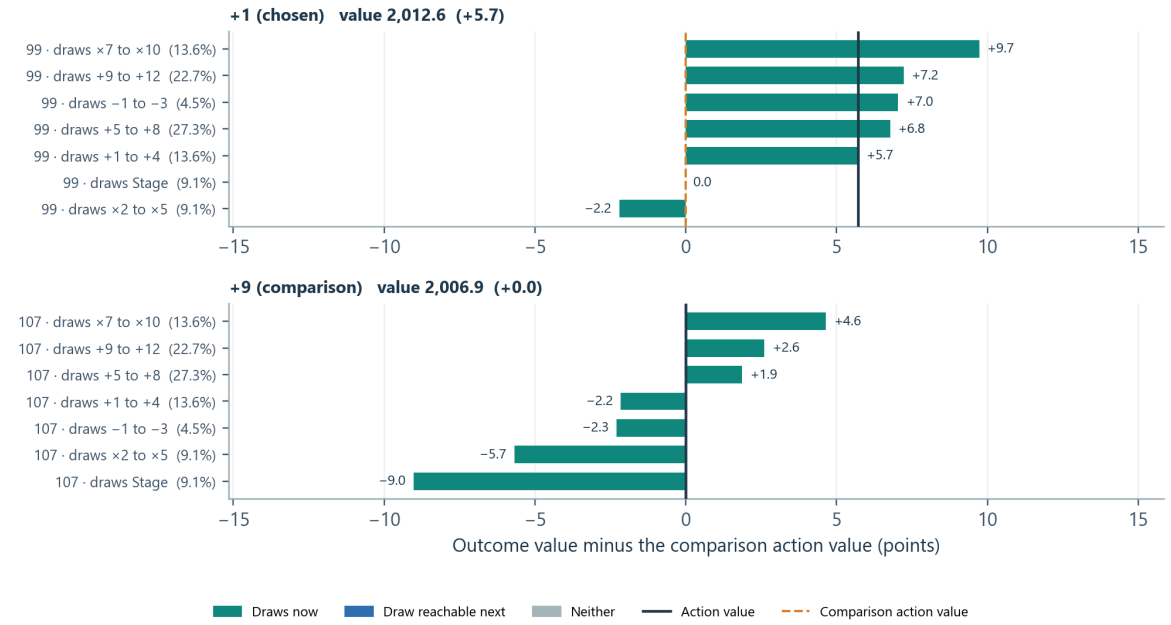


Figure 12: Outcome breakdown (reference: the value of the +9). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

After drawing at 99 with the +1, the +4 draws again at 103. In 88.6% of 1,000 play-outs the next action was the +4, drawing again (the rest played the card drawn at 99 first, or rolled). After drawing at 107 with the +9, no card in hand can draw right away.

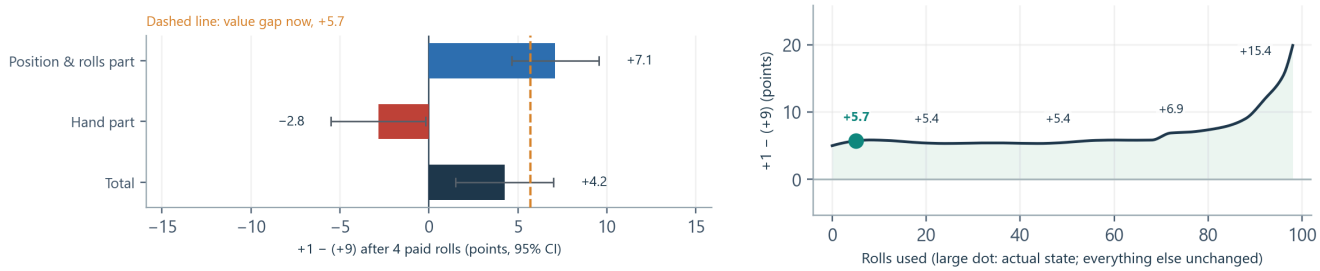


Figure 13: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (+1 − (+9)) when only the number of rolls used changes.

After four rolls the gap comes from the position part (+7.1 points). The +1 side had drawn 5.7 cards on average by then, more than the +9 side (4.3), and by playing those cards it is about 13 tiles further on. The hand part is almost even (−2.8 points): it played more cards but drew as many more. The counterfactuals agree. Removing another card (−1 or +4), replacing the deck with cards that cannot draw again from 99, or changing the rolls used to 20 (80 left) all leave the +1. The +4 following up helps, but even without it, trading the cheap +1 for a card beats rolling.

Evidence	Result (in brackets: value of +1 − (+9))
VELA v4.2, depth 1·2 (production)·3	+1 · +1 · +1
VELA v4.3 Normal·High·Max	+1 · +1 · +1
LYRA v2	+1 (+1 − (+9): 3.7)
Change only the rolls used	20 used: +1 (+5.4) · 70 used: +1 (+6.2) · 92 used: +1 (+11.9)
Remove one other card	no change (without −1: +1 · without +4: +1)
Replace the deck with a full 30	+1 (+5.5)
Gap after 4 rolls used	position part +7.1 (±2.5) · hand part −2.8 (±2.7) · total +4.2 (±2.7)
Progress over 4 rolls (mean)	+1: 107.6 tiles, 5.7 cards drawn, 6.0 played · +9: 94.8 tiles, 4.3 drawn, 4.4 played
Full game (then VELA v4.2)	+5.0 [+2.7, +7.2], 2,048 pairs
Full game (then v4.3 Normal)	+4.1 [+0.4, +7.8], 512 pairs
Change only the deck	backing deck: +1 (+6.1) · non-backing deck: +1 (+6.7)

2. Late on, spend several cards to draw one

In short

With few rolls left, the AI will play several cards in a row and draw only with the last one. In 1,926 such chains (two or more cards, a draw only at the end), changing only the rolls used to 20 (80 left) turns 72.3% into a roll. The fewer rolls are left, the fewer chances there are to use the hand, and turning cards into distance pays.

Case 2-1 · VELA v4.2 record

Four cards for 106 tiles and one draw

Position **1,686** · Rolls used **90** (10 left)

Deck (28): −3 · −2 · −1 · +1 · +2 · +3 · +4×2 · +5×2 · +6 · +7×2 · +8×2 · +9 · +10×2 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2





Figure 14: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

With ten rolls left, the AI plays the stage card, then the +9, the +7 and the +6, and draws one card at 1,792. The first three cards draw nothing, but the +9 and the +7 each ride a jump, so four cards cover 106 tiles. In 93% of 1,000 play-outs the four cards were played in this order; the rest moved once more with the card drawn.

Action	AI value	Mean move	Draws now	Draw possible by next action
Stage · chosen	**1,969.6**	**+38.0**	**0.0%**	**0.0%**
Roll · comparison	1,964.6	+8.2	8.3%	44.4%
+9	1,936.0	+9.0	0.0%	0.0%
+6	1,934.9	+6.0	0.0%	0.0%
+7	1,934.4	+7.0	0.0%	0.0%

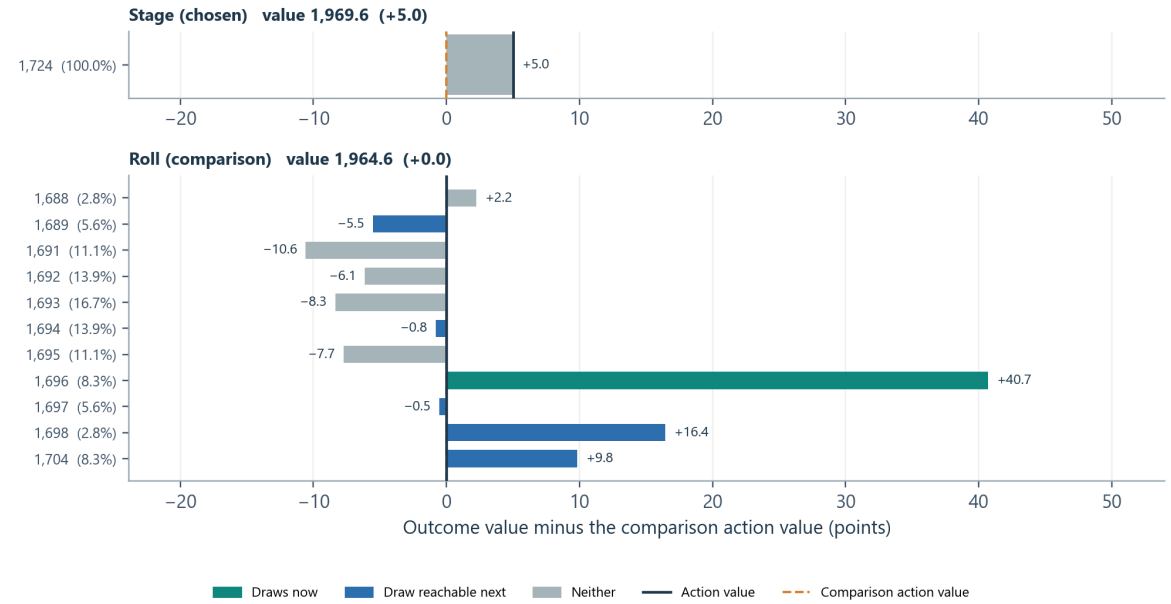


Figure 15: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

Taken one step at a time, the stage card spends a card and draws nothing. So VELA v4.2 at depth 1 rolls, and from depth 2, the production setting, it picks the stage card.

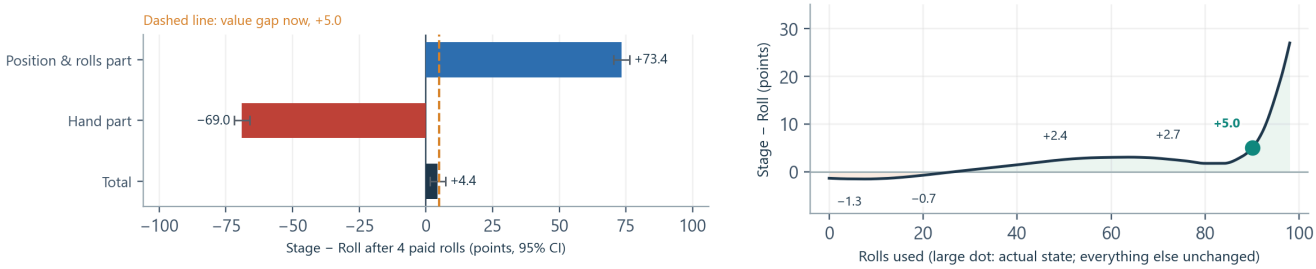


Figure 16: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Stage − Roll) when only the number of rolls used changes.

After four rolls the stage line is about 71 tiles ahead, 73.4 points up on position, and 69.0 points down on hand after spending almost every card (1.8 cards left on average against 3.6). The remaining rolls would not have been enough to use those cards anyway. With the rolls used changed to 20 (80 left), the roll edges ahead (−0.7 points): early in the game each of the four cards could buy its own draw.

Evidence	Result (in brackets: value of Stage − Roll)
VELA v4.2, depth 1·2 (production)·3	Roll · Stage · Stage
VELA v4.3 Normal·High·Max	Stage · Stage · Stage
LYRA v2	Stage (Stage − Roll: 1.3)
Change only the rolls used	20 used: Roll (−0.7) · 70 used: Stage (+2.8)
Remove one other card	flips in 1 of 3 (without +7: Stage · without +9: Roll · without +6: Stage)
Replace the deck with a full 30	Stage (+5.4)
Gap after 4 rolls used	position part +73.4 (±2.9) · hand part −69.0 (±2.9) · total +4.4 (±2.9)
Progress over 4 rolls (mean)	Stage: 149.1 tiles, 2.5 cards drawn, 4.7 played · Roll: 78.2 tiles, 2.7 drawn, 3.1 played
Full game (then VELA v4.2)	+3.7 [+3.3, +4.1], 2,048 pairs
Full game (then v4.3 Normal)	+3.2 [+2.6, +3.7], 512 pairs

Case 2-2 · VELA v4.2 record

The −1 rides a jump and the third card draws

Position **1,447** · Rolls used **91** (9 left)

Deck (19): −2 · +1 · +2 · +3 · +4 · +5×2 · +6 · +9×2 · +10×2 · +11 · ×2 · ×3 · ×5 · ×8 · Stage×2



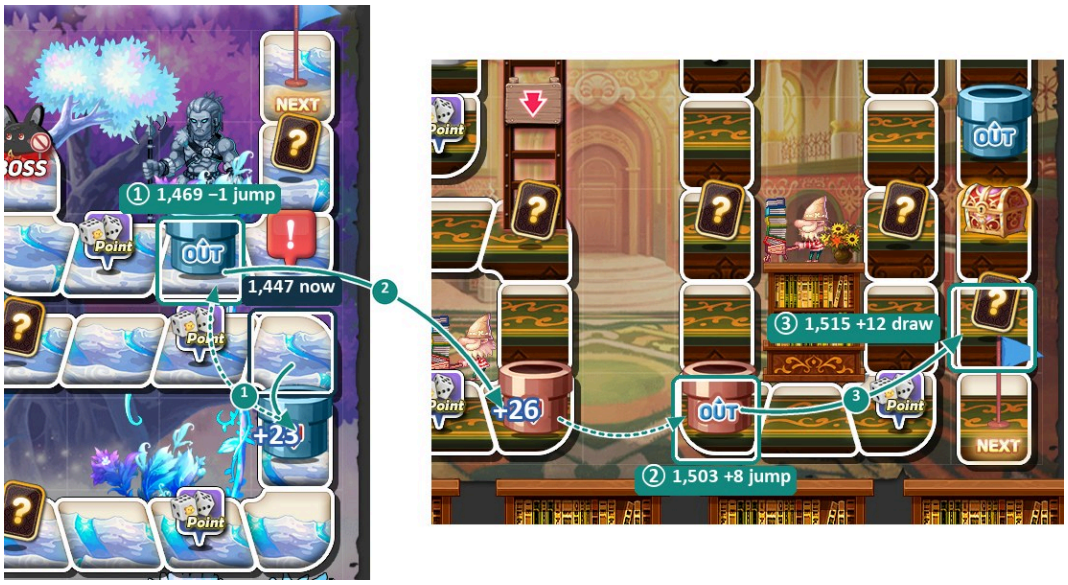


Figure 17: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

With nine rolls left, the AI plays the −1, the +8 and the +12, and draws with the third. The −1 rides the jump at 1,446 (+23) to 1,469. In 95% of 1,000 play-outs the three cards were played in this order.

Action	AI value	Mean move	Draws now	Draw possible by next action
−1 · chosen	**1,699.1**	**+22.0**	**0.0%**	**0.0%**
Roll · comparison	1,689.4	+7.8	11.1%	33.3%
+12	1,666.3	+12.0	0.0%	100.0%
+7	1,663.0	+7.0	0.0%	0.0%
+8	1,656.6	+8.0	0.0%	0.0%

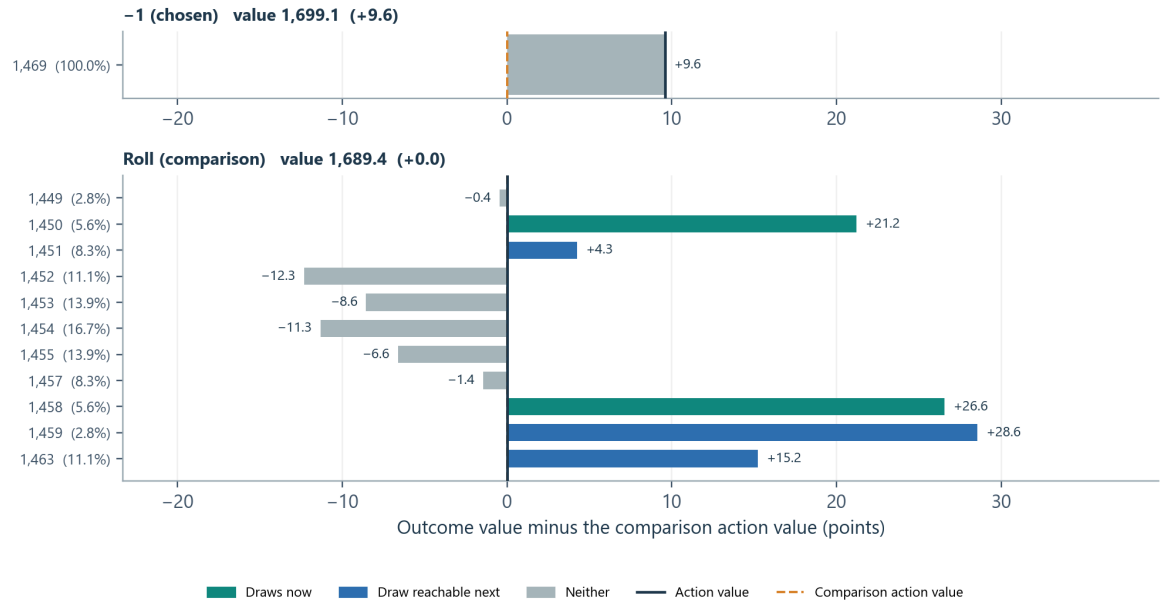


Figure 18: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

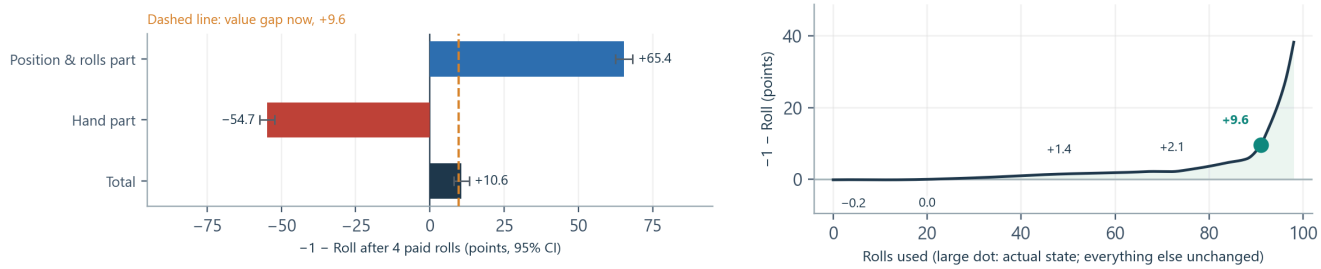


Figure 19: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−1 − Roll) when only the number of rolls used changes.

The structure matches 2-1: a position part of +65.4 points and a hand part of −54.7. With the rolls used changed to 20, the gap disappears (0.0 points). Removing the −1 that starts the chain turns the choice into a roll; removing any other card leaves the −1.

Evidence	Result (in brackets: value of −1 − Roll)
VELA v4.2, depth 1·2 (production)·3	−1 · −1 · −1
VELA v4.3 Normal·High·Max	−1 · −1 · −1
LYRA v2	−1 (−1 − Roll: 12.4)
Change only the rolls used	20 used: Roll (0.0) · 70 used: −1 (+2.3)
Remove one other card	no change (without +8: −1 · without +7: −1 · without +12: −1)
Replace the deck with a full 30	−1 (+10.6)
Gap after 4 rolls used	position part +65.4 (±2.9) · hand part −54.7 (±2.6) · total +10.6 (±2.7)
Progress over 4 rolls (mean)	−1: 132.6 tiles, 2.8 cards drawn, 4.8 played · Roll: 71.6 tiles, 2.8 drawn, 3.0 played
Full game (then VELA v4.2)	+10.8 [+10.3, +11.2], 2,048 pairs
Full game (then v4.3 Normal)	+10.9 [+10.2, +11.5], 512 pairs

3. With a full hand, play a card to clear a slot

In short

With five cards in hand, a roll that lands on a draw tile draws nothing. That lowers the roll's value, so playing a card tends to come out ahead. The evidence is counterfactual: remove one other card to leave four, and the AI rolls instead. Which card it plays is not settled by clearing a slot alone; whether that card rides a jump or leads to the next draw matters too (3-1, 3-2). With a full hand the AI sometimes still rolls (3-7, 3-8).

With a full hand, one more card is valued differently depending on the action. On the roll side the hand stays full, so removing a card frees a slot and costs little. On the card side a slot opens anyway, so the removed card's value is lost in full. The "loss when a card is removed" in this section's evidence tables and text shows this difference.

Case 3-1 · VELA v4.2 record

The −1 opens a slot and jumps to two draws

Position **785** · Rolls used **43** (57 left)
Deck (5): −2 · +4 · +5 · +8 · Stage





Figure 20: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The hand is +10, −1, +11, +4 and +3. The AI played the −1, the card with the lowest holding value (22.7 points). The −1 rides the jump at 784 to 794, and then the +11 and the +10 draw in turn. All 1,000 play-outs followed this order.

Action	AI value	Mean move	Draws now	Draw possible by next action
−1 · chosen	**1,970.6**	**+9.0**	**0.0%**	**100.0%**
Roll · comparison	1,960.8	+7.0	0.0%	47.2%
+10	1,948.3	+10.0	0.0%	100.0%
+3	1,948.3	+3.0	0.0%	100.0%
+11	1,943.8	+11.0	0.0%	0.0%
+4	1,942.7	+4.0	0.0%	0.0%

With a full hand the roll shows 0% under "draws now": a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 13.9%.

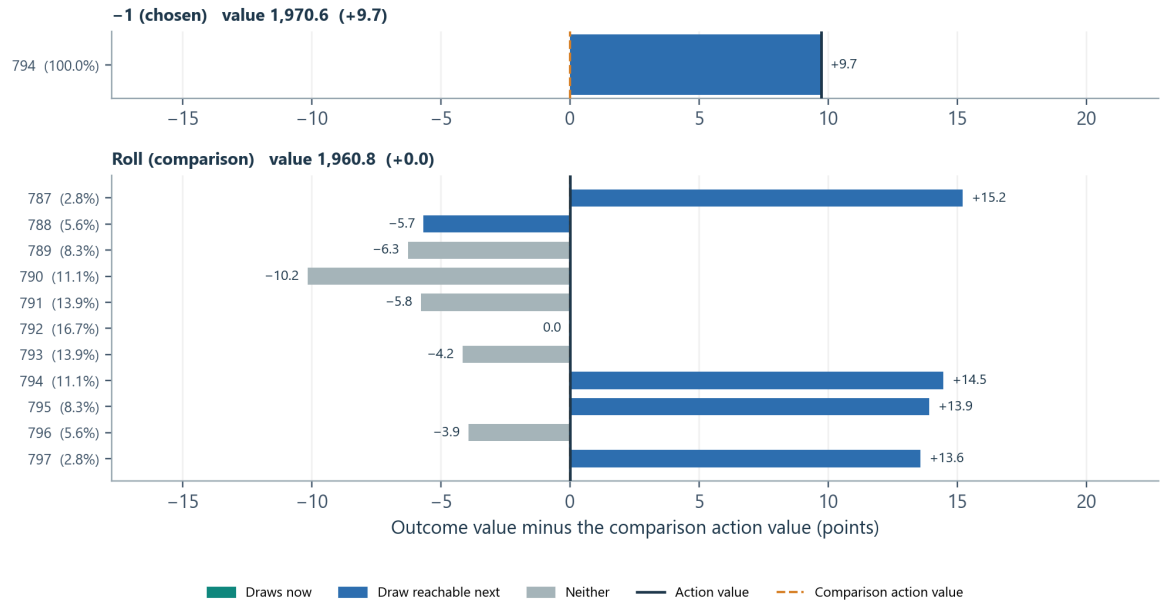


Figure 21: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

This choice needs two conditions at once. The first is the full hand: removing any card other than the −1 to leave four flips all four cases to a roll. The card side then loses more (removing the +11 costs it 60.9 points) than the roll side (29.5 points), because the roll side gains a free slot and can draw after the roll. The second is where the −1 lands. Keep the full hand but swap the −1 for a cheap card that does not ride a jump (+1, +2, −2 or −3), and playing that card falls 10–22 points below the roll, so the AI rolls. "Play the cheapest card" does not explain it on its own; the AI plays the −1 because it rides a jump to the draws of the +11 and the +10.

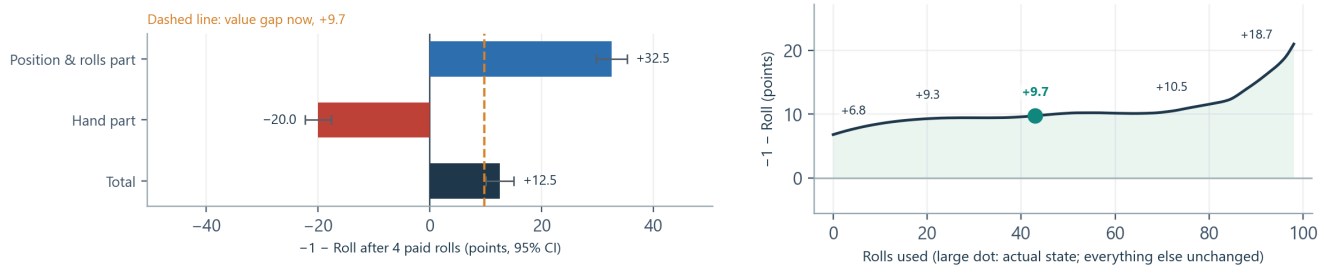


Figure 22: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−1 − Roll) when only the number of rolls used changes.

After four rolls the −1 side is about 30 tiles ahead, 32.5 points up on position, and 20.0 points down on hand after playing more cards. Changing the rolls used still leaves the −1.

Evidence	Result (in brackets: value of −1 − Roll)
VELA v4.2, depth 1·2 (production)·3	−1 · −1 · −1
VELA v4.3 Normal·High·Max	−1 · −1 · −1
LYRA v2	−1 (−1 − Roll: 9.5)
Change only the rolls used	20 used: −1 (+9.3) · 70 used: −1 (+10.3) · 92 used: −1 (+16.1)
Remove one other card	all flip (without +10: Roll · without +11: Roll · without +4: Roll · without +3: Roll)
Replace the deck with a full 30	−1 (+10.0)
Gap after 4 rolls used	position part +32.5 (±2.8) · hand part −20.0 (±2.4) · total +12.5 (±2.5)
Progress over 4 rolls (mean)	−1: 110.2 tiles, 4.6 cards drawn, 5.6 played · Roll: 80.5 tiles, 3.1 drawn, 3.7 played
Full game (then VELA v4.2)	+12.5 [+10.9, +14.0], 2,048 pairs
Full game (then v4.3 Normal)	+10.2 [+7.3, +13.0], 512 pairs

Case 3-2 · VELA v4.2 record

The −2 opens a slot and jumps to two draws

Position **895** · Rolls used **39** (61 left)

Deck (23): −3 · +1 · +2 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9 · +10 · +11 · +12 · ×5 · ×7 · ×8 · ×10 · Stage×2





Figure 23: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The hand is −1, +3, +9, −2 and +10. The AI played the −2 (23.2 points), which rides the jump at 893 to 902. The +9 and the +10 then each draw (in 98% of 1,000 play-outs in this order).

Action	AI value	Mean move	Draws now	Draw possible by next action
−2 · chosen	**2,187.1**	**+7.0**	**0.0%**	**100.0%**
Roll · comparison	2,177.9	+7.0	0.0%	58.3%
+3	2,167.6	+3.0	0.0%	100.0%
+9	2,167.6	+9.0	0.0%	100.0%
+10	2,157.4	+10.0	0.0%	0.0%
−1	2,153.6	−1.0	0.0%	0.0%

With a full hand the roll shows 0% under “draws now”: a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 2.8%.

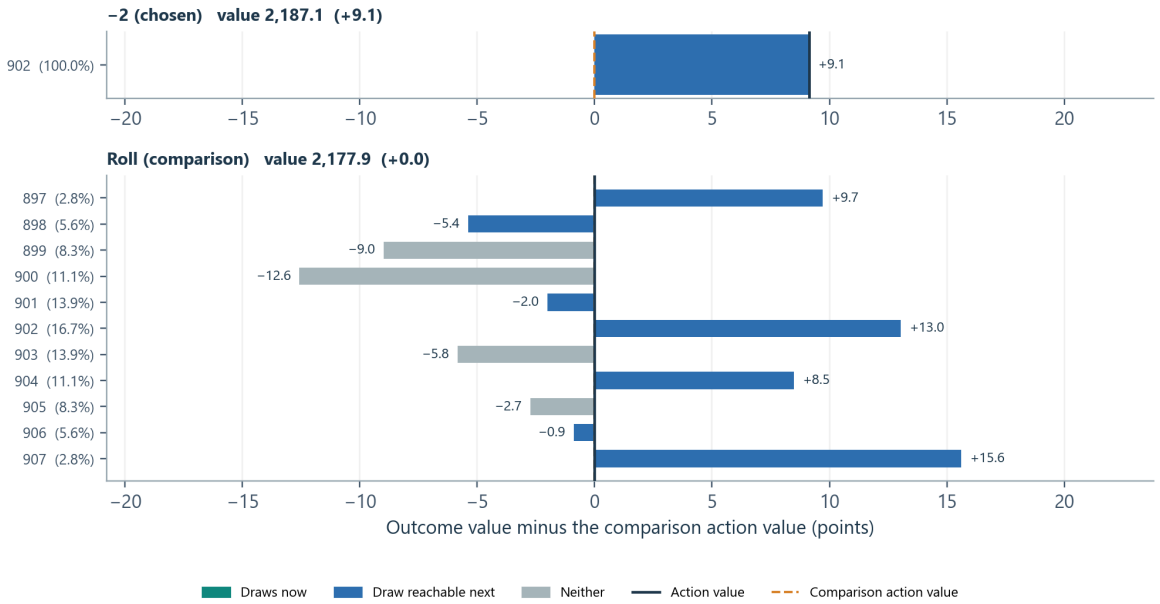


Figure 24: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

The structure matches 3-1. Removing any card other than the −2 flips all four cases to a roll, and swapping the −2 for a cheap card that does not ride a jump also gives a roll. The −1 in hand has almost the same holding value (22.9 points), but it lands on 894 and misses the jump. That is why the AI plays the −2 and not the −1.

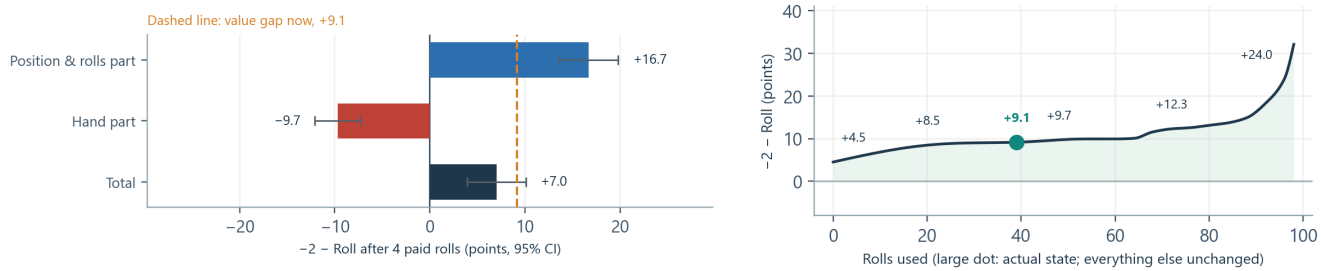


Figure 25: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−2 − Roll) when only the number of rolls used changes.

In both 3-1 and 3-2 the first card draws nothing. The draws come from the cards after it; the first card opens a slot in the hand and carries the token, by a jump, to where those draws are possible. After four rolls the −2 side is 16.7 points up on position and 9.7 points down on hand.

Evidence	Result (in brackets: value of −2 − Roll)
VELA v4.2, depth 1·2 (production)·3	−2 · −2 · −2
VELA v4.3 Normal·High·Max	−2 · −2 · −2
LYRA v2	−2 (−2 − Roll: 3.9)
Change only the rolls used	20 used: −2 (+8.5) · 70 used: −2 (+12.2) · 92 used: −2 (+18.3)
Remove one other card	all flip (without −1: Roll · without +3: Roll · without +9: Roll · without +10: Roll)
Replace the deck with a full 30	−2 (+8.7)
Gap after 4 rolls used	position part +16.7 (±3.1) · hand part −9.7 (±2.4) · total +7.0 (±3.1)
Progress over 4 rolls (mean)	−2: 125.7 tiles, 5.2 cards drawn, 6.3 played · Roll: 108.3 tiles, 4.4 drawn, 5.1 played
Full game (then VELA v4.2)	+4.9 [+3.3, +6.5], 2,048 pairs
Full game (then v4.3 Normal)	+5.0 [+2.1, +7.9], 512 pairs

Case 3-3 · VELA v4.2 record

A cheap ×3 clears a slot

Position **1,669** · Rolls used **82** (18 left)
Deck (2): +6 · Stage



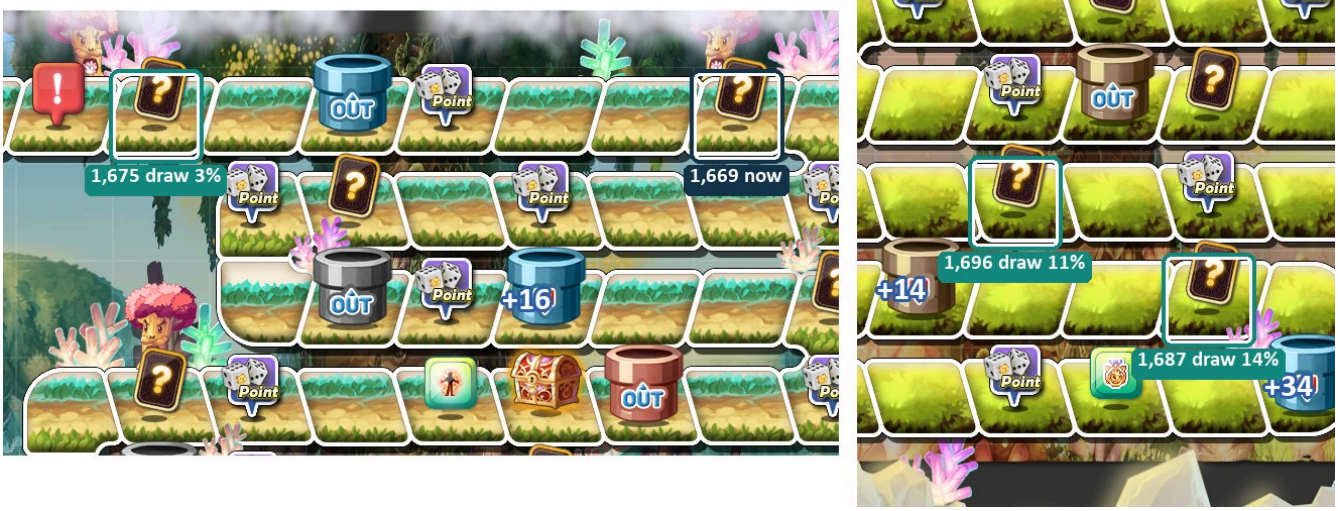


Figure 26: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile).

The hand is ×3, +10, +4, +3 and +4, with 18 rolls left. The AI played the ×3 (holding value 14.3 points, the lowest in hand). The ×3 uses a roll but moves 23.3 tiles on average and draws at once with probability 27.8%.

Action	AI value	Mean move	Draws now	Draw possible by next action
×3 · chosen	**2,117.7**	**+23.3**	**27.8%**	**77.8%**
Roll · comparison	2,106.1	+7.0	0.0%	38.9%
+4	2,088.9	+4.0	0.0%	0.0%
+3	2,084.2	+3.0	0.0%	0.0%
+10	2,083.1	+10.0	0.0%	0.0%

With a full hand the roll shows 0% under "draws now": a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 19.4%.

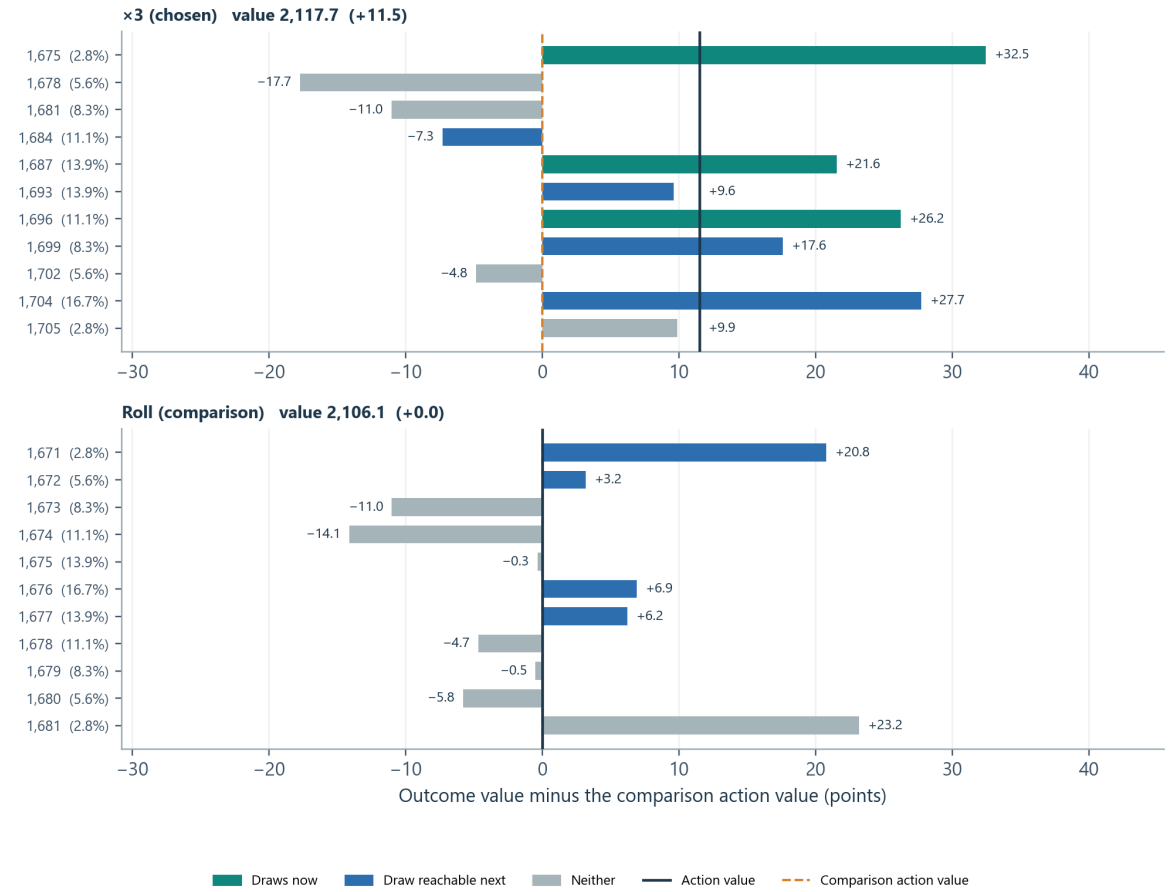


Figure 27: Outcome breakdown (reference: the value of the roll). 69.4% of ×3 outcomes are worth more than the roll.

Removing any card other than the ×3 flips all four cases to a roll. With four cards there is no reason to play the ×3. With the rolls used changed to 20 (80 left), the gap between the ×3 and the roll shrinks to +1.4 points: early in the game, even a full hand can afford to wait.

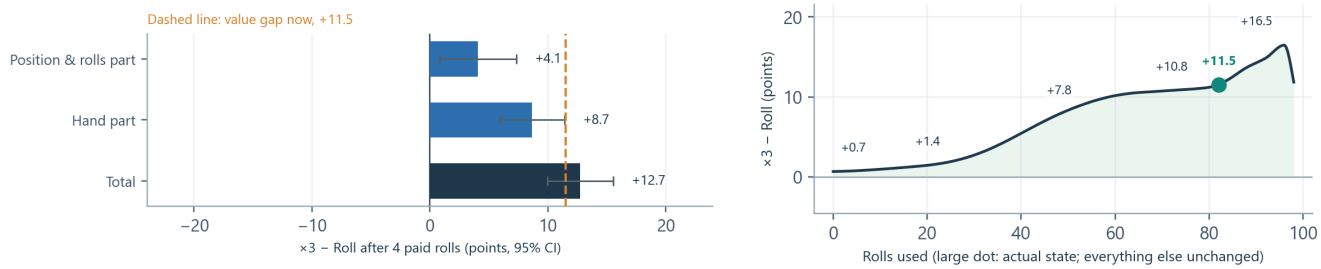


Figure 28: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×3 − Roll) when only the number of rolls used changes.

Here the gap comes from the hand rather than the position. After four rolls the position part is only +4.1 points and the hand part +8.7. The ×3 side spent the cheapest card and then drew 4.4 cards on average; the roll side, held back by its full hand, drew 3.0. In the play-outs, the ×3 was followed by three draws in a row (+3, +4, +10, or +10, +4, +4) in 31.7% of games. The roll side often spent the +10 without drawing, just to open a slot, or moved on with the stage card.

In the rolls-used chart the gap grows as more rolls are used: the fewer rolls are left, the larger the share of the remaining chances lost by one roll with a full hand. The 4.7-point drop at the very end (98 used, 2 left) is VELA's evaluation; LYRA v2 evaluates the same position with a slightly larger gap instead. Since the two AIs disagree there, read the rising trend before the end rather than the last point.

Evidence	Result (in brackets: value of ×3 − Roll)
VELA v4.2, depth 1·2 (production)·3	×3 · ×3 · ×3
VELA v4.3 Normal·High·Max	×3 · ×3 · ×3
LYRA v2	×3 (×3 − Roll: 12.5)
Change only the rolls used	20 used: ×3 (+1.4) · 70 used: ×3 (+10.7) · 92 used: ×3 (+14.8)
Remove one other card	all flip (without +10: Roll · without +4: Roll · without +3: Roll · without +4: Roll)
Replace the deck with a full 30	×3 (+12.2)
Gap after 4 rolls used	position part +4.1 (±3.3) · hand part +8.7 (±2.7) · total +12.7 (±2.8)
Progress over 4 rolls (mean)	×3: 98.2 tiles, 4.4 cards drawn, 5.6 played · Roll: 94.0 tiles, 3.0 drawn, 4.2 played
Full game (then VELA v4.2)	+12.5 [+11.7, +13.3], 2,048 pairs
Full game (then v4.3 Normal)	+12.0 [+10.7, +13.3], 512 pairs

Case 3-4 · VELA v4.2 record

The ×2 clears a slot, with a second reason

Position **87** · Rolls used **5** (95 left)
Deck (22): −3 · −2 · +1 · +2 · +3 · +4×2 · +5 · +6×2 · +7 · +8 · +9×2 · +11 · +12 · ×5 · ×7 · ×8 · ×10 · Stage×2







Figure 29: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile).

The hand is ×2, +7, −1, +5 and +8, with 95 rolls left. The AI played the ×2 (8.9 points). This time removing a card other than the ×2 flips the choice in only 2 of 4 cases (removing the +5 or the +8).

Action	AI value	Mean move	Draws now	Draw possible by next action
×2 · chosen	**2,024.1**	**+16.3**	**41.7%**	**75.0%**
Roll · comparison	2,014.1	+7.0	0.0%	80.6%
+7	1,995.3	+7.0	0.0%	100.0%
−1	1,995.3	−1.0	0.0%	100.0%
+5	1,990.9	+5.0	0.0%	100.0%
+8	1,988.8	+8.0	0.0%	0.0%

With a full hand the roll shows 0% under "draws now": a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 27.8%.

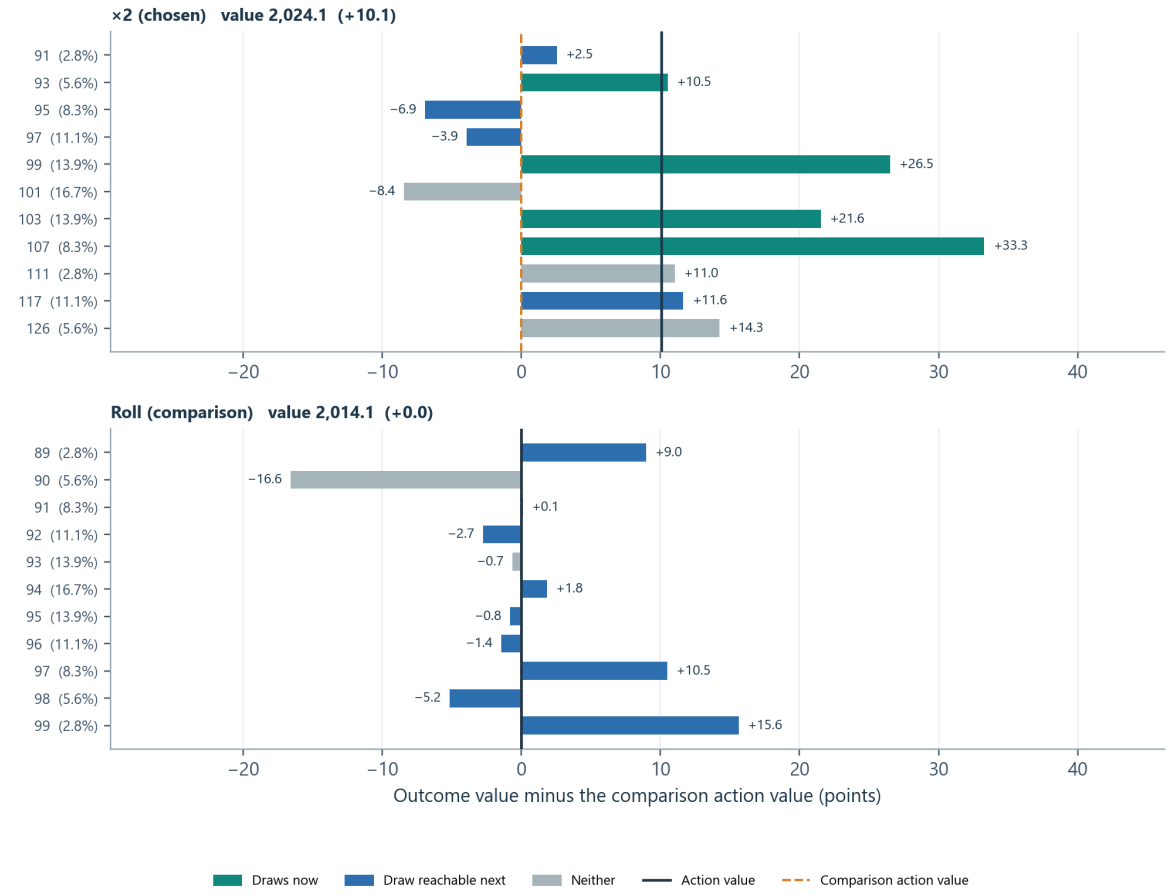


Figure 30: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

Clearing a slot does not explain this one fully. The ×2 is the cheapest card in hand and, from this tile, draws at once with probability 41.7%. With one other card removed, the ×2 and the roll come within −0.9 to +1.2 points of each other, so which card is removed decides the choice. Even with four cards the ×2 is worth about as much as a roll because a cheap card has a good chance to draw from here. Two factors work together: clearing a slot, and "a cheap card with a good chance to draw".

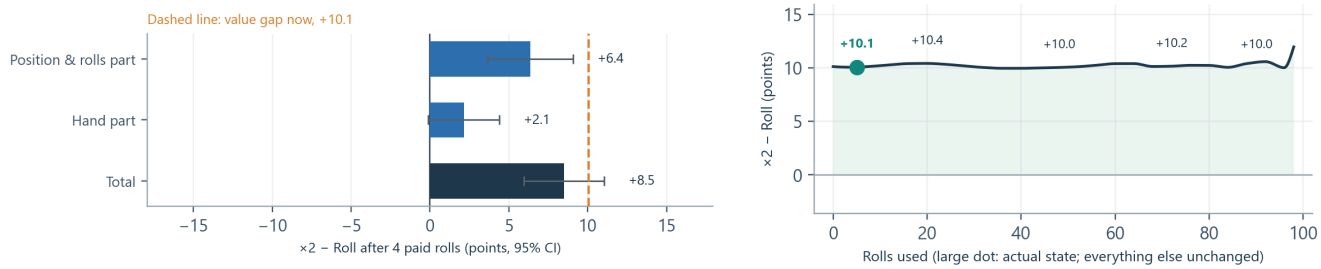


Figure 31: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×2 − Roll) when only the number of rolls used changes.

After four rolls the ×2 side is ahead by +6.4 points on position and +2.1 on hand. It spent the cheapest card, so it lost almost nothing in hand. The rolls-used chart is nearly flat (10–10.6 points) and rises by about 2 points only at the end (98 used). At the end of the game a ×2 in hand is worth almost nothing (Section 1.4), so using it to double a roll's move gains a little more.

Evidence	Result (in brackets: value of ×2 − Roll)
VELA v4.2, depth 1·2 (production)·3	×2 · ×2 · ×2
VELA v4.3 Normal·High·Max	×2 · ×2 · ×2
LYRA v2	×2 (×2 − Roll: 6.1)
Change only the rolls used	20 used: ×2 (+10.4) · 70 used: ×2 (+10.1) · 92 used: ×2 (+10.6)
Remove one other card	flips in 2 of 4 (without +7: ×2 · without −1: ×2 · without +5: Roll · without +8: Roll)
Replace the deck with a full 30	×2 (+9.7)
Gap after 4 rolls used	position part +6.4 (±2.7) · hand part +2.1 (±2.3) · total +8.5 (±2.5)
Progress over 4 rolls (mean)	×2: 101.3 tiles, 4.6 cards drawn, 5.4 played · Roll: 93.4 tiles, 4.2 drawn, 5.1 played
Full game (then VELA v4.2)	+11.6 [+9.5, +13.6], 2,048 pairs
Full game (then v4.3 Normal)	+11.7 [+8.0, +15.5], 512 pairs

Case 3-5 · VELA v4.2 record

A stage card clears a slot, then the +6 draws

Position **485** · Rolls used **28** (72 left)

Deck (13): −3 · −2 · −1 · +1 · +2 · +4 · +5 · +6 · +7 · ×3 · ×7 · ×8 · Stage





Figure 32: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The hand is +9, +6, +3, +10 and a stage card, with 72 rolls left. The stage card (28.4 points) is the cheapest. After the stage card takes the token to 499, the +6 and the +3 draw in turn (in all 1,000 play-outs), and the +10 drew once more in 86% of them.

Action	AI value	Mean move	Draws now	Draw possible by next action
Stage · chosen	**1,955.5**	**+14.0**	**0.0%**	**100.0%**
Roll · comparison	1,946.3	+7.0	0.0%	58.3%
+3	1,920.7	+3.0	0.0%	100.0%
+6	1,920.4	+6.0	0.0%	0.0%
+10	1,917.2	+10.0	0.0%	0.0%
+9	1,916.0	+9.0	0.0%	100.0%

With a full hand the roll shows 0% under “draws now”: a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 16.7%.

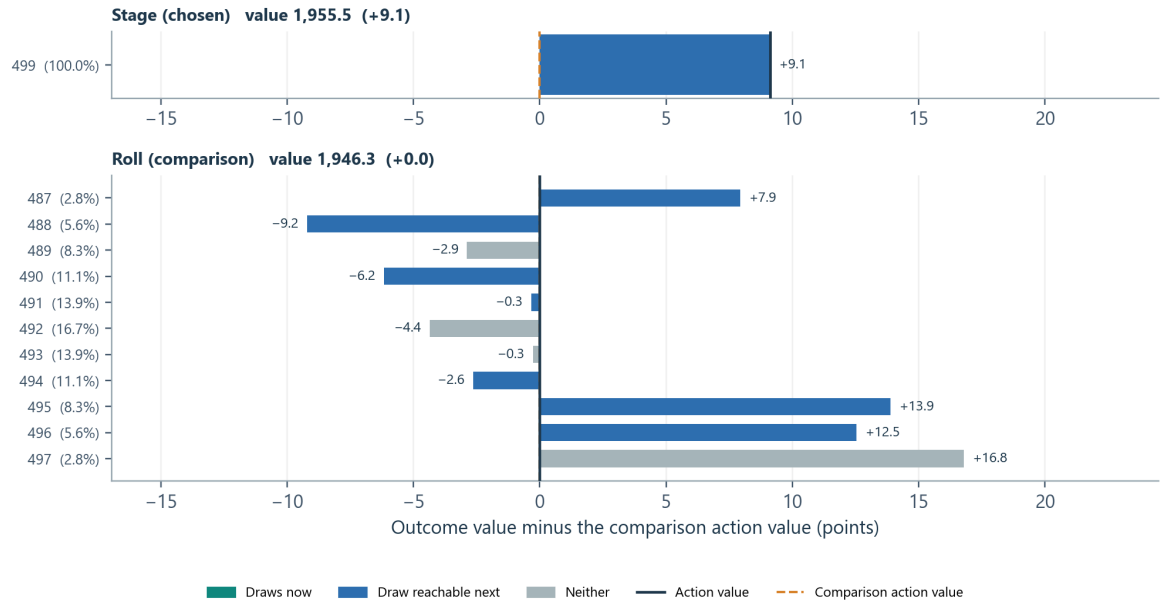


Figure 33: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

Removing any card other than the stage card flips all four cases to a roll. Again the card side loses more (57–62 points) than the roll side (32–42 points).

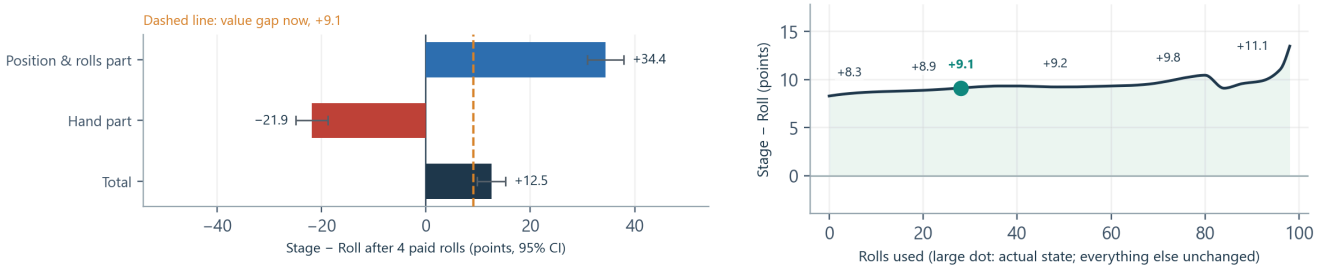


Figure 34: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Stage − Roll) when only the number of rolls used changes.

After four rolls the stage side is about 33 tiles ahead, 34.4 points up on position, and 21.9 points down on hand after playing more cards. The 1.4-point dip around 84 rolls used (16 left) in the rolls-used chart is one of the small mid-range wiggles described in Section 1.5. The gap stays above 8 points everywhere, so the choice never changes, and it grows at the end: with few rolls left, the +6, +3 and +10 that draw in turn are worth more as distance. Computed exactly with two rolls left (98 used), the stage card is 17.6 points above the roll: the same direction as the chart's end value (13.5), and an even larger gap.

Evidence	Result (in brackets: value of Stage − Roll)
VELA v4.2, depth 1·2 (production)·3	Stage · Stage · Stage
VELA v4.3 Normal·High·Max	Stage · Stage · Stage
LYRA v2	Stage (Stage − Roll: 9.2)
Change only the rolls used	70 used: Stage (+9.7) · 92 used: Stage (+9.9)
Remove one other card	all flip (without +9: Roll · without +6: Roll · without +3: Roll · without +10: Roll)
Replace the deck with a full 30	Stage (+9.7)
Gap after 4 rolls used	position part +34.4 (±3.5) · hand part −21.9 (±3.1) · total +12.5 (±2.7)
Progress over 4 rolls (mean)	Stage: 132.6 tiles, 5.7 cards drawn, 6.8 played · Roll: 99.9 tiles, 3.6 drawn, 4.4 played
Full game (then VELA v4.2)	+12.2 [+10.5, +13.9], 2,048 pairs
Full game (then v4.3 Normal)	+12.6 [+9.4, +15.8], 512 pairs

Case 3-6 · VELA v4.2 record

A stage card clears a slot; no draw follows

Position **717** · Rolls used **39** (61 left)

Deck (28): −3 · −2 · +1 · +3 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2

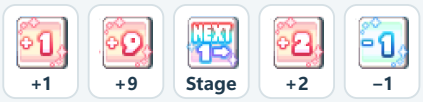




Figure 35: Board. Green: the chosen action. Arrows that cross between boards are moves into another stage.

The hand is +1, +9, a stage card, +2 and −1, with 61 rolls left. The stage card (19.1 points) takes the token to 732, where no card in hand can draw on the next action. The AI played it anyway. Removing any card other than the stage card flips all four cases to a roll.

Action	AI value	Mean move	Draws now	Draw possible by next action
Stage · chosen	**1,964.0**	**+15.0**	**0.0%**	**0.0%**
Roll · comparison	1,959.6	+7.0	0.0%	41.7%
+9	1,937.1	+9.0	0.0%	0.0%
−1	1,935.8	−1.0	0.0%	0.0%
+1	1,934.2	+1.0	0.0%	0.0%
+2	1,932.8	+2.0	0.0%	0.0%

With a full hand the roll shows 0% under “draws now”: a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 19.4%.

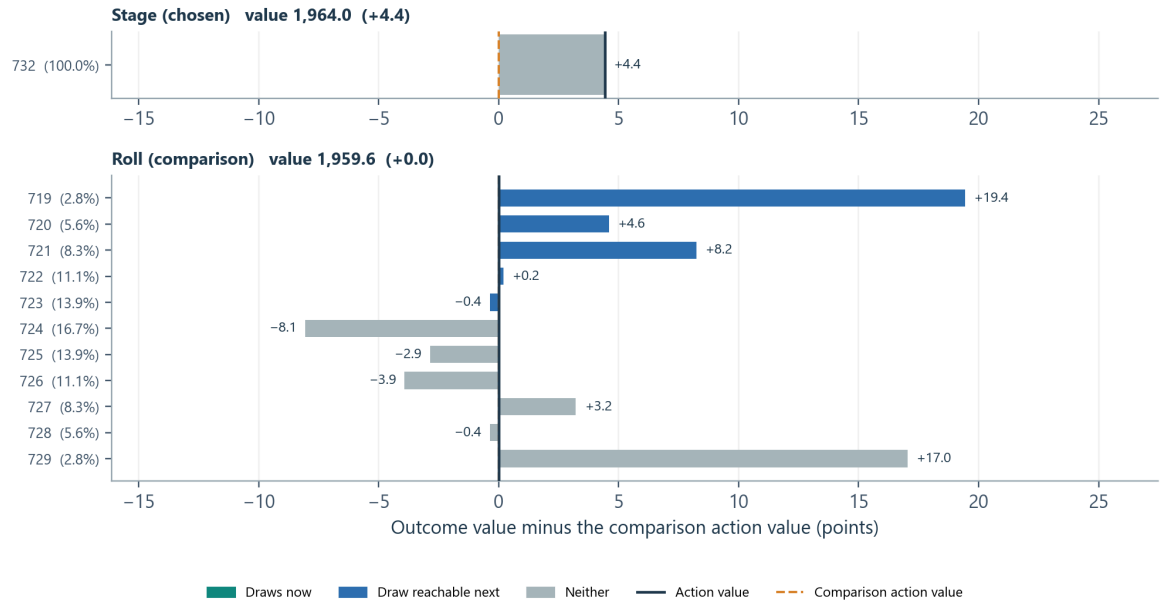


Figure 36: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

Unlike 3-5, no draw follows the stage card here. It buys 15 tiles and one free slot, nothing more. Rolling with a full hand risks landing on a draw tile and drawing nothing. With the rolls used changed to 92 (8 left) the choice flips to a roll: near the end there are fewer chances to use the free slot.

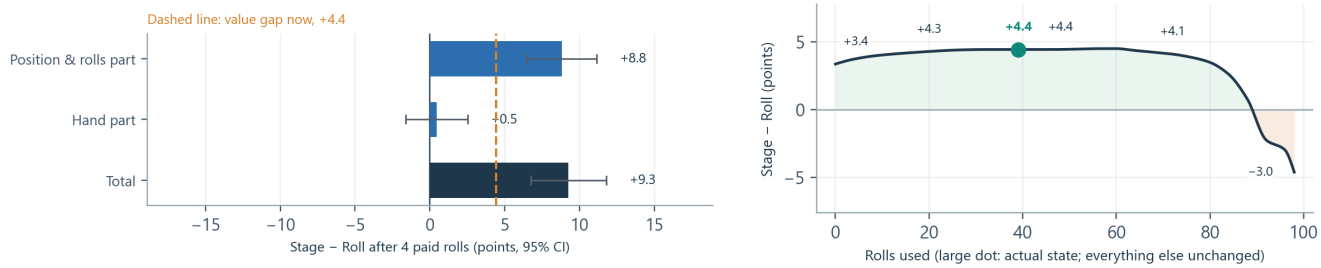


Figure 37: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Stage − Roll) when only the number of rolls used changes.

After four rolls the stage side is +8.8 points ahead on position, and the hand part is even (+0.5 points). Thanks to the free slot it drew 3.4 cards in those four rolls against 2.3 on the roll side, which made up for the stage card it spent. The roll side plays the stage card too, sooner or later: 29.7% of its play-outs went roll, roll, stage card. The difference is only when the card is played, and playing it first means the next two rolls are made with a free slot.

Evidence	Result (in brackets: value of Stage − Roll)
VELA v4.2, depth 1·2 (production)·3	Stage · Stage · Stage
VELA v4.3 Normal·High·Max	Stage · Stage · Stage
LYRA v2	Stage (Stage − Roll: 4.4)
Change only the rolls used	20 used: Stage (+4.3) · 70 used: Stage (+4.2) · 92 used: Roll (−2.3)
Remove one other card	all flip (without +1: Roll · without +9: Roll · without +2: Roll · without −1: Roll)
Replace the deck with a full 30	Stage (+4.4)
Gap after 4 rolls used	position part +8.8 (±2.3) · hand part +0.5 (±2.1) · total +9.3 (±2.5)
Progress over 4 rolls (mean)	Stage: 84.4 tiles, 3.4 cards drawn, 4.1 played · Roll: 74.8 tiles, 2.3 drawn, 3.0 played
Full game (then VELA v4.2)	+4.5 [+2.9, +6.1], 2,048 pairs
Full game (then v4.3 Normal)	+5.9 [+2.9, +8.9], 512 pairs

Case 3-7 · VELA v4.2 record

A full hand, yet a roll beats the +10 draw

Position **1,701** · Rolls used **77** (23 left)
Deck (7): +1 · +4 · +5 · +6 · ×7 · Stage×2

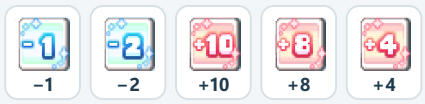




Figure 38: Board. Orange dashed: the comparison action.

The hand is −1, −2, +10, +8 and +4. The +10 could draw at 1,711, but the AI rolled (by 8.5 points). The +10 leaves the hand as it is played, so it can draw at 1,711. Removing any one card leaves the roll in all four cases, so the full hand is not the issue.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**2,243.9**	**+7.0**	**0.0%**	**77.8%**
+10 · comparison	2,235.4	+10.0	100.0%	100.0%
−2	2,233.3	−2.0	0.0%	100.0%
+4	2,233.3	+4.0	0.0%	100.0%
+8	2,231.8	+8.0	0.0%	100.0%
−1	2,220.8	+13.0	0.0%	0.0%

With a full hand the roll shows 0% under “draws now”: a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 25.0%.

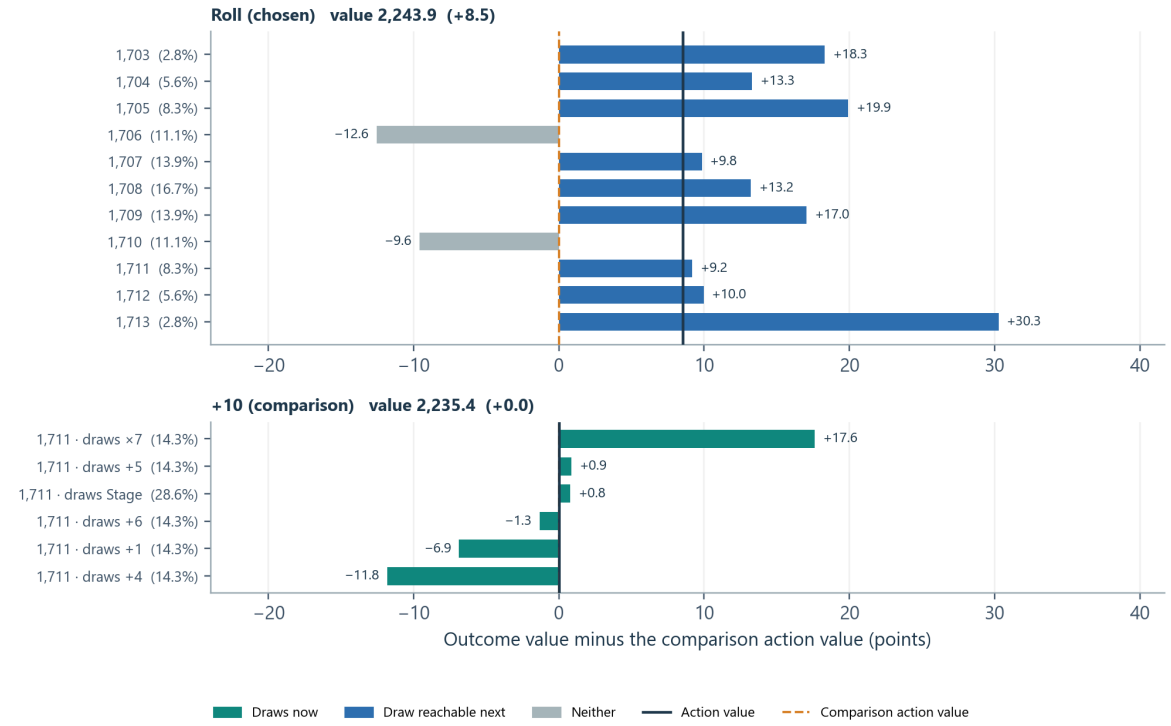


Figure 39: Outcome breakdown (reference: the value of the +10). 77.8% of roll outcomes are worth more than the +10. Most of them are “draw reachable next” (blue).

Playing the +10 opens a slot too, so the difference is not the slot but what follows. The draw tiles around here are 1,707, 1,711 and 1,721, and the cards in hand link them: the +4 goes from 1,707 to 1,711 and the +10 from 1,711 to 1,721, drawing each time. Playing the +10 now draws at 1,711, but from 1,711 no card in hand reaches the next draw tile (the −1 goes to 1,710, the −2 to 1,709, the +4 to 1,715, the +8 to 1,719). So the next action is a roll, and after that a card comes only when a roll lands on a draw tile.

A roll lands between 1,703 and 1,713, and from most of those tiles one of the −1, −2, +4 or +8 leads to 1,707 or 1,711; the +4 and the +10 then draw in turn. With a sum of 7 to 1,708, for example, the −1 draws at 1,707, the +4 at 1,711 and the +10 at 1,721. In 52.6% of 1,000 play-outs the roll was followed by three draws in a row, and in 70.3% by at least two. The +10 is the last link of this chain, which is why it is worth more kept (holding value 40.3 points).

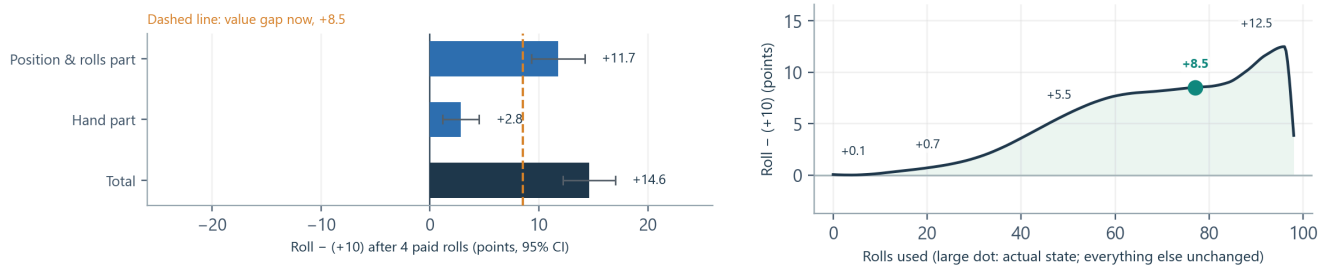


Figure 40: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+10)) when only the number of rolls used changes.

After four rolls the roll side is ahead by 11.7 points on position, and by +2.8 on hand. In those four rolls it drew 5.3 cards, more than the +10 side (3.9), and moved further by playing them.

Evidence	Result (in brackets: value of Roll − (+10))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+10): 10.1)
Change only the rolls used	20 used: Roll (+0.7) · 92 used: Roll (+11.7)
Remove one other card	no change (without −1: Roll · without −2: Roll · without +8: Roll · without +4: Roll)
Replace the deck with a full 30	Roll (+7.5)
Gap after 4 rolls used	position part +11.7 (±2.4) · hand part +2.8 (±1.7) · total +14.6 (±2.4)
Progress over 4 rolls (mean)	Roll: 102.2 tiles, 5.3 cards drawn, 6.2 played · +10: 88.4 tiles, 3.9 drawn, 4.8 played
Full game (then VELA v4.2)	+12.4 [+11.6, +13.2], 2,048 pairs
Full game (then v4.3 Normal)	+11.9 [+10.4, +13.4], 512 pairs

Case 3-8 · VELA v4.2 record

A full hand, yet a roll beats the +7 draw

Position **486** · Rolls used **33** (67 left)

Deck (9): −1 · +6×2 · +8 · +10 · +11 · ×2 · ×8 · Stage

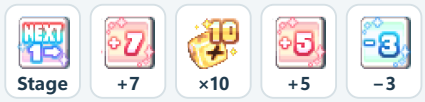




Figure 41: Board. Orange dashed: the comparison action.

The hand is a stage card, +7, ×10, +5 and −3, with 67 rolls left. The +7 would draw at 493, but the AI rolled. The margin is small: 3.0 points.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,877.8**	**+7.0**	**0.0%**	**36.1%**
+7 · comparison	1,874.8	+7.0	100.0%	100.0%
×10	1,870.1	+70.3	33.3%	58.3%
Stage	1,854.2	+13.0	0.0%	0.0%
+5	1,852.2	+5.0	0.0%	0.0%
−3	1,846.8	−3.0	0.0%	0.0%

With a full hand the roll shows 0% under “draws now”: a roll that lands on a draw tile draws nothing. With a free slot, the roll would land on a draw tile with probability 22.2%.

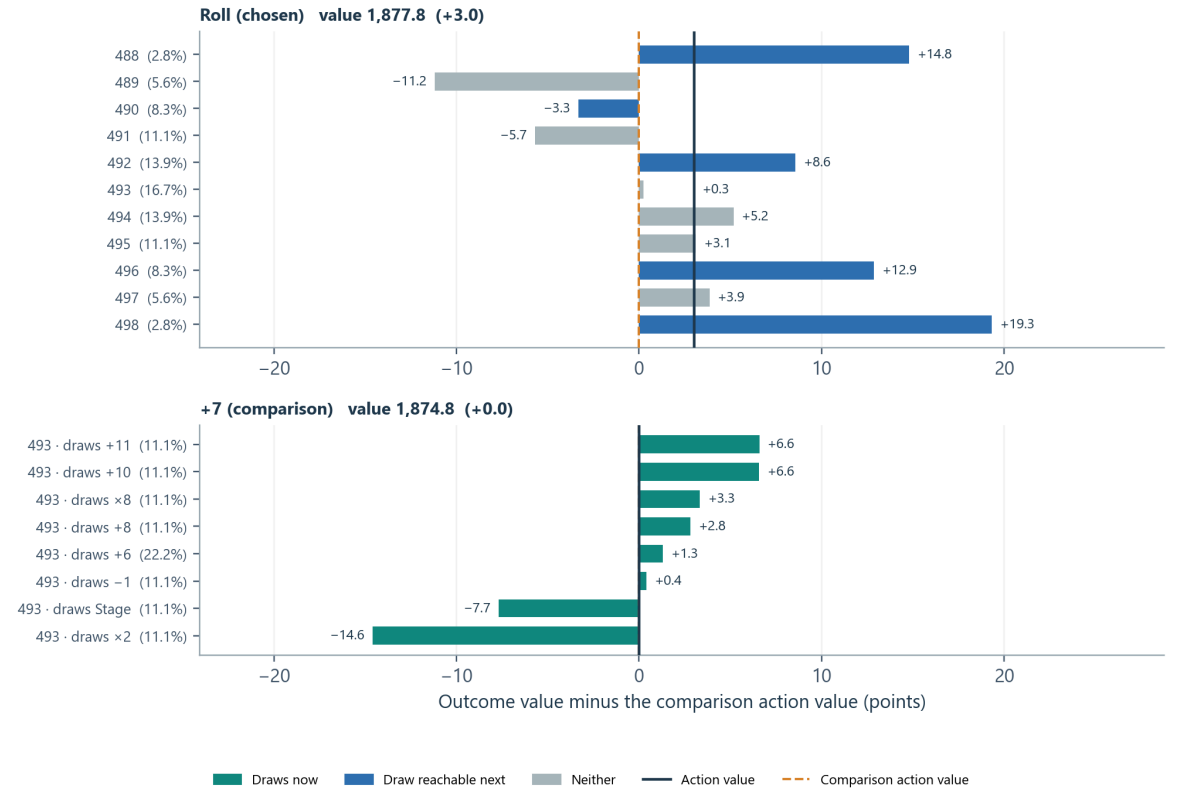


Figure 42: Outcome breakdown (reference: the value of the +7). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

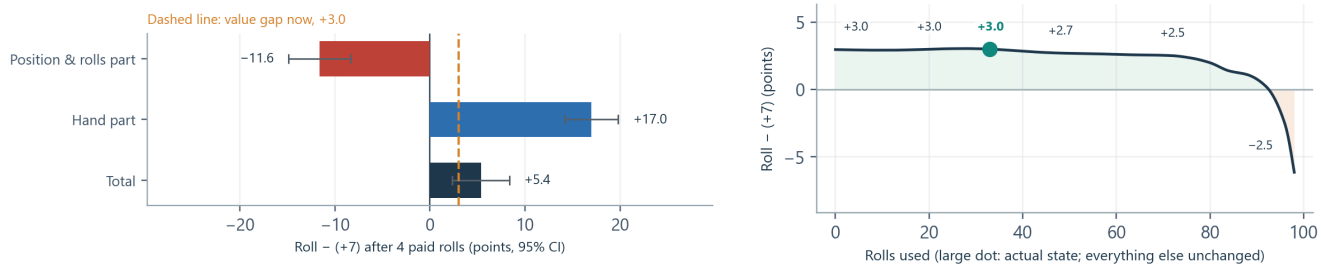


Figure 43: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+7)) when only the number of rolls used changes.

After four rolls the +7 side is ahead on position (11.6 points) and the roll side on hand (17.0 points). Rolling while holding the +7 (47.8 points) and the ×10 (58.8 points) is worth slightly more than one card drawn now.

The counterfactuals go two ways. Removing the ×10, the +5 or the −3 widens the roll's lead, while removing the stage card flips the choice to the +7. The stage card is part of the roll side's plan: 47.8% of the play-outs that started with a roll went roll, roll, stage card, then +7 (draw). After two rolls, the stage card takes the token into the next stage, where the saved +7 draws. Without it the roll side loses 36.7 points and the +7 side 31.8, which reverses the order. The gap is small, but in the full-game comparison the roll side still came out ahead (evidence table).

Evidence	Result (in brackets: value of Roll − (+7))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+7): 4.4)
Change only the rolls used	20 used: Roll (+3.0) · 70 used: Roll (+2.5) · 92 used: Roll (+0.2)
Remove one other card	flips in 1 of 4 (without Stage: +7 · without ×10: Roll · without +5: Roll · without −3: Roll)
Replace the deck with a full 30	Roll (+1.0)
Gap after 4 rolls used	position part −11.6 (±3.3) · hand part +17.0 (±2.8) · total +5.4 (±3.0)
Progress over 4 rolls (mean)	Roll: 111.6 tiles, 1.9 cards drawn, 2.9 played · +7: 121.9 tiles, 2.8 drawn, 3.8 played
Full game (then VELA v4.2)	+6.2 [+4.3, +8.1], 2,048 pairs
Full game (then v4.3 Normal)	+7.3 [+3.9, +10.7], 512 pairs

4. Before a stop tile, keep move cards even when the roll is blocked

In short

Right before a stop tile (boss tile), a roll can only move one or two tiles. Even so, the AI does not spend a move card to go a few tiles further. One card is worth far more (about 40–60 points) than a few tiles.

Case 4-1 · VELA v4.2 record

A roll that moves 2 tiles against a +5 that moves 5



Position **358** · Rolls used **30** (70 left)

Deck (25): −3 · −2 · −1 · +1 · +2 · +4 · +5 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · +12 · ×2 · ×5 · ×8 · ×10 · Stage×2

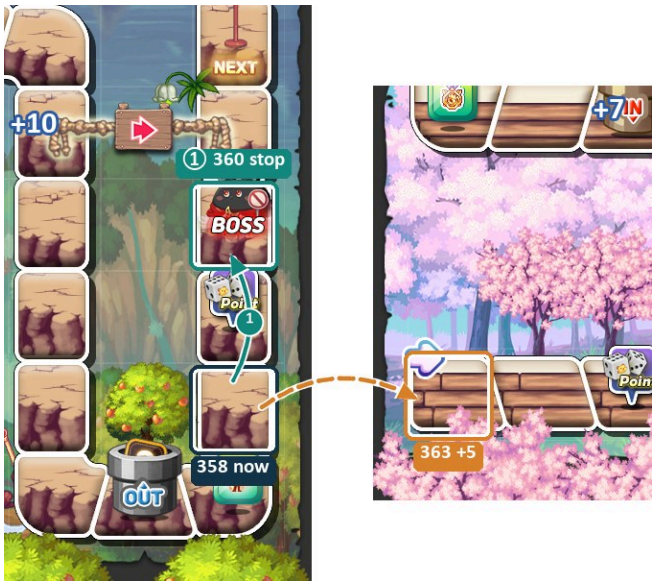


Figure 44: Board. Green: the chosen action. Orange dashed: the comparison action. Arrows that cross between boards are moves into another stage.

From 358 every roll stops at 360 (a boss tile). The +5 goes to 363. Neither draws. The AI rolled, by 40.5 points.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,597.7**	**+2.0**	**0.0%**	**0.0%**
+5 · comparison	1,557.2	+5.0	0.0%	0.0%

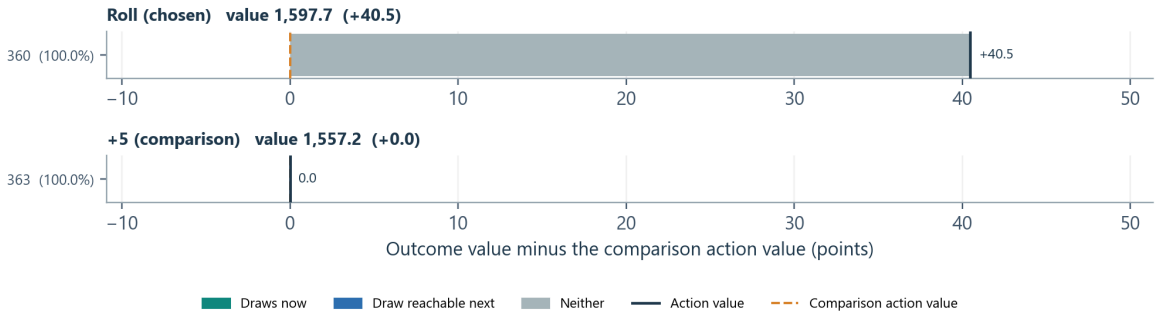


Figure 45: Outcome breakdown (reference: the value of the +5). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

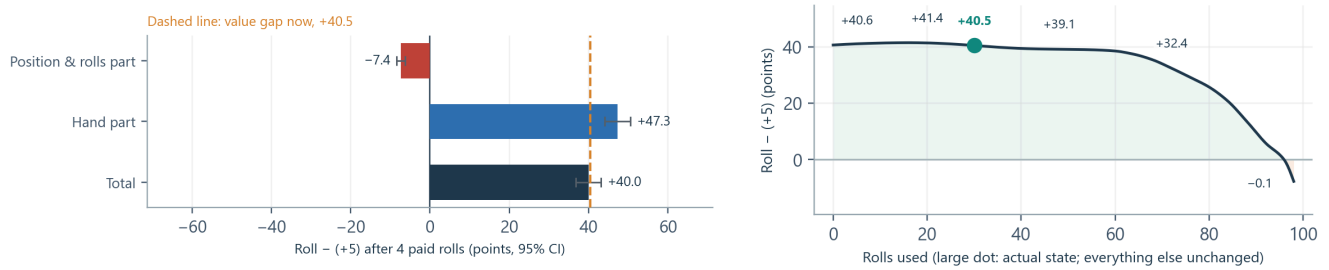


Figure 46: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+5)) when only the number of rolls used changes.

After four rolls nearly all of the gap is in the hand part (+47.3 points). The +5 side is three tiles further on but has an empty hand; the roll side keeps the +5 for the next draw tile. With the rolls used changed to 92 (8 left) the AI still rolls, but the gap shrinks to +5.9 points.

Evidence	Result (in brackets: value of Roll − (+5))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+5): 44.0)
Change only the rolls used	20 used: Roll (+41.4) · 70 used: Roll (+34.0) · 92 used: Roll (+5.9)
Replace the deck with a full 30	Roll (+40.4)
Gap after 4 rolls used	position part −7.4 (±1.1) · hand part +47.3 (±3.3) · total +40.0 (±3.2)
Progress over 4 rolls (mean)	Roll: 34.6 tiles, 0.6 cards drawn, 0.1 played · +5: 44.3 tiles, 0.7 drawn, 1.1 played
Full game (then VELA v4.2)	+45.6 [+44.0, +47.2], 2,048 pairs
Full game (then v4.3 Normal)	+42.1 [+39.2, +45.0], 512 pairs

Case 4-2 · VELA v4.2 record

Stop on the stop tile, then draw with the +8

Position **761** · Rolls used **39** (61 left)

Deck (8): −3 · −1 · +1 · +5 · +8 · +9 · +10 · ×8



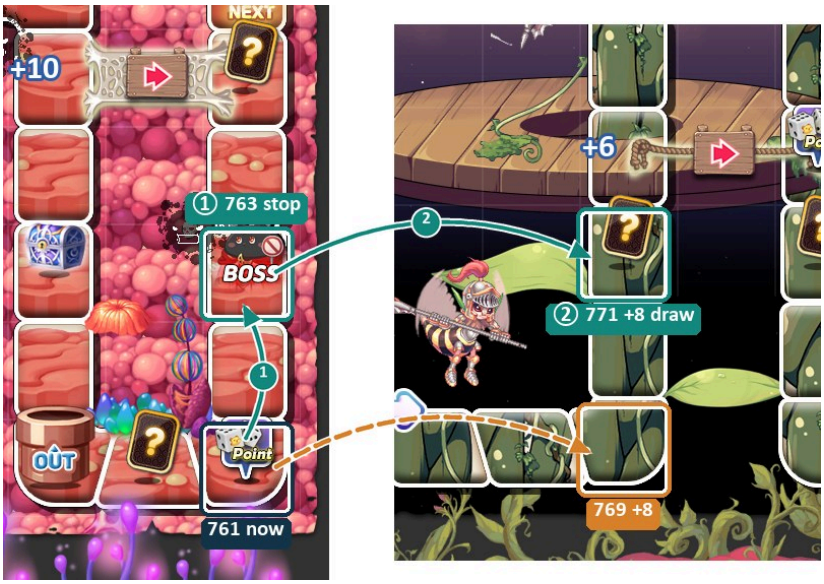


Figure 47: Board. Green arrows: the chosen action and what follows; numbers give the order. Orange dashed: the comparison action. Arrows that cross between boards are moves into another stage.

From 761 a roll stops at 763 (a boss tile); the +8 goes to 769. In this case, playing the +8 from 763 draws at 771. In all 1,000 play-outs the action after the roll was the +8, drawing at 771.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,839.0**	**+2.0**	**0.0%**	**100.0%**
+8 · comparison	1,791.9	+8.0	0.0%	0.0%

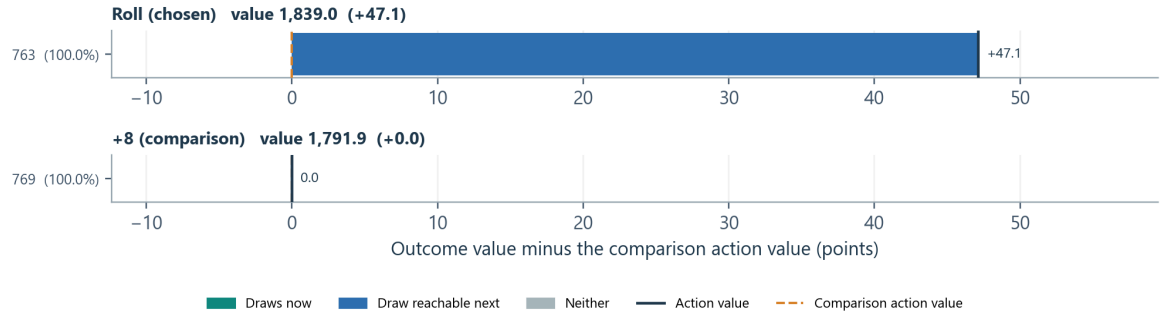


Figure 48: Outcome breakdown (reference: the value of the +8). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

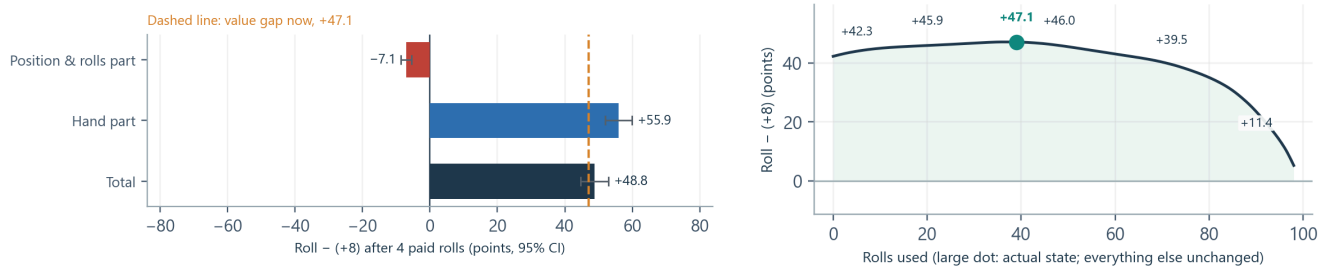


Figure 49: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Roll − (+8)) when only the number of rolls used changes.

The principle of 4-1 with a bonus: the blocked roll puts the token on the stop tile, and the saved +8 then draws at once. The hand part is +55.9 points.

Evidence	Result (in brackets: value of Roll − (+8))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+8): 48.3)
Change only the rolls used	20 used: Roll (+45.9) · 70 used: Roll (+40.3) · 92 used: Roll (+19.9)
Replace the deck with a full 30	Roll (+45.3)
Gap after 4 rolls used	position part −7.1 (±1.6) · hand part +55.9 (±4.0) · total +48.8 (±4.1)
Progress over 4 rolls (mean)	Roll: 43.8 tiles, 2.0 cards drawn, 1.5 played · +8: 50.5 tiles, 0.8 drawn, 1.1 played
Full game (then VELA v4.2)	+48.7 [+47.3, +50.1], 2,048 pairs
Full game (then v4.3 Normal)	+48.4 [+46.1, +50.8], 512 pairs

5. Before a stop tile, jump it with a multiplier

In short

Right before a stop tile, with a multiplier card in hand, the AI jumps the stop tile with the multiplier even when the chance of drawing is low. One roll that covers dozens of tiles beats one roll that moves a tile or two.

Case 5-1 · VELA v4.2 record

Jumping the boss tile with the ×8

Position **865** · Rolls used **54** (46 left)
Deck (30): −3 · −2 · −1 · +1 · +2 · +3 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2





Figure 50: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the comparison action.

From 865 a roll stops at 867 (a boss tile). The AI played the ×8, which moves 60.8 tiles on average but draws at once with a low probability of 16.7%.

Action	AI value	Mean move	Draws now	Draw possible by next action
×8 · chosen	**1,827.5**	**+60.8**	**16.7%**	**52.8%**
Roll · comparison	1,811.8	+2.0	0.0%	0.0%
+11	1,799.5	+23.0	0.0%	0.0%
−1	1,791.5	−1.0	0.0%	100.0%
+5	1,757.8	−15.0	0.0%	0.0%

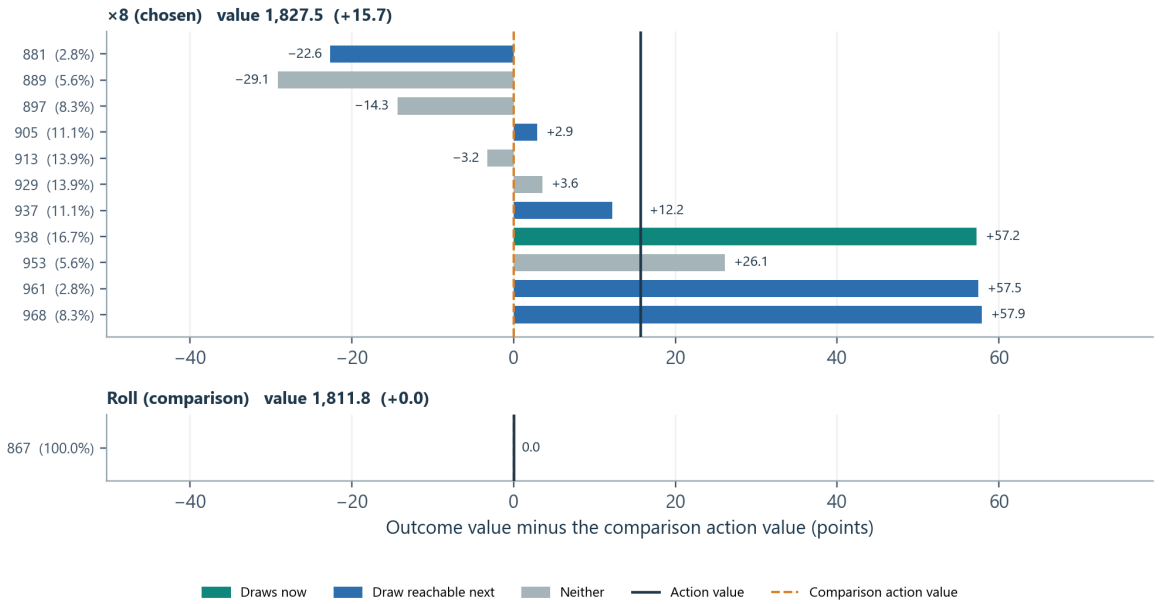


Figure 51: Outcome breakdown (reference: the value of the roll). The roll has a single outcome (867). 69.4% of ×8 outcomes are worth more.

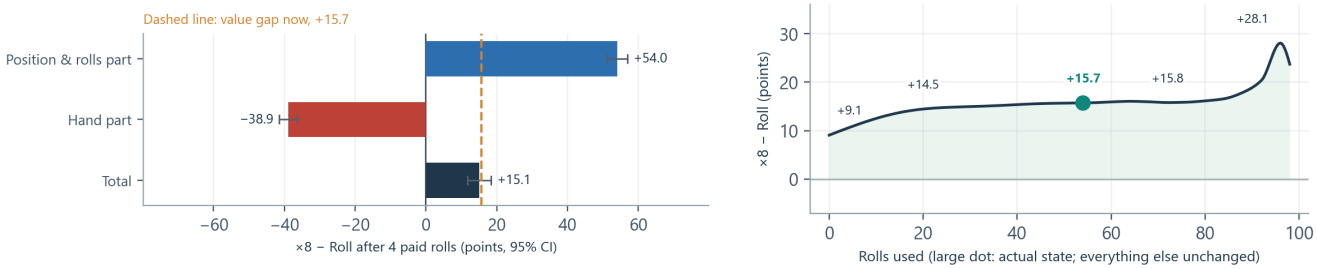


Figure 52: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×8 − Roll) when only the number of rolls used changes.

After four rolls the ×8 side is 54.0 points ahead on position and 38.9 behind on hand. The ×8 is the most valuable card in hand (66.3 points) and still worth playing, because a roll wasted on a blocked tile costs that much. Removing any other card leaves the ×8; hand size does not matter here. In the play-outs, the ×8 was followed by two draws in a row from the hand (+11 then +5, or −1 then +5) in 14.1% of games. On the roll side, 38.6% of games spent their first four actions all on rolls: after stopping on the boss tile, no draw tile was within reach of the cards in hand.

Evidence	Result (in brackets: value of ×8 − Roll)
VELA v4.2, depth 1·2 (production)·3	×8 · ×8 · ×8
VELA v4.3 Normal·High·Max	×8 · ×8 · ×8
LYRA v2	×8 (×8 − Roll: 15.2)
Change only the rolls used	20 used: ×8 (+14.5) · 70 used: ×8 (+15.9) · 92 used: ×8 (+20.4)
Remove one other card	no change (without +5: ×8 · without −1: ×8 · without +11: ×8)
Replace the deck with a full 30	×8 (+15.7)
Gap after 4 rolls used	position part +54.0 (±2.9) · hand part −38.9 (±2.6) · total +15.1 (±3.3)
Progress over 4 rolls (mean)	×8: 111.3 tiles, 2.5 cards drawn, 3.0 played · Roll: 53.7 tiles, 1.3 drawn, 1.2 played
Full game (then VELA v4.2)	+16.4 [+15.0, +17.8], 2,048 pairs
Full game (then v4.3 Normal)	+15.8 [+13.2, +18.4], 512 pairs

Case 5-2 · VELA v4.2 record

Jumping the boss tile with the ×3

Position **968** · Rolls used **62** (38 left)

Deck (6): +4 · +6 · +12 · ×2 · ×7 · ×10





Figure 53: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile).

From 968 a roll stops at the boss tile three tiles ahead. The AI played the ×3 (25.0 tiles on average), which draws at once with probability 13.9%.

Action	AI value	Mean move	Draws now	Draw possible by next action
×3 · chosen	**1,747.5**	**+25.0**	**13.9%**	**44.4%**
Roll · comparison	1,740.2	+3.0	0.0%	0.0%
+7	1,715.1	+7.0	0.0%	0.0%
+8	1,712.3	+8.0	0.0%	0.0%
+3	1,711.5	+3.0	0.0%	0.0%

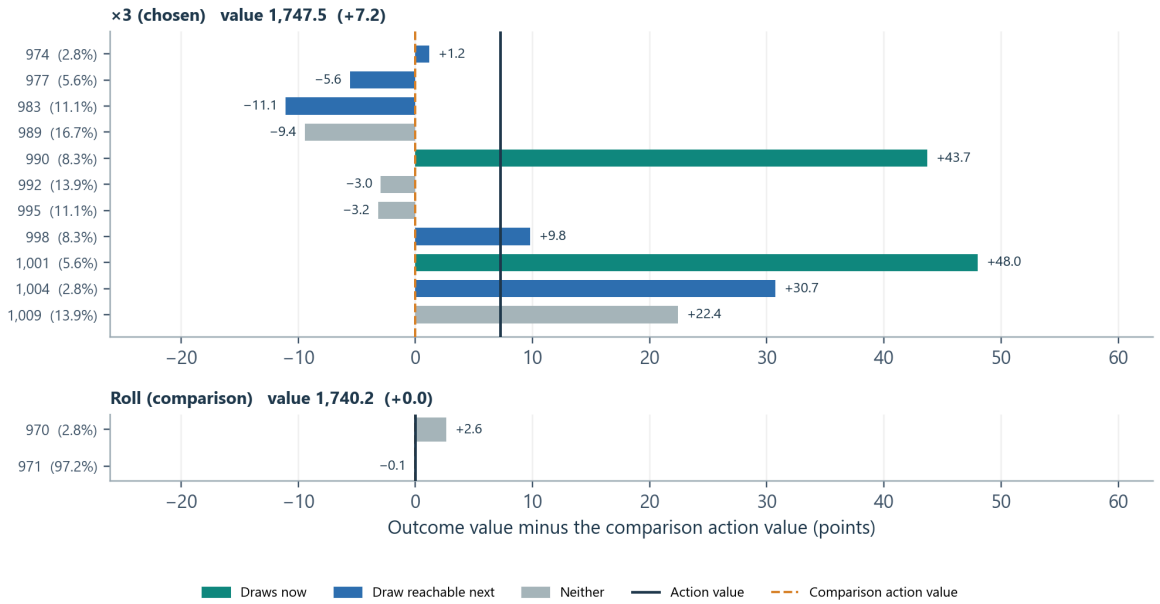


Figure 54: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

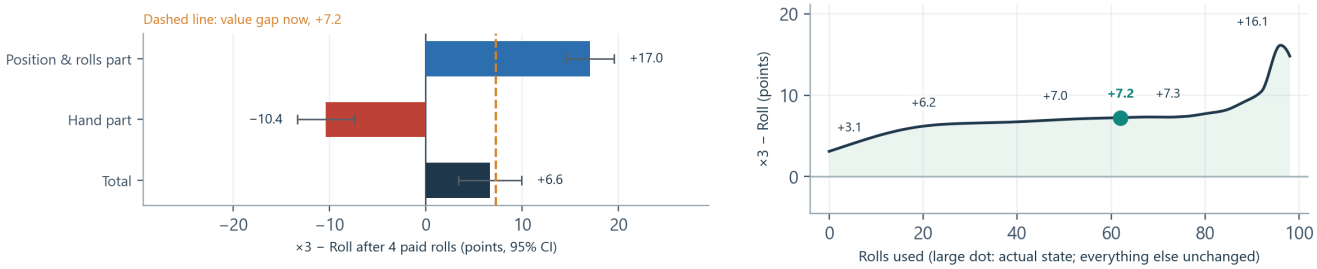


Figure 55: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×3 − Roll) when only the number of rolls used changes.

The structure matches 5-1: a position part of +17.0 points and a hand part of −10.4. This time the AI spends the cheapest multiplier in hand (×3, 22.7 points), so the hand loses little.

Evidence	Result (in brackets: value of ×3 − Roll)
VELA v4.2, depth 1·2 (production)·3	×3 · ×3 · ×3
VELA v4.3 Normal·High·Max	×3 · ×3 · ×3
LYRA v2	×3 (×3 − Roll: 5.6)
Change only the rolls used	20 used: ×3 (+6.2) · 92 used: ×3 (+10.5)
Remove one other card	no change (without +7: ×3 · without +3: ×3 · without +8: ×3)
Replace the deck with a full 30	×3 (+6.5)
Gap after 4 rolls used	position part +17.0 (±2.5) · hand part −10.4 (±3.0) · total +6.6 (±3.3)
Progress over 4 rolls (mean)	×3: 70.7 tiles, 2.2 cards drawn, 2.7 played · Roll: 52.8 tiles, 2.0 drawn, 1.9 played
Full game (then VELA v4.2)	+7.7 [+6.4, +8.9], 2,048 pairs
Full game (then v4.3 Normal)	+8.0 [+5.9, +10.1], 512 pairs

6. A move card that rides a jump is played even without a draw

In short

A small move card that lands on a jump ahead covers dozens of tiles for free, even without drawing. When a card in hand can draw from where the jump lands, the AI plays the small card before rolling.

Case 6-1 · VELA v4.2 record

The +1 rides a jump and the +6 draws



Position **845** · Rolls used **40** (60 left)
Deck (25): −3 · −2 · −1 · +3 · +4×2 · +5×2 · +6 · +7×2 · +8×2 · +9 · +10 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2



Figure 56: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The +1 rides the jump at 846 (+23) to 869. It draws nothing, but from 869 the +6 draws at 875. Neither action involves chance, so all 1,000 play-outs went +1, then +6 (draw); only the rolls after that differed.

Action	AI value	Mean move	Draws now	Draw possible by next action
+1 · chosen	**2,036.1**	**+24.0**	**0.0%**	**100.0%**
Roll · comparison	2,029.1	+7.4	16.7%	58.3%
+6	2,002.5	+6.0	0.0%	100.0%
+2	1,992.7	+2.0	0.0%	0.0%

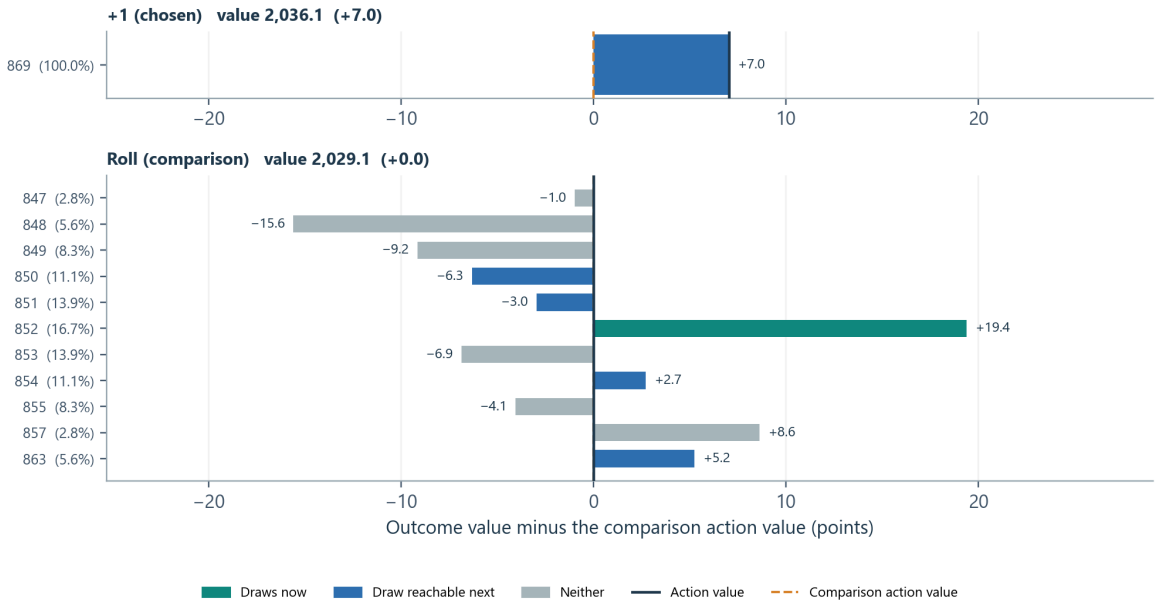


Figure 57: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

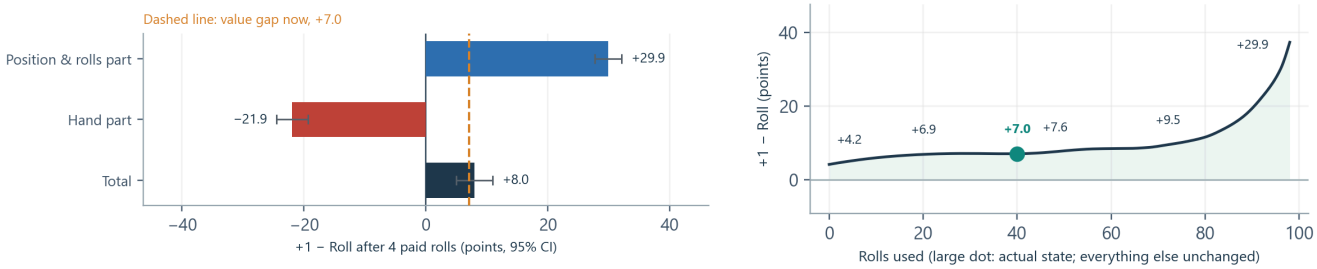


Figure 58: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (+1 − Roll) when only the number of rolls used changes.

After four rolls the +1 side is about 33 tiles ahead, 29.9 points up on position, and 21.9 points down on hand after playing two cards. With the rolls used changed to 92 (8 left) the gap grows to +22.3 points: 24 tiles without a roll are worth more late in the game.

In the counterfactuals, removing any card other than the +1 turns the choice into a roll. With two +6 cards, removing one still leaves the +1 → +6 chain, and the +2 is not part of the chain at all. The reason is how many cards are left after each action. As Section 1.4 showed, one card is worth more the fewer cards are held with it. The +1 side plays two cards and draws one, leaving three (+6, +2 and the card drawn); the roll side keeps all four. So removing the same +6 costs the +1 side 52.2 points and the roll side 35.4. Removing the +2 costs 47.2 against 37.3. The difference in losses exceeds the original gap (7.0 points), so the choice flips.

Evidence	Result (in brackets: value of +1 − Roll)
VELA v4.2, depth 1·2 (production)·3	+1 · +1 · +1
VELA v4.3 Normal·High·Max	+1 · +1 · +1
LYRA v2	+1 (+1 − Roll: 2.7)
Change only the rolls used	20 used: +1 (+6.9) · 70 used: +1 (+9.1) · 92 used: +1 (+22.3)
Remove one other card	all flip (without +6: Roll · without +2: Roll · without +6: Roll)
Replace the deck with a full 30	+1 (+7.4)
Gap after 4 rolls used	position part +29.9 (±2.2) · hand part −21.9 (±2.6) · total +8.0 (±2.9)
Progress over 4 rolls (mean)	+1: 85.4 tiles, 2.9 cards drawn, 3.4 played · Roll: 52.1 tiles, 2.3 drawn, 2.1 played
Full game (then VELA v4.2)	+3.7 [+2.2, +5.3], 2,048 pairs
Full game (then v4.3 Normal)	+3.8 [+0.8, +6.8], 512 pairs

Case 6-2 · VELA v4.2 record

The −1 rides a jump behind

Position **1,166** · Rolls used **56** (44 left)

Deck (17): +2 · +3 · +4 · +5×2 · +6 · +7 · +8 · +10 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×10 · Stage



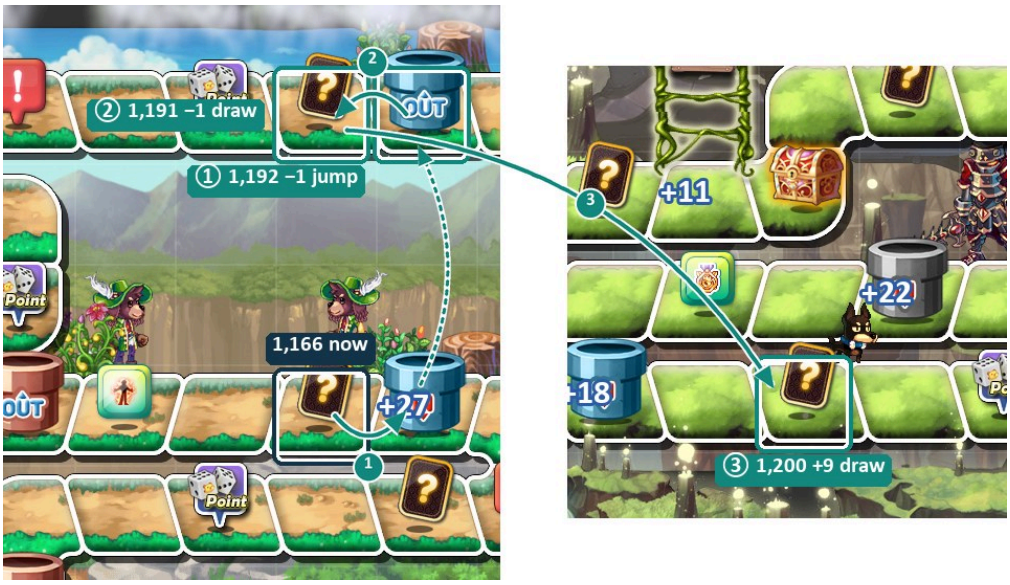


Figure 59: Board. Green arrows: the chosen action and what follows; numbers give the order. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The −1 rides the jump at 1,165 (+27) to 1,192. The second −1 then draws at 1,191, and the +9 draws again at 1,200. None of the three actions involves chance, so all 1,000 play-outs followed this order; in 81% of them the next action was a roll (the rest played a drawn card first).

Action	AI value	Mean move	Draws now	Draw possible by next action
−1 · chosen	**2,053.5**	**+26.0**	**0.0%**	**100.0%**
Roll · comparison	2,046.4	+6.3	5.6%	30.6%
−3	2,021.8	−3.0	0.0%	100.0%
+9	2,017.7	+9.0	0.0%	0.0%

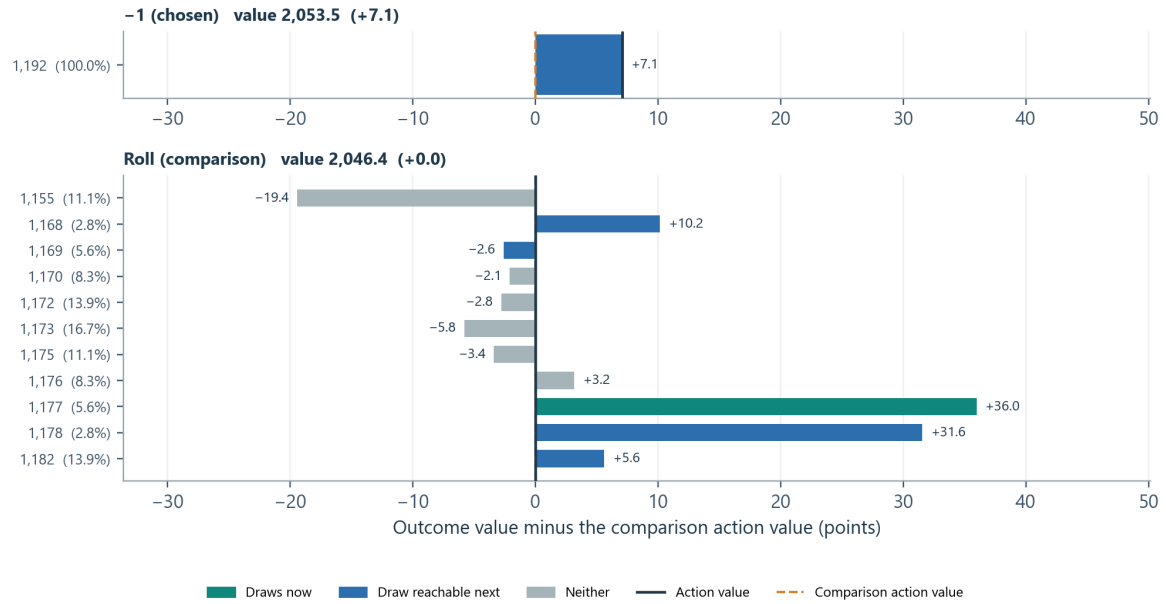


Figure 60: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

This time a backward card rides the jump. Removing any card other than the first −1 flips the choice to a roll. Removing the second −1 or the +9 breaks the chain, so that flip is expected. Removing the −3, which the chain does not use, flips it too, for the same reason as in 6-1. The −1 side plays three cards and draws two, leaving three (−3 and the two cards drawn); the roll side holds four. In this position the −3 alone was worth 49.7 points with no other card in hand and 37.8 points with three. Removing the −3 costs the −1 side 42.0 points and the roll side 32.9, and that 9.1-point difference exceeds the original gap of 7.1 points.

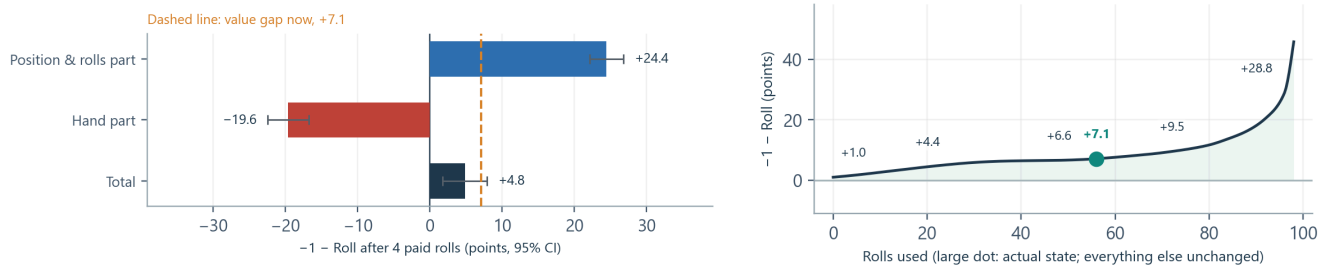


Figure 61: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−1 − Roll) when only the number of rolls used changes.

After four rolls the −1 side is about 26 tiles ahead, 24.4 points up on position, and 19.6 points down on hand. The full-game comparison found a small gap. With VELA v4.2 playing on, the −1 led by +1.4 points over 16,384 pairs; with v4.3 Normal playing on, the gap was +0.6 points over 2,048 pairs, not distinguishable from 0. The two actions are close to even.

Evidence	Result (in brackets: value of −1 − Roll)
VELA v4.2, depth 1·2 (production)·3	−1 · −1 · −1
VELA v4.3 Normal·High·Max	−1 · −1 · −1
LYRA v2	−1 (−1 − Roll: 1.6)
Change only the rolls used	20 used: −1 (+4.4) · 70 used: −1 (+9.1) · 92 used: −1 (+20.5)
Remove one other card	all flip (without −3: Roll · without −1: Roll · without +9: Roll)
Replace the deck with a full 30	−1 (+7.7)
Gap after 4 rolls used	position part +24.4 (±2.3) · hand part −19.6 (±2.9) · total +4.8 (±3.0)
Progress over 4 rolls (mean)	−1: 86.1 tiles, 4.1 cards drawn, 4.8 played · Roll: 60.2 tiles, 3.0 drawn, 2.9 played
Full game (then VELA v4.2)	+1.4 [+0.9, +1.9], 16,384 pairs
Full game (then v4.3 Normal)	+0.6 [−0.6, +1.8], 2,048 pairs · not distinguishable

7. Drawing with a negative card when a positive one could draw too

In short

When both a negative and a positive move card can draw, the AI sometimes picks the one that moves backward. Which cards are kept, and whether a card can draw right after, matter more than a few tiles lost.

Case 7-1 · VELA v4.2 record

−1 over +9 when both draw

Position **799** · Rolls used **44** (56 left)
Deck (8): +3 · +4 · +5 · +10 · +11 · ×3 · ×8 · Stage





Figure 62: Board. Green arrows: the chosen action and what follows; numbers give the order. Orange dashed: the comparison action. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

The −1 draws at 798 and the +9 at 808. The AI picked the −1, by 17.0 points. Both draw again on the next move: after the −1, the +7 rides the jump at 805 and draws at 820; after the +9, the +12 draws at 820. Both lines draw two cards and reach the same tile (820). Up to those two draws nothing depends on chance; in 79% of 1,000 play-outs the next action was a roll (the rest played a drawn card first).

Action	AI value	Mean move	Draws now	Draw possible by next action
−1 · chosen	**1,962.1**	**−1.0**	**100.0%**	**100.0%**
+9 · comparison	1,945.1	+9.0	100.0%	100.0%
Roll	1,944.9	+10.9	25.0%	55.6%
+7	1,914.6	+7.0	0.0%	100.0%
+12	1,896.7	+12.0	0.0%	100.0%

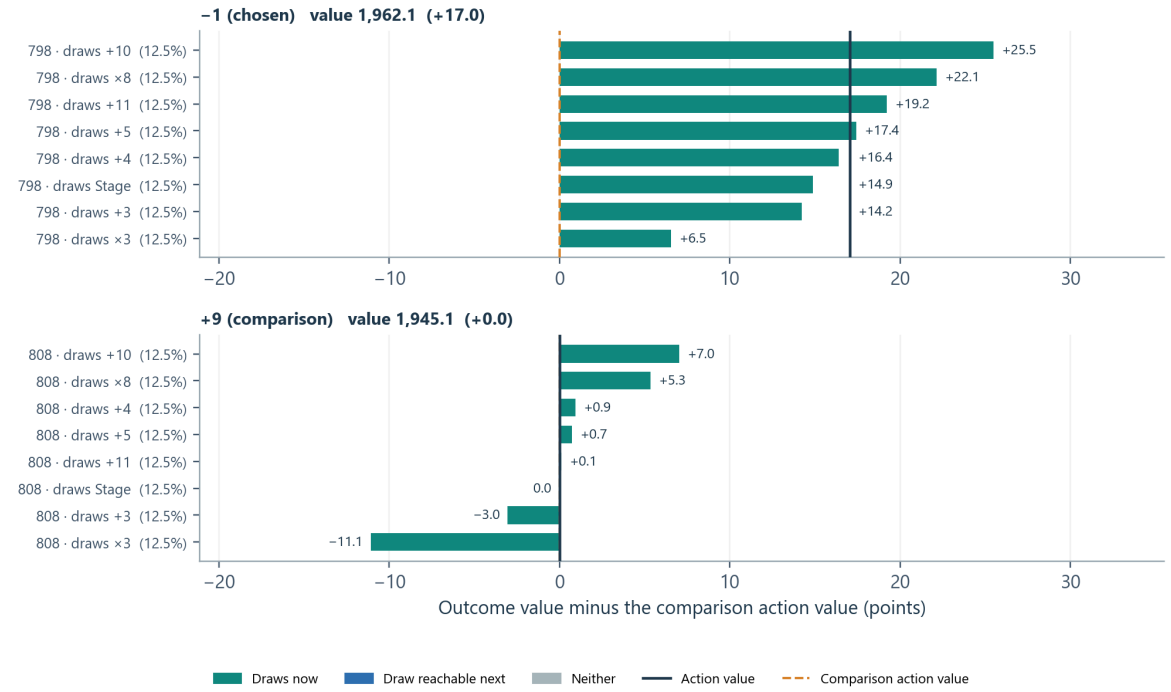


Figure 63: Outcome breakdown (reference: the value of the +9). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

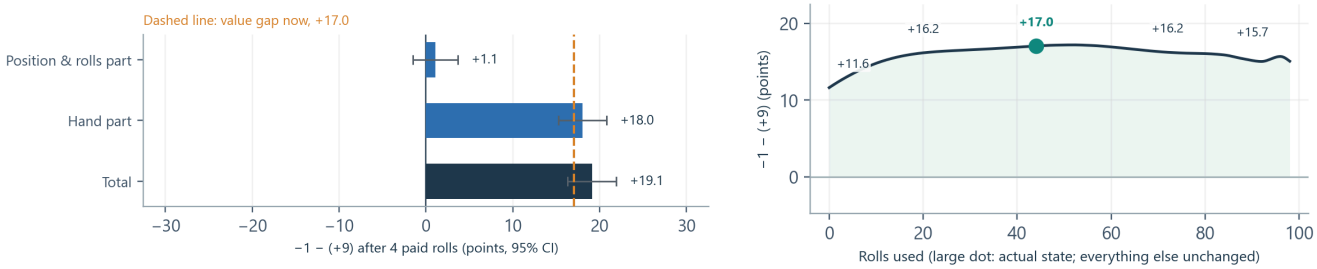


Figure 64: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−1 − (+9)) when only the number of rolls used changes.

Since both lines reach the same tile, the position parts after four rolls are almost equal (+1.1 points), and the gap comes from the hand part (+18.0 points). The −1 line plays the −1 and the +7 and keeps the +12 and the +9. The +9 line plays the +9 and the +12 and keeps the +7 and the −1. For the same two draws, keeping the large move cards is better: the +12 reaches a wider range of draw tiles, while a one-tile step back has few places to be useful. Changing the rolls used still leaves the −1.

Evidence	Result (in brackets: value of −1 − (+9))
VELA v4.2, depth 1·2 (production)·3	−1 · −1 · −1
VELA v4.3 Normal·High·Max	−1 · −1 · −1
LYRA v2	−1 (−1 − (+9): 18.9)
Change only the rolls used	20 used: −1 (+16.2) · 70 used: −1 (+16.3) · 92 used: −1 (+15.0)
Remove one other card	no change (without +7: −1 · without +12: −1)
Replace the deck with a full 30	−1 (+17.3)
Gap after 4 rolls used	position part +1.1 (±2.6) · hand part +18.0 (±2.8) · total +19.1 (±2.8)
Progress over 4 rolls (mean)	−1: 111.0 tiles, 4.7 cards drawn, 4.9 played · +9: 109.8 tiles, 4.7 drawn, 5.2 played
Full game (then VELA v4.2)	+18.8 [+17.2, +20.3], 2,048 pairs
Full game (then v4.3 Normal)	+18.6 [+16.2, +21.1], 512 pairs

Case 7-2 · VELA v4.2 record

−3 over +4 when both draw

Position **307** · Rolls used **18** (82 left)

Deck (18): −2 · −1 · +2 · +3 · +4 · +6 · +7 · +8 · +9×2 · +10×2 · +11 · +12 · ×2 · ×5 · ×8 · ×10

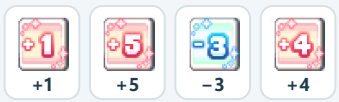




Figure 65: Board. Green arrows: the chosen action and what follows; numbers give the order. Orange dashed: the comparison action. Thin dotted lines are jumps.

The −3 draws at 304 and the +4 at 311. The AI picked the −3 (by 23.4 points). This time the +5 and the +4 draw in turn after the −3. All 1,000 play-outs shared these three draws, and in 71% of them the next action was a roll (the rest drew once more with a +9 they had drawn). After drawing with the +4, no card in hand can draw right away, so the next action is a roll.

Action	AI value	Mean move	Draws now	Draw possible by next action
−3 · chosen	**1,949.3**	**−3.0**	**100.0%**	**100.0%**
+4 · comparison	1,925.9	+4.0	100.0%	100.0%
Roll	1,923.4	+9.3	13.9%	66.7%
+5	1,902.6	+5.0	0.0%	100.0%
+1	1,882.8	+1.0	0.0%	0.0%

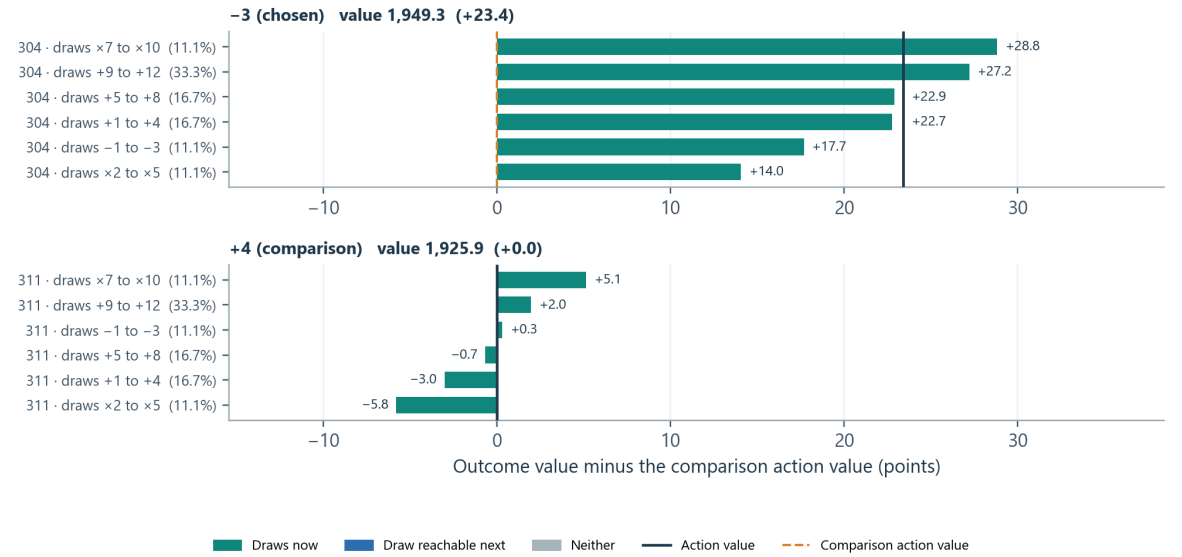


Figure 66: Outcome breakdown (reference: the value of the +4). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

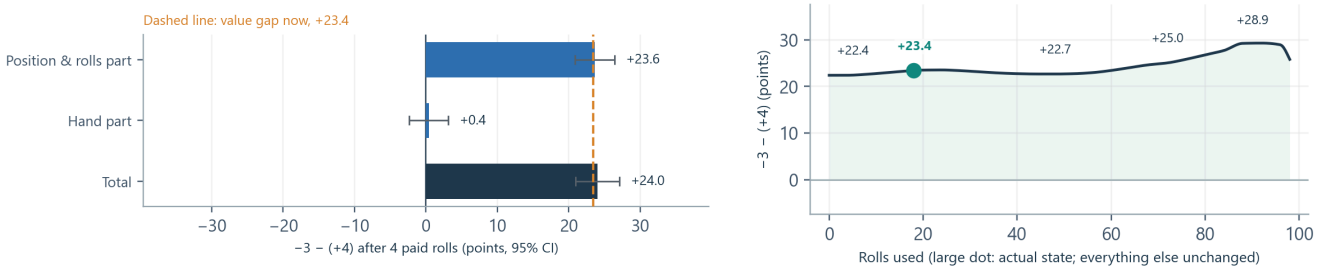


Figure 67: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (−3 − (+4)) when only the number of rolls used changes.

The reason differs from 7-1. The gap comes from the position part (+23.6 points), and the hand part is +0.4. The chain that starts three tiles back carries the token 96.1 tiles on average over four rolls; starting with the +4, it is 71.7. A negative card can be the right pick either because it keeps the better cards (7-1) or because it puts the token at the start of a chain (7-2).

Evidence	Result (in brackets: value of −3 − (+4))
VELA v4.2, depth 1·2 (production)·3	−3 · −3 · −3
VELA v4.3 Normal·High·Max	−3 · −3 · −3
LYRA v2	−3 (−3 − (+4): 21.7)
Change only the rolls used	70 used: −3 (+24.9) · 92 used: −3 (+29.3)
Remove one other card	no change (without +1: −3 · without +5: −3)
Replace the deck with a full 30	−3 (+22.9)
Gap after 4 rolls used	position part +23.6 (±2.8) · hand part +0.4 (±2.7) · total +24.0 (±3.1)
Progress over 4 rolls (mean)	−3: 96.1 tiles, 6.0 cards drawn, 6.2 played · +4: 71.7 tiles, 4.2 drawn, 4.1 played
Full game (then VELA v4.2)	+22.4 [+20.3, +24.5], 2,048 pairs
Full game (then v4.3 Normal)	+19.3 [+15.6, +23.1], 512 pairs

8. A multiplier even when a move card would draw

In short

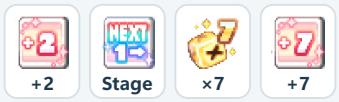
On an ordinary stretch with no stop tile or jump, the AI plays a multiplier even when a move card would draw for certain. From this tile the multiplier itself has a high chance of drawing, and it covers dozens of tiles.

Case 8-1 · VELA v4.2 record

×7 when the +2 would draw

Position **690** · Rolls used **49** (51 left)

Deck (17): −3 · −1 · +1 · +3 · +4 · +5×2 · +6×2 · +7 · +8 · +9 · +10 · +11 · +12 · ×2 · ×5



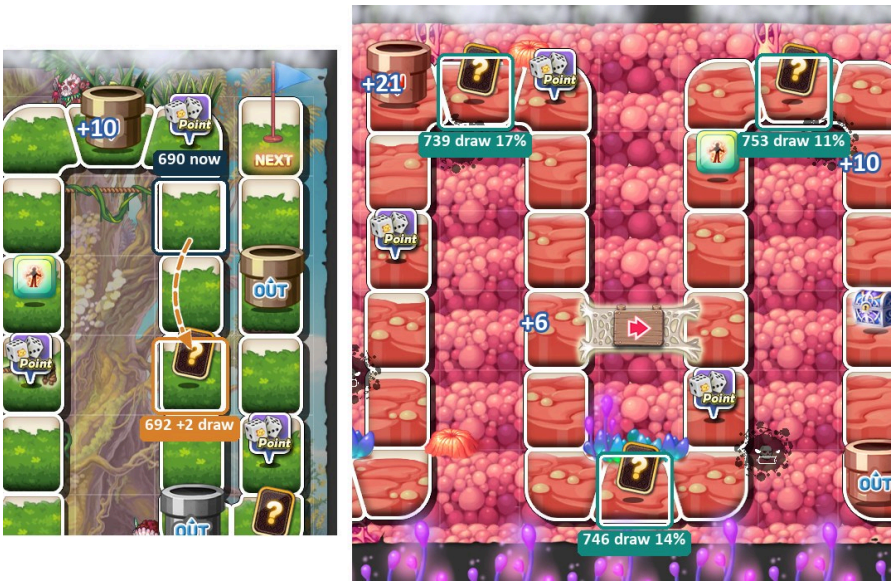


Figure 68: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the comparison action.

The +2 would draw at 692. The AI played the ×7, which moves 49.6 tiles on average and itself draws with probability 50.0%.

Action	AI value	Mean move	Draws now	Draw possible by next action
×7 · chosen	**1,728.0**	**+49.6**	**50.0%**	**63.9%**
+2 · comparison	1,717.5	+2.0	100.0%	100.0%
Roll	1,715.8	+7.0	13.9%	41.7%
Stage	1,688.5	+11.0	0.0%	100.0%
+7	1,680.0	+7.0	0.0%	0.0%

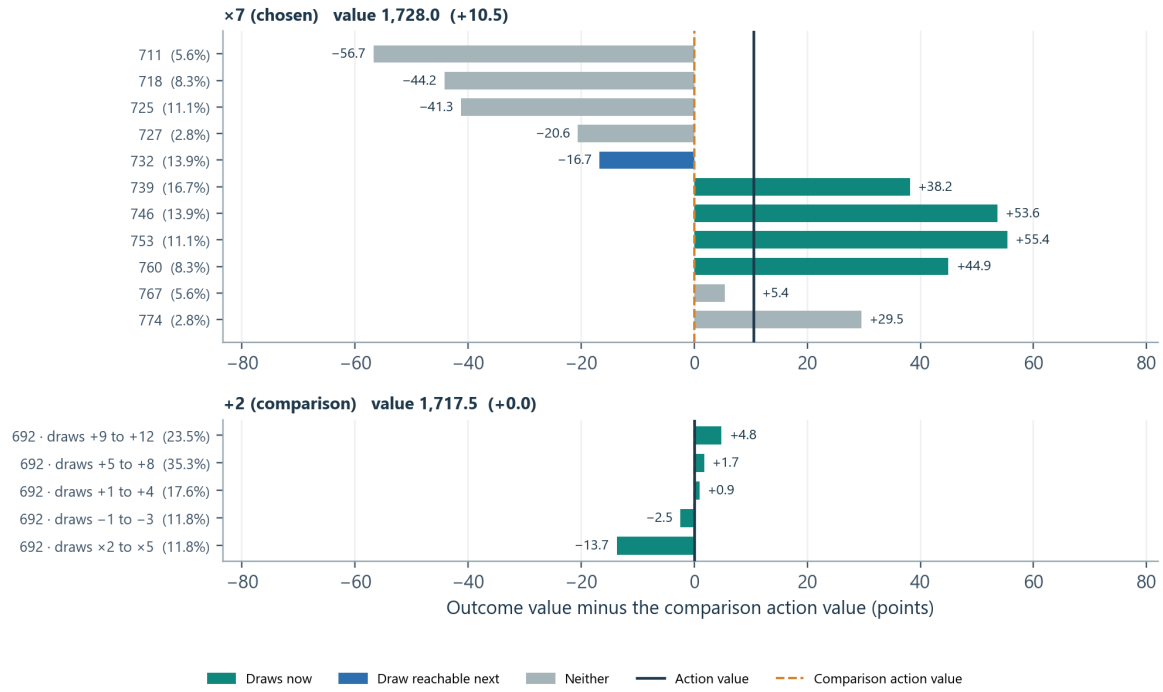


Figure 69: Outcome breakdown (reference: the value of the +2). The ×7's drawing outcomes (green, 50.0%) average +47.4 points; the rest average −26.4.

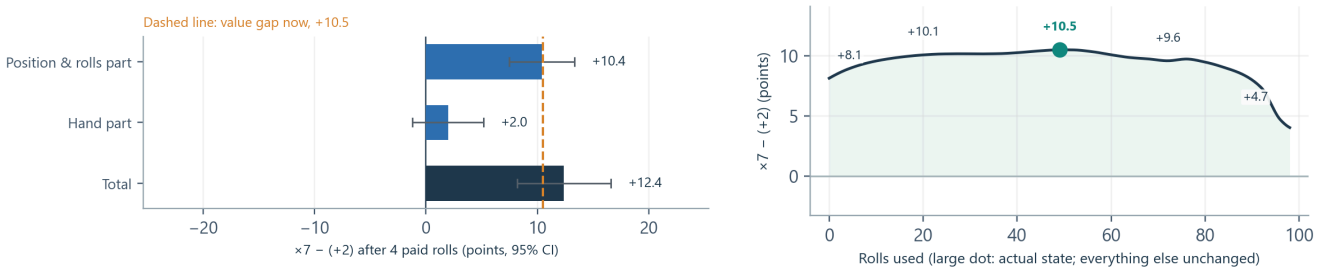


Figure 70: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×7 − (+2)) when only the number of rolls used changes.

The ×7 spends a roll and an expensive card (61.6 points). In return it draws about half the time and moves close to 50 tiles. The +2 moves two tiles and draws one card. After four rolls the ×7 side is 10.4 points ahead on position. Changing the rolls used still leaves the ×7.

Evidence	Result (in brackets: value of ×7 − (+2))
VELA v4.2, depth 1·2 (production)·3	×7 · ×7 · ×7
VELA v4.3 Normal·High·Max	×7 · ×7 · ×7
LYRA v2	×7 (×7 − (+2): 10.8)
Change only the rolls used	20 used: ×7 (+10.1) · 70 used: ×7 (+9.7) · 92 used: ×7 (+7.3)
Remove one other card	no change (without Stage: ×7 · without +7: ×7)
Replace the deck with a full 30	×7 (+10.3)
Gap after 4 rolls used	position part +10.4 (±2.9) · hand part +2.0 (±3.2) · total +12.4 (±4.2)
Progress over 4 rolls (mean)	×7: 101.6 tiles, 3.2 cards drawn, 3.5 played · +2: 90.8 tiles, 3.6 drawn, 4.0 played
Full game (then VELA v4.2)	+11.7 [+10.2, +13.2], 2,048 pairs
Full game (then v4.3 Normal)	+13.3 [+10.6, +16.1], 512 pairs

Case 8-2 · VELA v4.2 record

×7 when the +5 would draw

Position **154** · Rolls used **7** (93 left)

Deck (26): −3 · −2 · −1 · +2 · +3 · +4×2 · +5 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · ×2 · ×3 · ×5 · ×8 · ×10 · Stage×2

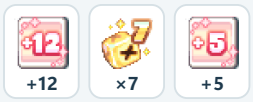




Figure 71: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the comparison action.

The +5 would draw at 159. The AI played the ×7, which draws with probability 41.7% and moves 49.0 tiles on average. The gap is 8.8 points.

Action	AI value	Mean move	Draws now	Draw possible by next action
×7 · chosen	**1,970.1**	**+49.0**	**41.7%**	**50.0%**
+5 · comparison	1,961.4	+5.0	100.0%	100.0%
Roll	1,960.6	+7.0	11.1%	30.6%
+12	1,923.6	+12.0	0.0%	0.0%

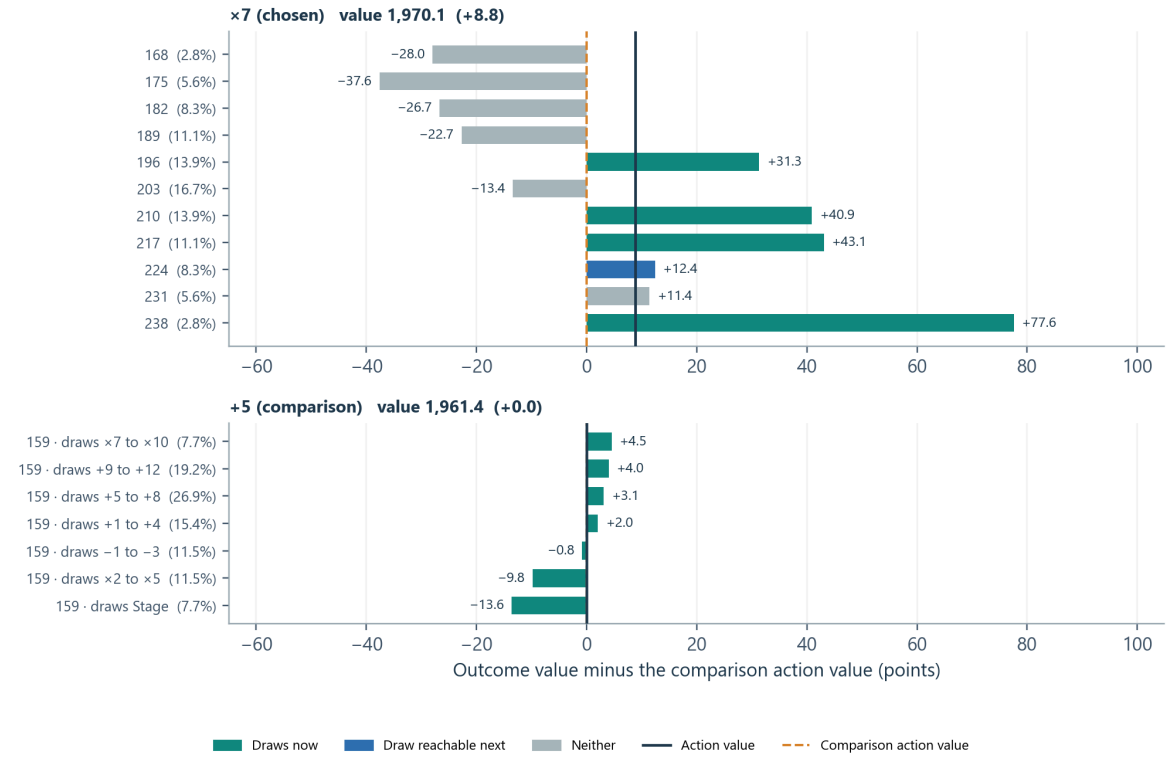


Figure 72: Outcome breakdown (reference: the value of the +5). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

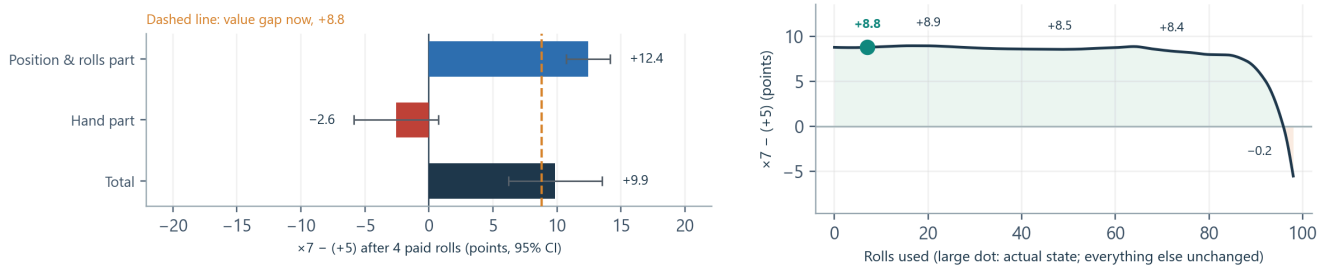


Figure 73: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×7 − (+5)) when only the number of rolls used changes.

The structure matches 8-1: a position part of +12.4 points and a hand part of −2.6. In both cases the multiplier draws at once with a probability of 40% or more from that tile.

Evidence	Result (in brackets: value of ×7 − (+5))
VELA v4.2, depth 1·2 (production)·3	×7 · ×7 · ×7
VELA v4.3 Normal·High·Max	×7 · ×7 · ×7
LYRA v2	×7 (×7 − (+5): 9.5)
Change only the rolls used	20 used: ×7 (+8.9) · 70 used: ×7 (+8.5) · 92 used: ×7 (+5.1)
Remove one other card	no change (without +12: ×7)
Replace the deck with a full 30	×7 (+8.7)
Gap after 4 rolls used	position part +12.4 (±1.7) · hand part −2.6 (±3.3) · total +9.9 (±3.7)
Progress over 4 rolls (mean)	×7: 95.3 tiles, 2.8 cards drawn, 2.6 played · +5: 75.4 tiles, 2.7 drawn, 2.5 played
Full game (then VELA v4.2)	+9.9 [+7.8, +12.1], 2,048 pairs
Full game (then v4.3 Normal)	+8.7 [+4.5, +12.8], 512 pairs

9. A stage card even without a stop stretch ahead

In short

The stage card is not only for skipping a run of stop tiles. When a card in hand can draw right away from the start tile of the next stage, the AI plays it on an ordinary stretch too.

Case 9-1 · VELA v4.2 record

Stage card to 1,517, then the −2 draws

Position **1,482** · Rolls used **88** (12 left)
Deck (11): +5 · +6×2 · +7 · +9 · +10×2 · ×3 · ×7 · ×8 · Stage



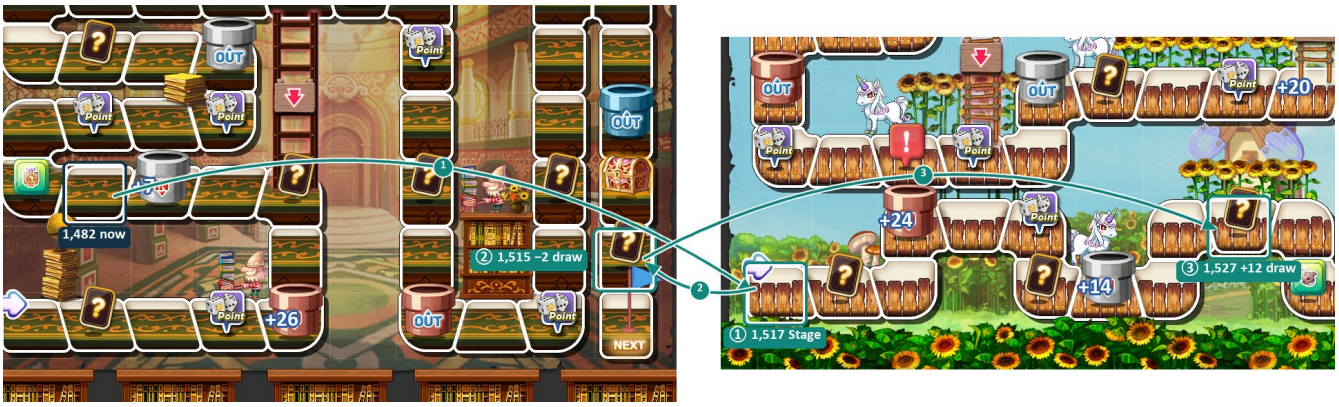


Figure 74: Board. Green arrows: the chosen action and what follows; numbers give the order. Arrows that cross between boards are moves into another stage.

With twelve rolls left, the stage card goes to 1,517 (35 tiles ahead). From there the −2 draws at 1,515, and the +12 draws again at 1,527. These three actions were the same in all 1,000 play-outs; in about half of them a drawn card then moved the token once more.

Action	AI value	Mean move	Draws now	Draw possible by next action
Stage · chosen	**1,820.4**	**+35.0**	**0.0%**	**100.0%**
Roll · comparison	1,810.5	+7.0	13.9%	50.0%
+4	1,779.5	+4.0	0.0%	0.0%
+12	1,779.5	+12.0	0.0%	0.0%
−2	1,773.8	−2.0	0.0%	0.0%

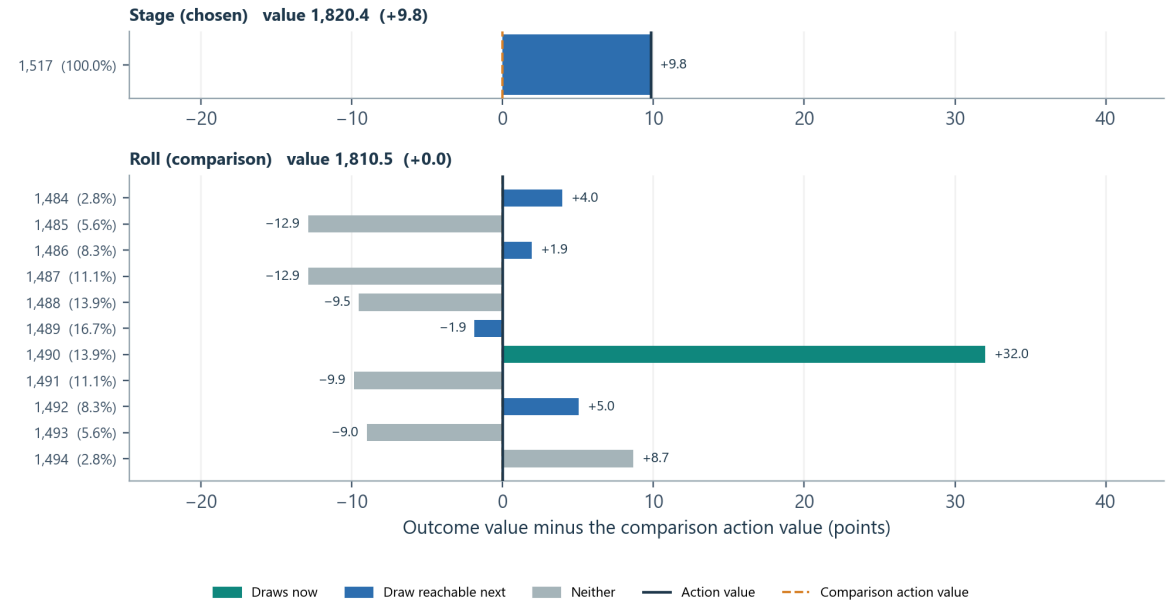


Figure 75: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

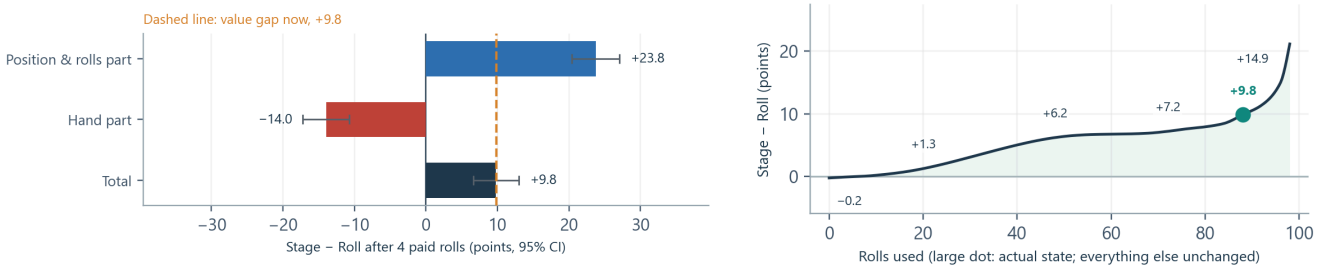


Figure 76: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Stage − Roll) when only the number of rolls used changes.

Removing the −2 turns the choice into a roll; removing any other card leaves the stage card. Its value lies in the card the −2 draws right away on landing. With the rolls used changed to 20 (80 left) the AI still plays the stage card, but the gap shrinks to +1.3 points.

Evidence	Result (in brackets: value of Stage − Roll)
VELA v4.2, depth 1·2 (production)·3	Stage · Stage · Stage
VELA v4.3 Normal·High·Max	Stage · Stage · Stage
LYRA v2	Stage (Stage − Roll: 8.2)
Change only the rolls used	20 used: Stage (+1.3) · 70 used: Stage (+7.0)
Remove one other card	flips in 1 of 3 (without −2: Roll · without +12: Stage · without +4: Stage)
Replace the deck with a full 30	Stage (+7.6)
Gap after 4 rolls used	position part +23.8 (±3.3) · hand part −14.0 (±3.3) · total +9.8 (±3.2)
Progress over 4 rolls (mean)	Stage: 123.5 tiles, 4.5 cards drawn, 5.7 played · Roll: 100.6 tiles, 4.0 drawn, 4.7 played
Full game (then VELA v4.2)	+7.6 [+6.8, +8.5], 2,048 pairs
Full game (then v4.3 Normal)	+7.8 [+6.3, +9.3], 512 pairs

Case 9-2 · VELA v4.2 record

Stage card to 1,556, then the +8 draws

Position **1,536** · Rolls used **75** (25 left)

Deck (7): −3 · −1 · +1 · +7 · +10 · ×10 · Stage

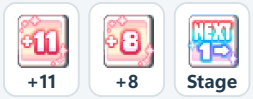




Figure 77: Board. Green arrows: the chosen action and what follows; numbers give the order. Arrows that cross between boards are moves into another stage.

With 25 rolls left, the stage card goes 20 tiles ahead to 1,556, and the +8 then draws at 1,564 (in all 1,000 play-outs). The gap is 3.3 points.

Action	AI value	Mean move	Draws now	Draw possible by next action
Stage · chosen	**2,072.5**	**+20.0**	**0.0%**	**100.0%**
Roll · comparison	2,069.2	+7.0	8.3%	13.9%
+8	2,035.2	+8.0	0.0%	0.0%
+11	2,033.0	+11.0	0.0%	0.0%

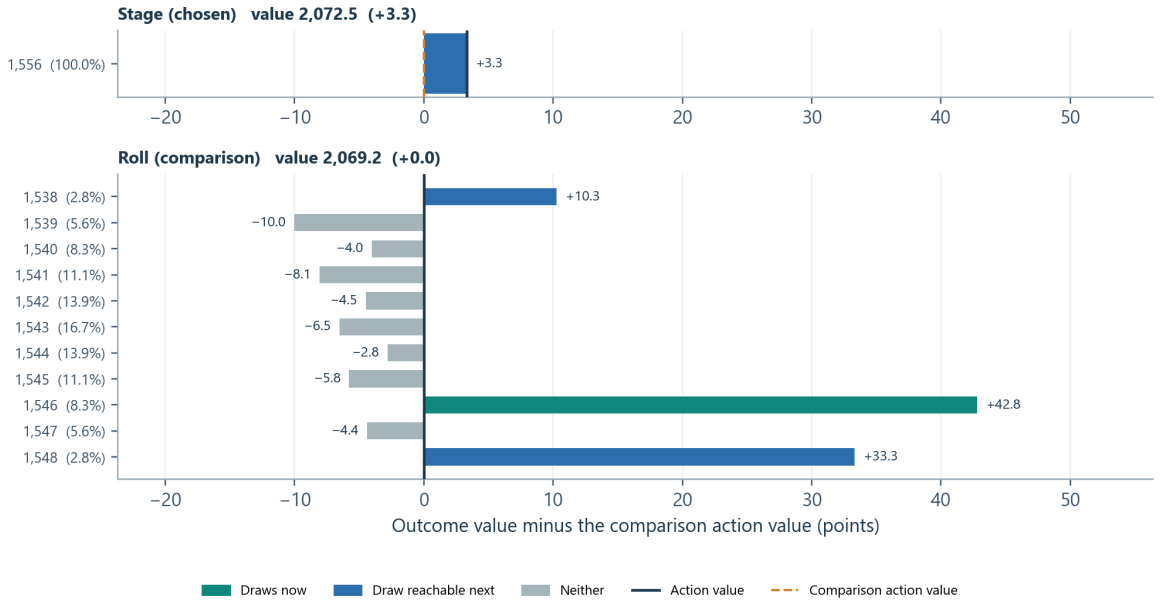


Figure 78: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

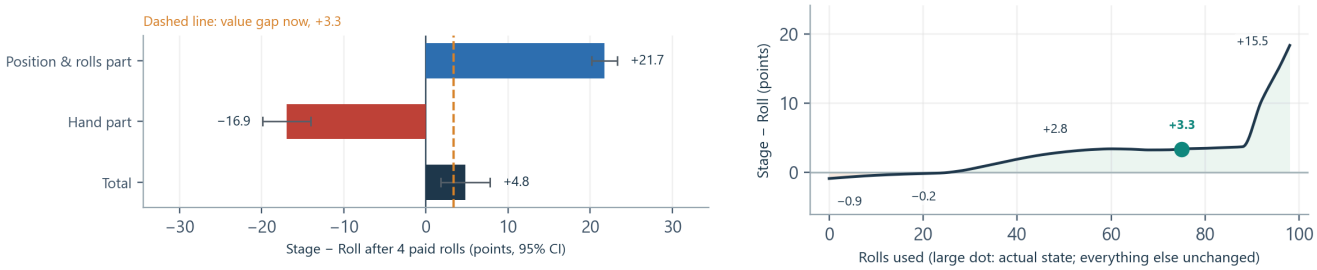


Figure 79: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (Stage − Roll) when only the number of rolls used changes.

Removing the +8 turns the choice into a roll. With the rolls used changed to 20 (80 left), the roll edges ahead (−0.2 points). This choice holds when few rolls are left.

Evidence	Result (in brackets: value of Stage − Roll)
VELA v4.2, depth 1·2 (production)·3	Stage · Stage · Stage
VELA v4.3 Normal·High·Max	Stage · Stage · Stage
LYRA v2	Stage (Stage − Roll: 2.5)
Change only the rolls used	20 used: Roll (−0.2) · 92 used: Stage (+10.3)
Remove one other card	flips in 1 of 2 (without +11: Stage · without +8: Roll)
Replace the deck with a full 30	Stage (+1.9)
Gap after 4 rolls used	position part +21.7 (±1.6) · hand part −16.9 (±2.9) · total +4.8 (±3.0)
Progress over 4 rolls (mean)	Stage: 72.5 tiles, 2.9 cards drawn, 3.1 played · Roll: 40.6 tiles, 1.1 drawn, 0.7 played
Full game (then VELA v4.2)	+4.0 [+3.0, +5.0], 2,048 pairs
Full game (then v4.3 Normal)	+4.8 [+3.0, +6.5], 512 pairs

10. With several multipliers, the small one first

In short

With two or more multipliers in hand, the AI often plays the smallest first (Section 2.3). A small multiplier is worth little in hand, and its shorter move makes it easier for a card in hand to draw from where it lands.

Case 10-1 · VELA v4.2 record

×3 over ×5

Position **722** · Rolls used **32** (68 left)
Deck (29): −3 · −2 · −1 · +1 · +2 · +3 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · +12 · ×2 · ×5 · ×7 · ×8 · ×10 · Stage×2



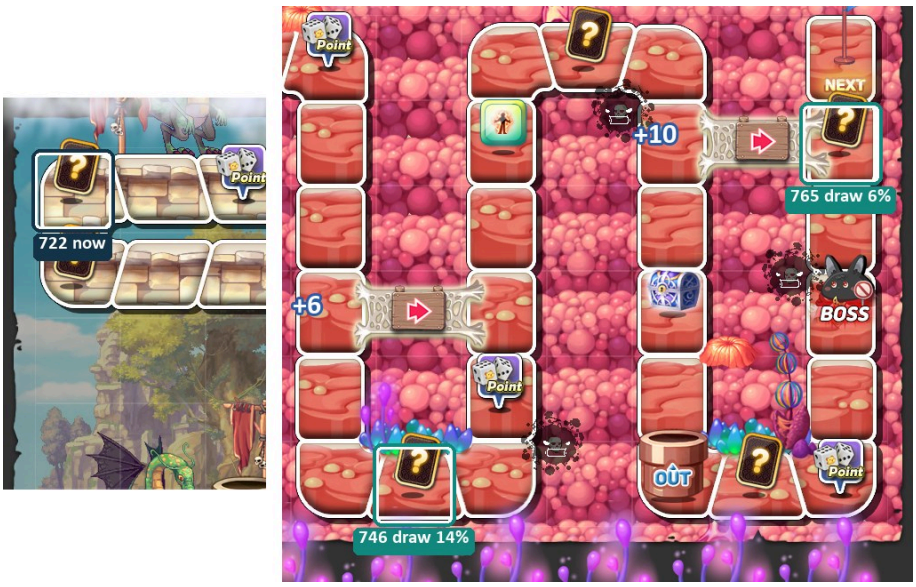


Figure 80: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile).

The hand is +6, +2, ×5 and ×3. The ×3 moves 22.6 tiles on average and the ×5 35.5. The AI picked the ×3, valuing it 21.3 points above the ×5. The chance that a draw is possible by the next action is 75.0% for the ×3 and 22.2% for the ×5.

Action	AI value	Mean move	Draws now	Draw possible by next action
×3 · chosen	**2,064.6**	**+22.6**	**19.4%**	**75.0%**
Roll	2,055.1	+7.0	0.0%	5.6%
×5 · comparison	2,043.3	+35.5	0.0%	22.2%
+6	2,012.3	+6.0	0.0%	0.0%
+2	2,010.4	+2.0	0.0%	0.0%

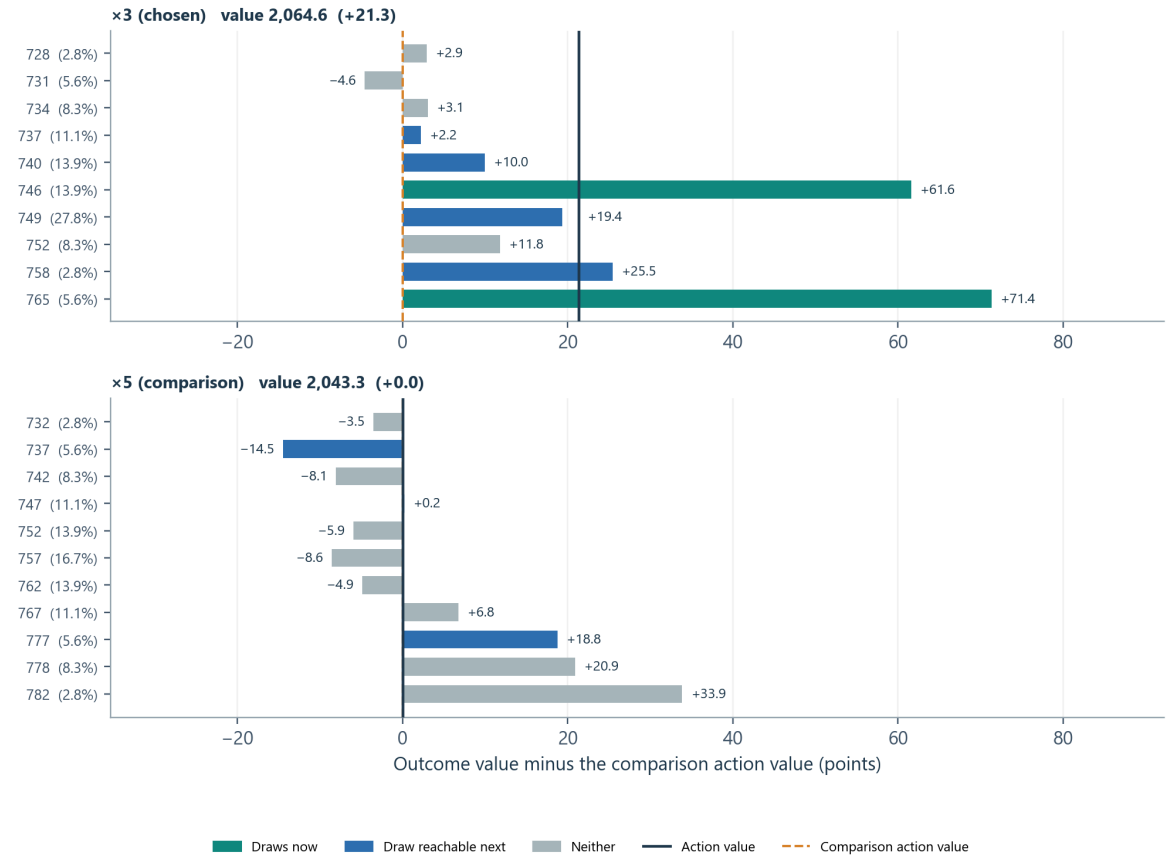


Figure 81: Outcome breakdown (reference: the value of the ×5). 94.4% of ×3 outcomes are worth more than the ×5, and 2.8% of ×5 outcomes are worth more than the ×3.

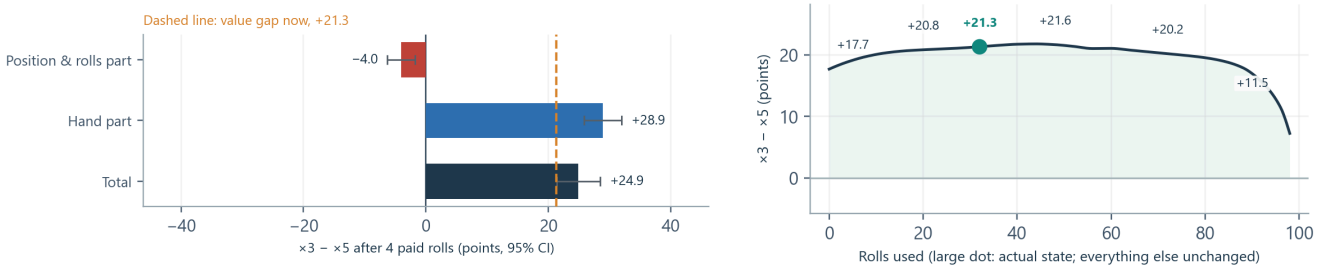


Figure 82: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (×3 − ×5) when only the number of rolls used changes.

After four rolls the gap is mostly in the hand part (+28.9 points). Playing the ×3 keeps the ×5 (36.6 points); playing the ×5 keeps the ×3 (26.1 points). The large multiplier is saved for a later chance. The play-outs differ too. After the ×3, 20.2% of games drew right away with the +6. The ×5 goes further but often lands where no card in hand reaches a draw tile; 26.0% of games had to roll once more before drawing with the +2 and then the +6.

The comparison action here (the ×5) is not the runner-up. The runner-up is the roll, 9.5 points below the ×3. Playing the ×3 and the roll on with the same dice until four rolls are used gives a total gap of +6.8 points (±3.2), with a position part of +9.1 and a hand part of −2.2. The ×3 beats the roll on distance: the cheapest multiplier triples one roll's move at a small cost in hand.

Evidence	Result (in brackets: value of ×3 − ×5)
VELA v4.2, depth 1·2 (production)·3	×3 · ×3 · ×3
VELA v4.3 Normal·High·Max	×3 · ×3 · ×3
LYRA v2	×3 (×3 − ×5: 22.1)
Change only the rolls used	20 used: ×3 (+20.8) · 70 used: ×3 (+20.3) · 92 used: ×3 (+15.6)
Remove one other card	no change (without +6: ×3 · without +2: ×3)
Replace the deck with a full 30	×3 (+21.1)
Gap after 4 rolls used	position part −4.0 (±2.3) · hand part +28.9 (±3.1) · total +24.9 (±3.5)
Progress over 4 rolls (mean)	×3: 75.4 tiles, 2.8 cards drawn, 3.3 played · ×5: 79.0 tiles, 2.5 drawn, 3.1 played
Full game (then VELA v4.2)	+21.4 [+19.7, +23.1], 2,048 pairs
Full game (then v4.3 Normal)	+20.4 [+17.4, +23.4], 512 pairs

Case 10-2 · VELA v4.2 record

×3 over ×8, with 15 rolls left

Position **1,576** · Rolls used **85** (15 left)
Deck (5): +4 · +7 · +9×2 · +12





Figure 83: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile).

The hand is −1, ×8, ×3 and +5. The ×3 draws at once with probability 41.7%, the ×8 with 8.3%. The AI valued the ×3 19.8 points above the ×8.

Action	AI value	Mean move	Draws now	Draw possible by next action
×3 · chosen	**1,964.6**	**+22.2**	**41.7%**	**75.0%**
Roll	1,954.9	+7.0	2.8%	27.8%
−1	1,947.7	+23.0	0.0%	0.0%
×8 · comparison	1,944.8	+57.0	8.3%	41.7%
+5	1,929.7	+5.0	0.0%	0.0%

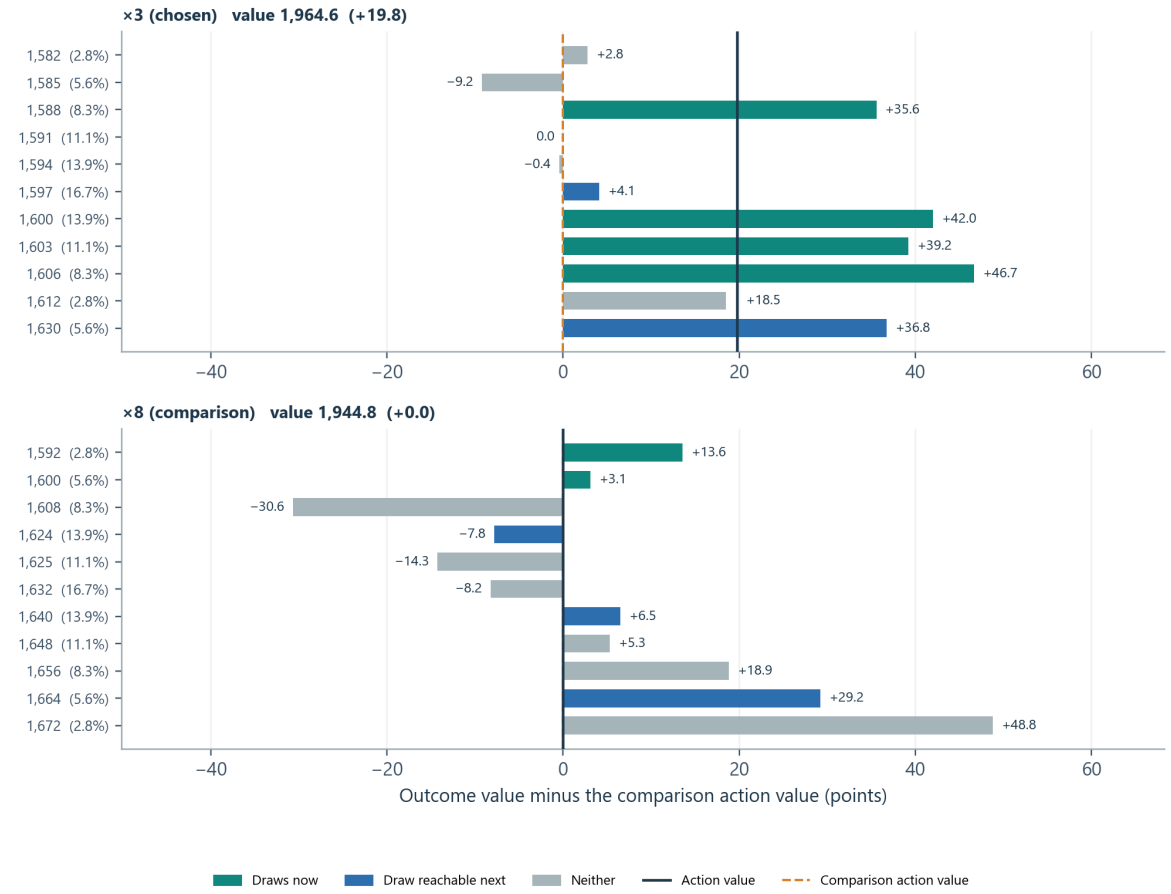


Figure 84: Outcome breakdown (reference: the value of the ×8). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

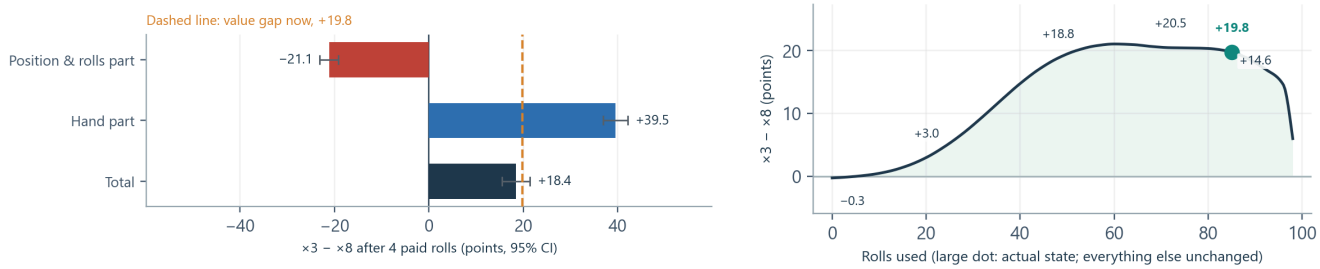


Figure 85: Left: the gap after four rolls (×3 − ×8). Right: the value gap (×3 − ×8) when only the number of rolls used changes.

The ×8 side leads by 21.1 points on position, but the ×3 side leads by 39.5 on hand. Keeping the ×8 means it can be played once more in the remaining rolls. With the rolls used changed to 20 (80 left) the gap shrinks to +3.0 points: how much this choice matters depends on the rolls left.

Here too the runner-up is the roll (9.8 points below the ×3). Against the roll, the four-roll comparison gives a total of +8.9 points (±2.9), a position part of +8.1 and a hand part of +0.7: as in 10-1, the ×3 wins on distance.

Evidence	Result (in brackets: value of ×3 – ×8)
VELA v4.2, depth 1·2 (production)·3	×3 · ×3 · ×3
VELA v4.3 Normal·High·Max	×3 · ×3 · ×3
LYRA v2	×3 (×3 – ×8: 21.8)
Change only the rolls used	20 used: ×3 (+3.0) · 70 used: ×3 (+20.5)
Remove one other card	no change (without −1: ×3 · without +5: ×3)
Replace the deck with a full 30	×3 (+19.4)
Gap after 4 rolls used	position part −21.1 (±2.0) · hand part +39.5 (±2.6) · total +18.4 (±2.9)
Progress over 4 rolls (mean)	×3: 76.4 tiles, 3.3 cards drawn, 3.6 played · ×8: 97.9 tiles, 2.2 drawn, 2.9 played
Full game (then VELA v4.2)	+21.5 [+20.9, +22.1], 2,048 pairs
Full game (then v4.3 Normal)	+20.6 [+19.6, +21.5], 512 pairs

11. The cards left in the deck decide

In short

When only two or three cards are left in the deck, the same position and hand can lead to different choices depending on the deck. In 600 random positions, swapping only the deck for another deck of the same size changed the choice in 2.4% of cases: 21.0% of close calls (gap under 1 point) and 0% of decisions with a gap over 10 points. The deck mostly settles close calls, but when it is nearly empty it changes the decision itself.

Case 11-1 · VELA v4.2 record

Two cards in the deck: roll or +4?

Position **804** · Rolls used **42** (58 left)
Deck (2): +6 · +8





Figure 86: Board. Green tiles: where the chosen action can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the comparison action.

The +4 draws at 808. With the actual deck (+6 · +8), the roll leads by 2.0 points. Trying all 435 possible two-card decks, the AI rolls with 53% of them and plays the +4 with the rest.

Action	AI value	Mean move	Draws now	Draw possible by next action
Roll · chosen	**1,871.8**	**+11.7**	**8.3%**	**19.4%**
+4 · comparison	1,869.9	+4.0	100.0%	100.0%
+6	1,818.1	+6.0	0.0%	0.0%

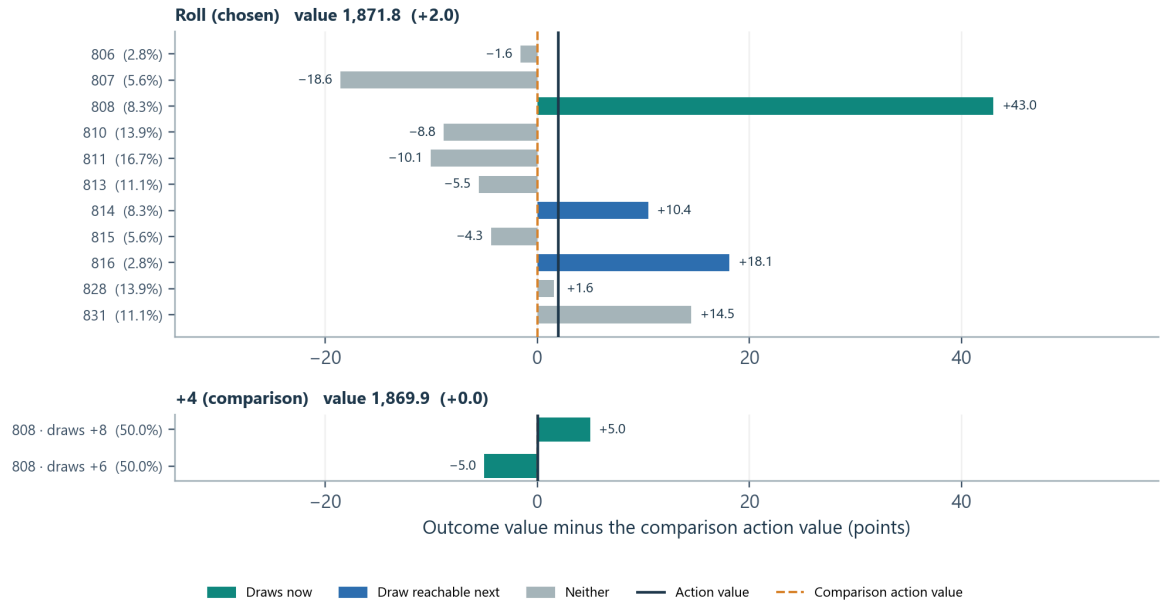


Figure 87: Outcome breakdown (reference: the value of the +4). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

Which way it goes depends on the card to be drawn at 808. If the deck holds a card such as the −3 or the +12 that sends the token from 808 to the draw tile at 820, the +4 comes out ahead; the deck that favors it most is −3 · ×10 (17.7 points). If the deck holds only small multipliers that cannot draw from 808, the roll comes out ahead; the deck that favors it most is ×2 · ×3 (15.5 points).

Removing the +6 from the hand flips the choice to the +4. The original gap is a close 2.0 points, so a small difference in what the +6 is worth under each action is enough. Removing it costs the roll side 57.4 points and the +4 side 55.0. The +6 is worth a little more on the roll side for two reasons. After the roll, it draws right away from the landing tile with probability 8.3%, against 2.8% for the +4. And on the +4 side the card drawn at 808 is the +6 or the +8 from the deck, so half the time a second +6 arrives and overlaps with the one in hand.

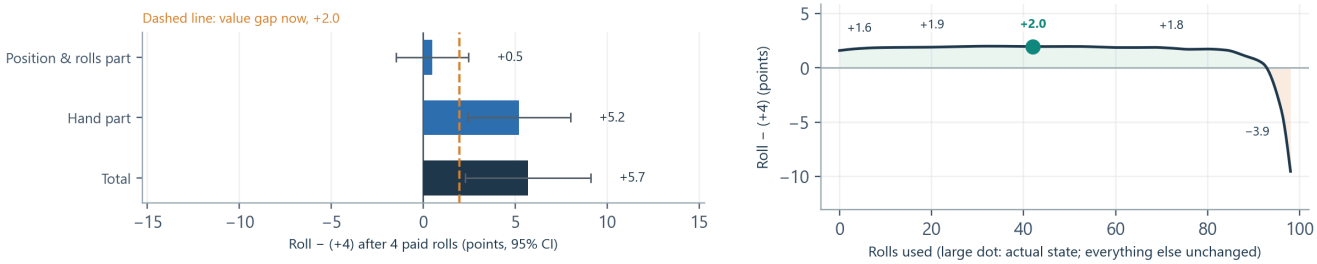


Figure 88: Left: the gap after four rolls with the actual deck. Right: the value gap (roll − (+4)) when only the number of rolls used changes.

The full-game comparison also favors the roll (evidence table). The value gap is a close 2.0 points, but with the actual deck rolling is the right choice.

Evidence	Result (in brackets: value of Roll − (+4))
VELA v4.2, depth 1·2 (production)·3	Roll · Roll · Roll
VELA v4.3 Normal·High·Max	Roll · Roll · Roll
LYRA v2	Roll (Roll − (+4): 4.7)
Change only the rolls used	20 used: Roll (+1.9) · 70 used: Roll (+1.9) · 92 used: Roll (+0.4)
Remove one other card	all flip (without +6: +4)
Replace the deck with a full 30	Roll (+0.4)
Gap after 4 rolls used	position part +0.5 (±2.0) · hand part +5.2 (±2.8) · total +5.7 (±3.4)
Progress over 4 rolls (mean)	Roll: 46.6 tiles, 1.8 cards drawn, 1.4 played · +4: 46.2 tiles, 2.4 drawn, 2.1 played
Full game (then VELA v4.2)	+2.9 [+1.3, +4.4], 2,048 pairs
Full game (then v4.3 Normal)	+3.9 [+1.2, +6.7], 512 pairs

Case 11-2 · VELA v4.2 record

Three cards in the deck: +10 or roll?

Position **810** · Rolls used **47** (53 left)
Deck (3): +9 · ×2 · ×8





Figure 89: Board. Green arrows: the chosen action and what follows; numbers give the order.

The +10 draws at 820. With the actual deck (+9 · ×2 · ×8), the +10 leads by 1.1 points. Of the 4,060 possible three-card decks, the AI plays the +10 with 60%.

Action	AI value	Mean move	Draws now	Draw possible by next action
+10 · chosen	**1,885.8**	**+10.0**	**100.0%**	**100.0%**
Roll · comparison	1,884.7	+7.4	8.3%	63.9%
+2	1,847.5	+18.0	0.0%	0.0%
+6	1,845.1	+6.0	0.0%	100.0%
+5	1,840.2	+5.0	0.0%	0.0%

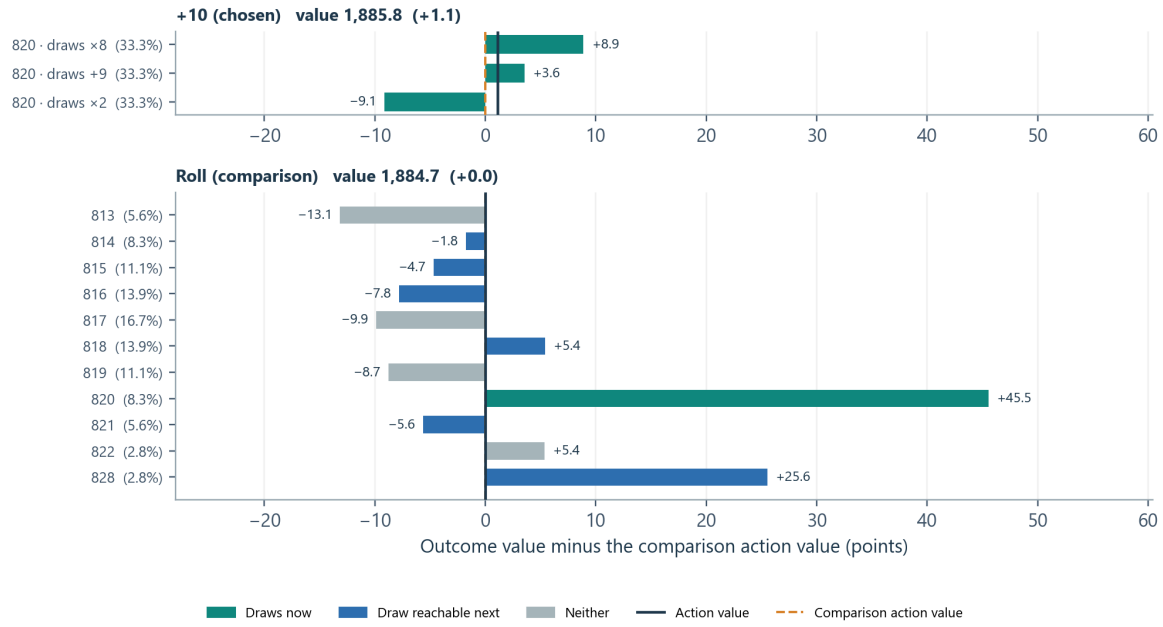


Figure 90: Outcome breakdown (reference: the value of the roll). Each bar is the outcome's value minus that reference; the percentage is the outcome's probability.

After the +10 draws at 820, the +6 in hand draws again at 826. The fork comes next. If the deck holds a +10 (820 to 830) or a +4 (826 to 830), one more draw follows, and a card that is valuable in itself, such as the ×10, also tilts things toward the +10. A +10 in the deck tilts the choice toward the +10 by 8.0 points on average, a ×10 by 8.8 and a +4 by 4.2. The deck that favors the +10 most is +10 · +10 · ×10 (23.6 points). If the deck holds only a +6 that duplicates the one in hand and small multipliers, such as +6 · +6 · ×3, the roll leads by up to 7.0 points.

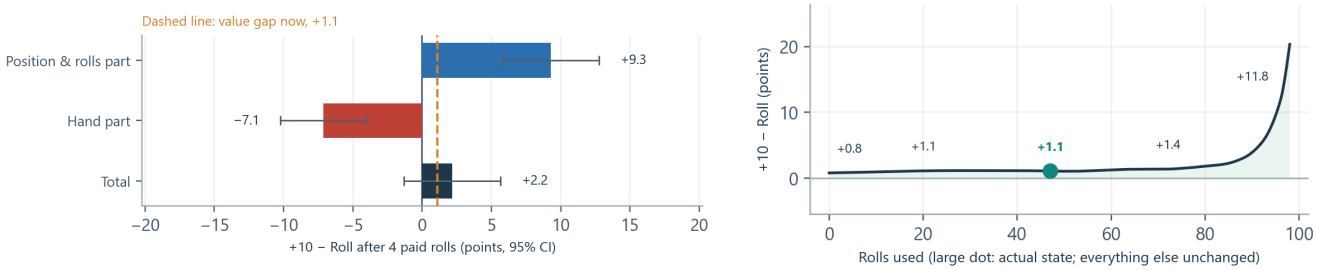


Figure 91: Left: the gap between the two actions after 4 paid rolls in 1,000 paired play-outs with the same dice, split into the position-and-rolls part and the hand part. Right: the value gap (+10 − Roll) when only the number of rolls used changes.

Both cases follow the same rule: play the card when the card drawn can draw again right away or is valuable in itself, and roll otherwise. In the full-game comparison 11-2 also favored the +10 (evidence table).

Evidence	Result (in brackets: value of +10 − Roll)
VELA v4.2, depth 1·2 (production)·3	+10 · +10 · +10
VELA v4.3 Normal·High·Max	+10 · +10 · +10
LYRA v2	+10 (+10 − Roll: 3.0)
Change only the rolls used	20 used: +10 (+1.1) · 70 used: +10 (+1.4) · 92 used: +10 (+5.4)
Remove one other card	no change (without +2: +10 · without +6: +10 · without +5: +10)
Replace the deck with a full 30	+10 (+2.0)
Gap after 4 rolls used	position part +9.3 (±3.5) · hand part −7.1 (±3.1) · total +2.2 (±3.5)
Progress over 4 rolls (mean)	+10: 94.0 tiles, 3.2 cards drawn, 3.6 played · Roll: 85.7 tiles, 3.1 drawn, 3.3 played
Full game (then VELA v4.2)	+4.3 [+2.6, +5.9], 2,048 pairs
Full game (then v4.3 Normal)	+4.2 [+1.4, +7.0], 512 pairs

PART 4

Endgame: order checked against exact values

With 0–2 rolls left, the exact value (the expected final score over every remaining possibility) can be computed. This part checks two judgments that come up often in the endgame.

The exact values in this part come from the production code's exact solver. Drawn cards come from the deck without replacement, an empty deck refills to 30, and the five-card hand limit applies. The values match the exact-value catalog to six decimal places in the same positions. Each case shows the board, the exact value of every action, an exact outcome breakdown (built like the outcome breakdowns of Part 3, with exact values in place of AI values) and the three AIs' values. Because the timing of the last roll matters here, the text counts the rolls left.

12. Order of the free cards before the last roll

In short

With one roll left, move and stage cards should normally all be played before the last roll. What decides the score is the order: a later card builds on an earlier one only if the earlier card reaches a jump or a draw tile.

Case 12-1 · Exact-value catalog

Play the −1 first so the +7 and +8 ride jumps

Position **1,734** · Rolls used **99** (1 left)
Deck (2): +12 · ×3

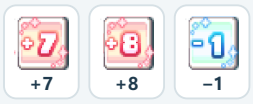




Figure 92: Board. Green arrows: the best order and what follows; numbers give the order. Orange dashed: the next-best order (+7 first). Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

First action	Exact value	Loss vs best	Chosen by
−1 · best	1,804.13	0	v4.2, v4.3 Normal, LYRA v2
+7	1,756.33	−47.8	
+8	1,756.33	−47.8	
Roll	1,748.72	−55.4	

The hand is +7, +8 and −1, with two cards in the deck. Playing the −1 first rides the jump at 1,733 to 1,755; the +7 then rides the jump at 1,762 to 1,786, and the +8 goes to 1,794. Playing the +7 or the +8 first rides no jump at all and stops at 1,749. The same three cards in a different order are 47.8 points apart.

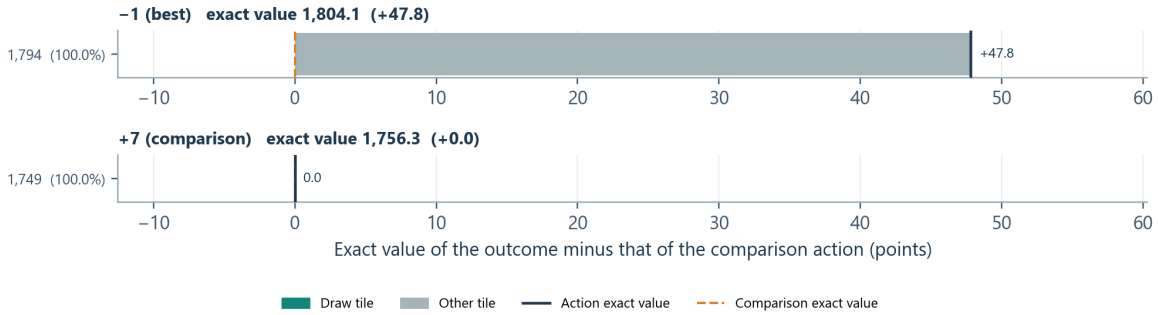


Figure 93: Exact outcome breakdown (reference: playing the +7 first). Each row is the tile reached after all free cards, just before the last roll, with the exact value there.

The −1 is the key because of where the jumps are. From 1,734 the −1 reaches 1,733 just behind, and 1,733 is a jump that climbs 22 tiles. From 1,755 the +7 reaches the jump at 1,762. Moving forward with the +7 or the +8 first passes both jumps. Every AI found this order.

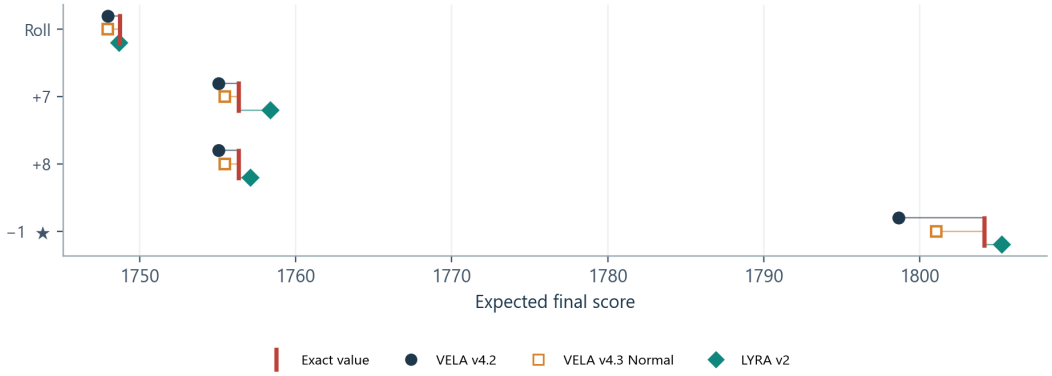


Figure 94: Exact value of each action (red vertical line) and the three AIs' values. ★ marks the best action.

Case 12-2 · Exact-value catalog

Play the +6 first to draw one more card

Position **1,842** · Rolls used **99** (1 left)

Deck (16): +1 · +3 · +4 · +6 · +7×2 · +8 · +10×2 · +11 · +12 · ×2 · ×8 · ×10 · Stage×2

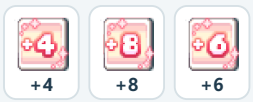




Figure 95: Board. Green arrows: the best order and what follows; numbers give the order. Orange dashed: the next-best order (+4 first). Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

First action	Exact value	Loss vs best	Chosen by
+6 · best	1,920.31	0	v4.2, v4.3 Normal, LYRA v2
+4	1,869.93	−50.4	
+8	1,869.93	−50.4	
Roll	1,860.48	−59.8	

The hand is +4, +8 and +6. Playing the +6 first rides the jump at 1,848 to 1,872, and the +8 then draws at 1,880. The drawn card can be played right away before the last roll, so the result depends on which card comes (a ×10 would be kept for the last roll). Playing the +4 or the +8 first draws nothing and ends at 1,860. The gap is 50.4 points.

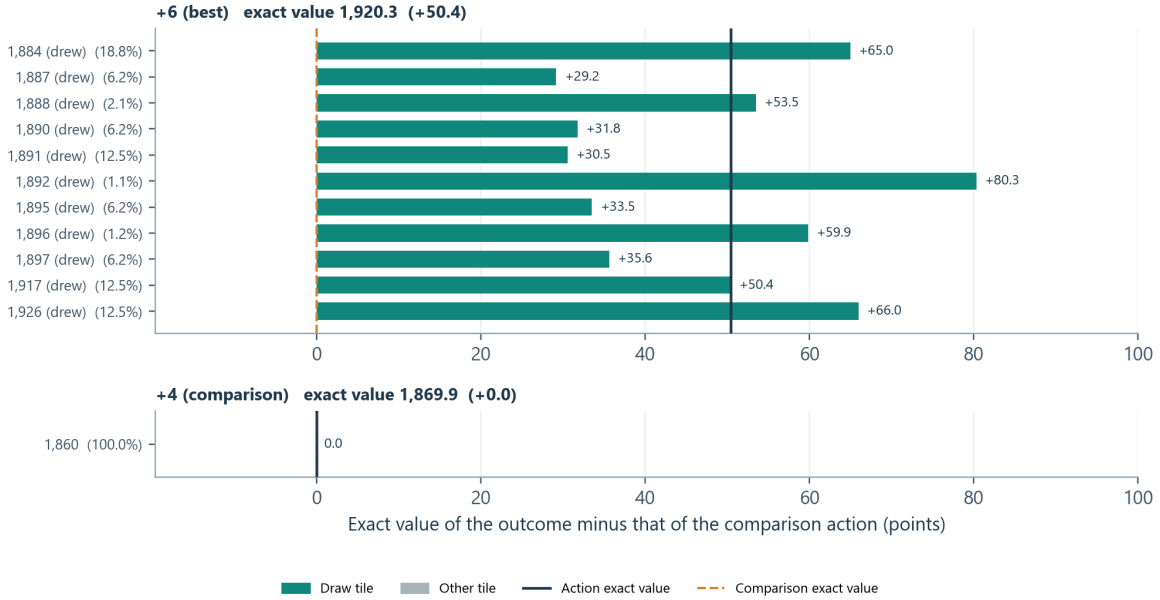


Figure 96: Exact outcome breakdown (reference: playing the +4 first). With the +6 first, the result depends on the card drawn at 1,880.

VELA v4.2 picks the +6 even with a one-step search, though at that depth it values the +6 9.1 points too low; at two steps, the production setting, the error falls to 2.5 points. A one-step search does not see the +8 drawing after the +6 and reads the position at 1,872 from its value table.

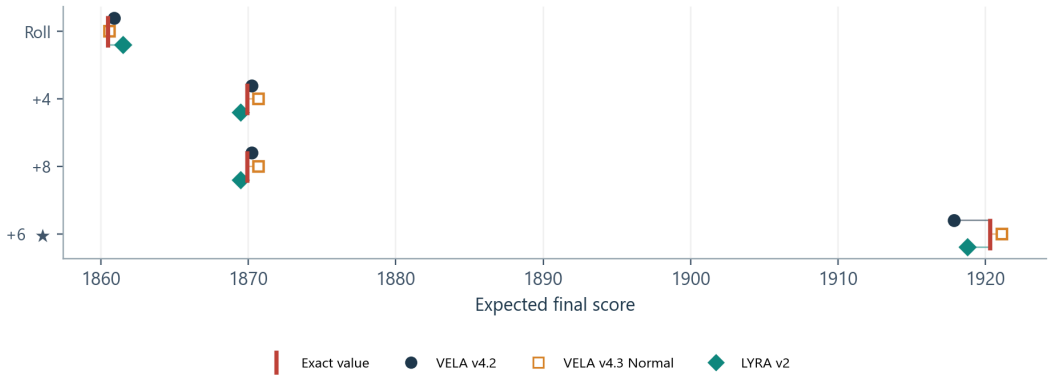


Figure 97: Exact value of each action (red vertical line) and the three AIs' values. ★ marks the best action.

13. Two rolls left: play cards first, or roll first?

In short

With two rolls left, free cards can wait until the dice have been seen. The exception is a jump or draw tile that only the current tile can reach (13-2). When two multipliers will both be used on the two rolls, the order is set by where the first roll lands, not by size (13-3).

Case 13-1 · Exact-value catalog

Roll before the move cards

Position **1,932** · Rolls used **98** (2 left)
Deck (9): +1 · +6×2 · +8 · +12 · ×5 · ×8 · ×10 · Stage





Figure 98: Board. Green tiles: where the best action (roll) can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange dashed: the next-best action (+9).

First action	Exact value	Loss vs best	Chosen by
Roll · best	2,009.47	0	v4.2, v4.3 Normal, LYRA v2
+9	1,996.96	−12.5	
+4	1,993.88	−15.6	
+3	1,990.55	−18.9	

The hand is +3, +9 and +4, with two rolls left. By exact value, rolling first is better by 12.5 points or more. Move cards use no roll, so they can be played at any time. Played now, the +9 goes to 1,941, the +4 to 1,936 and the +3 to 1,935, and none of those tiles draws or has a jump.

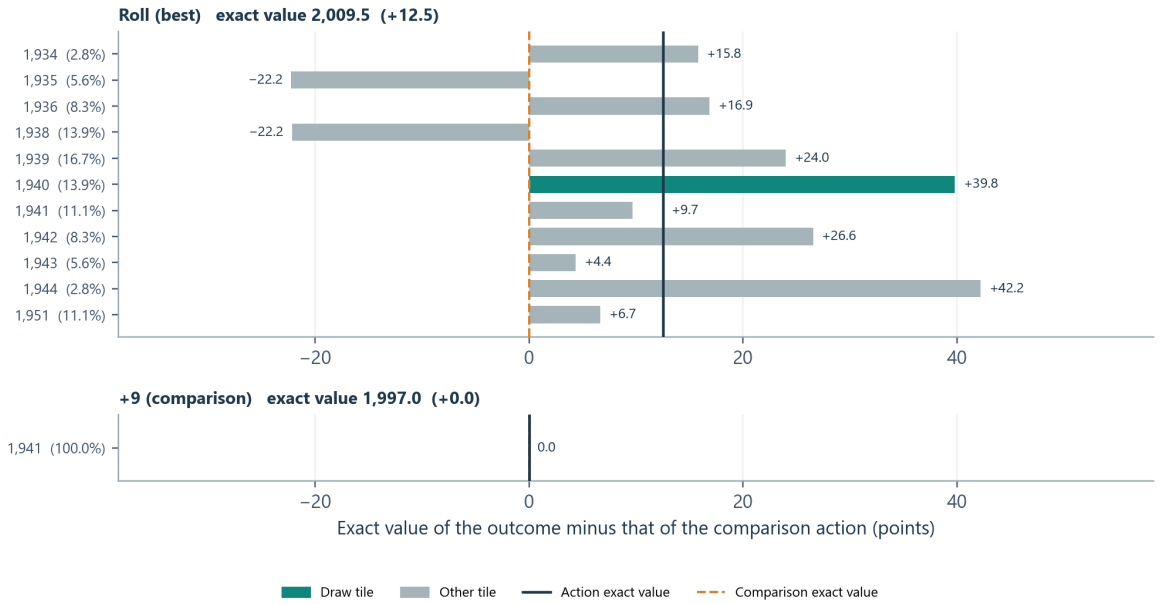


Figure 99: Exact outcome breakdown (reference: playing the +9 first). The roll is split by landing tile; the +9 has a single tile.

Rolling first lets the player see the landing tile and then fit the three cards to jumps and draw tiles. The exact outcome breakdown shows most landing tiles of the roll above the +9: every landing tile opens a new set of tiles the three cards can reach. Playing a card first gives that up. Every AI rolled.

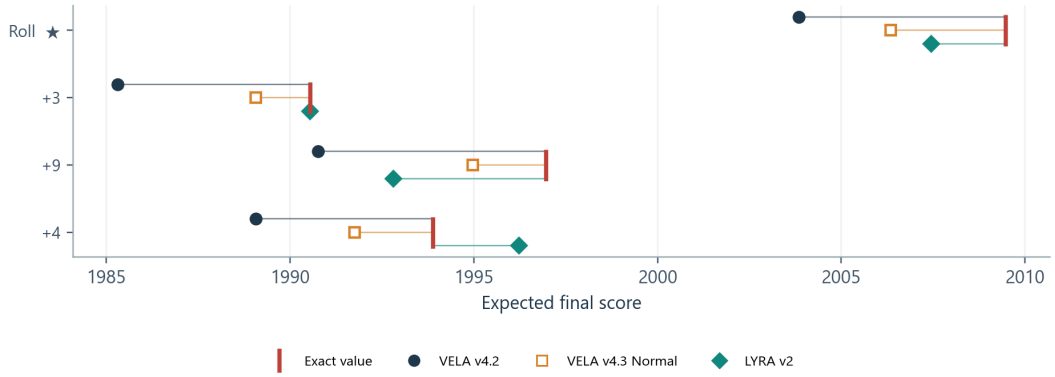


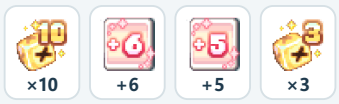
Figure 100: Exact value of each action (red vertical line) and the three AIs' values. ★ marks the best action.

Case 13-2 · Exact-value catalog

Contrast: a jump within reach now, so the +5 first

Position **1,680** · Rolls used **98** (2 left)

Deck (6): +4 · +7 · +11 · ×2 · ×5 · Stage



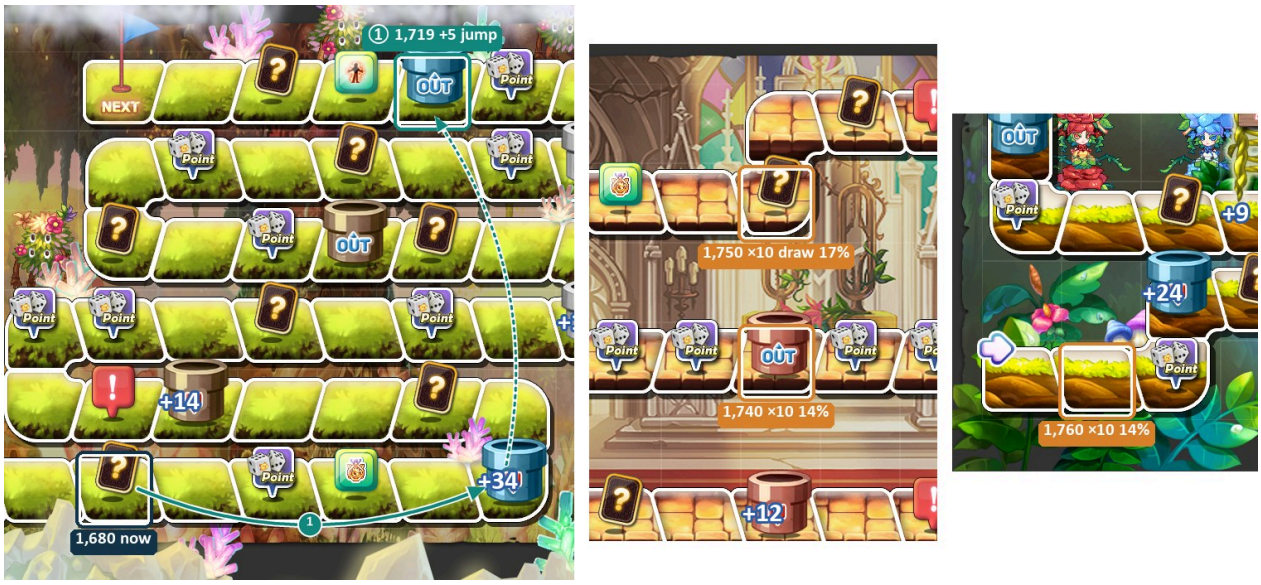


Figure 101: Board. Green: the best action (+5). A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange tiles: where the next-best action (×10) lands. Thin dotted lines are jumps.

First action	Exact value	Loss vs best	Chosen by
+5 · best	1,827.87	0	v4.2, v4.3 Normal, LYRA v2
×10	1,797.36	−30.5	
×3	1,796.41	−31.5	
+6	1,791.55	−36.3	
Roll	1,789.51	−38.4	

The hand is ×10, +6, +5 and ×3, with two rolls left. The +5 rides the jump at 1,685 (+34) to 1,719. It draws nothing, but the +5 is the only card that reaches this jump for certain from here. A roll rides it only on a sum of 5 (11.1%), and the ×10 and the ×3 cannot land on 1,685. Once passed, it cannot be ridden again. After the +5, the ×10 and the ×3 go on the two rolls.

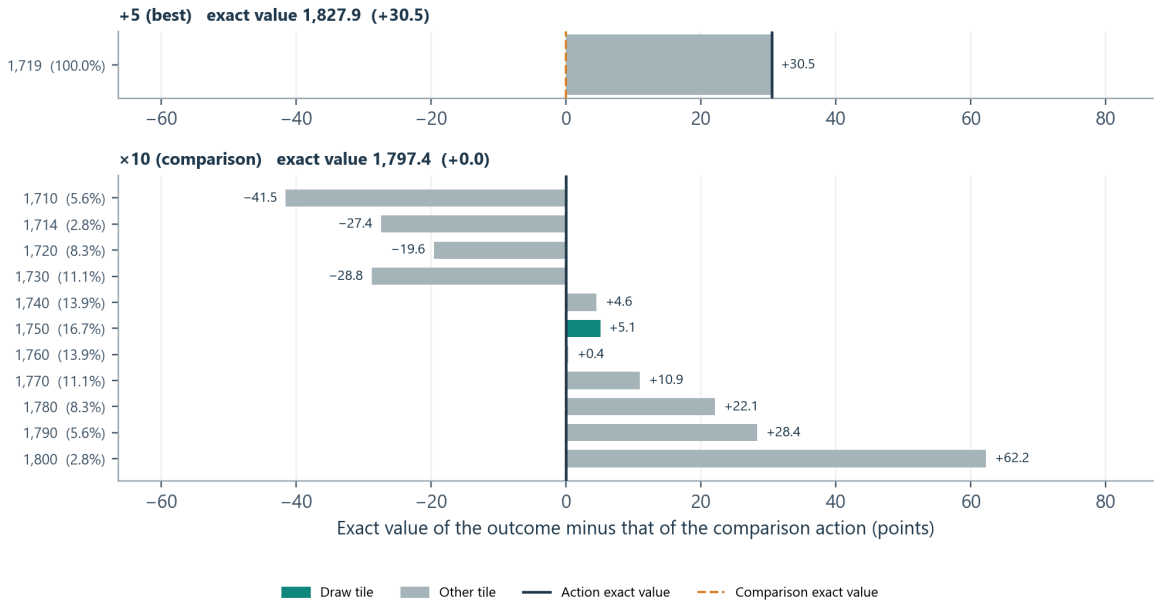


Figure 102: Exact outcome breakdown (reference: playing the ×10 first). The +5 starts the last two rolls from 1,719.

By exact value the +5 beats the next-best ×10 by 30.5 points and the roll by 38.4 points. What separates this case from 13-1 is a single 34-tile jump that the +5 reaches for certain from the current tile. “Roll first” holds only when the cards lose nothing by waiting. Every AI picked the +5.

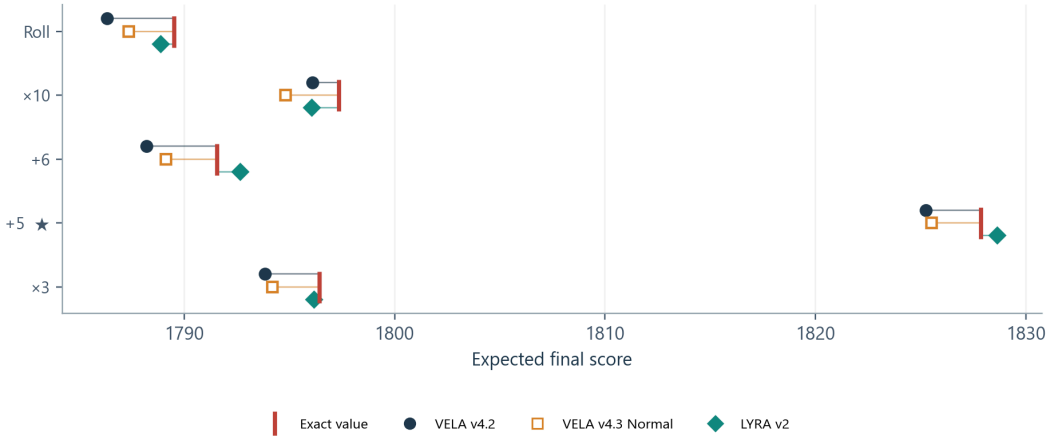


Figure 103: Exact value of each action (red vertical line) and the three AIs’ values. ★ marks the best action.

Case 13-3 · Exact-value catalog

Both the ×8 and the ×7 get used, but the ×7 goes first

Position **1,684** · Rolls used **98** (2 left)

Deck (7): −3 · −1 · +5 · +6 · +7 · +9 · ×3





Figure 104: Board. Green tiles: where the best action (×7) can land. A roll or multiplier can land on many tiles, so only some are marked (% = chance of landing there; "draw" = draw tile). Orange tiles: where the next-best action (×8) lands.

First action	Exact value	Loss vs best	Chosen by
×7 · best	1,798.69	0	v4.2, v4.3 Normal, LYRA v2
×8	1,793.95	−4.7	
Roll	1,766.42	−32.3	

The hand is ×8 and ×7, with two rolls left. Whichever goes first, both cards are used on the two rolls. Even so, the ×7 first is 4.7 points better, against the intuition of using the larger multiplier first.

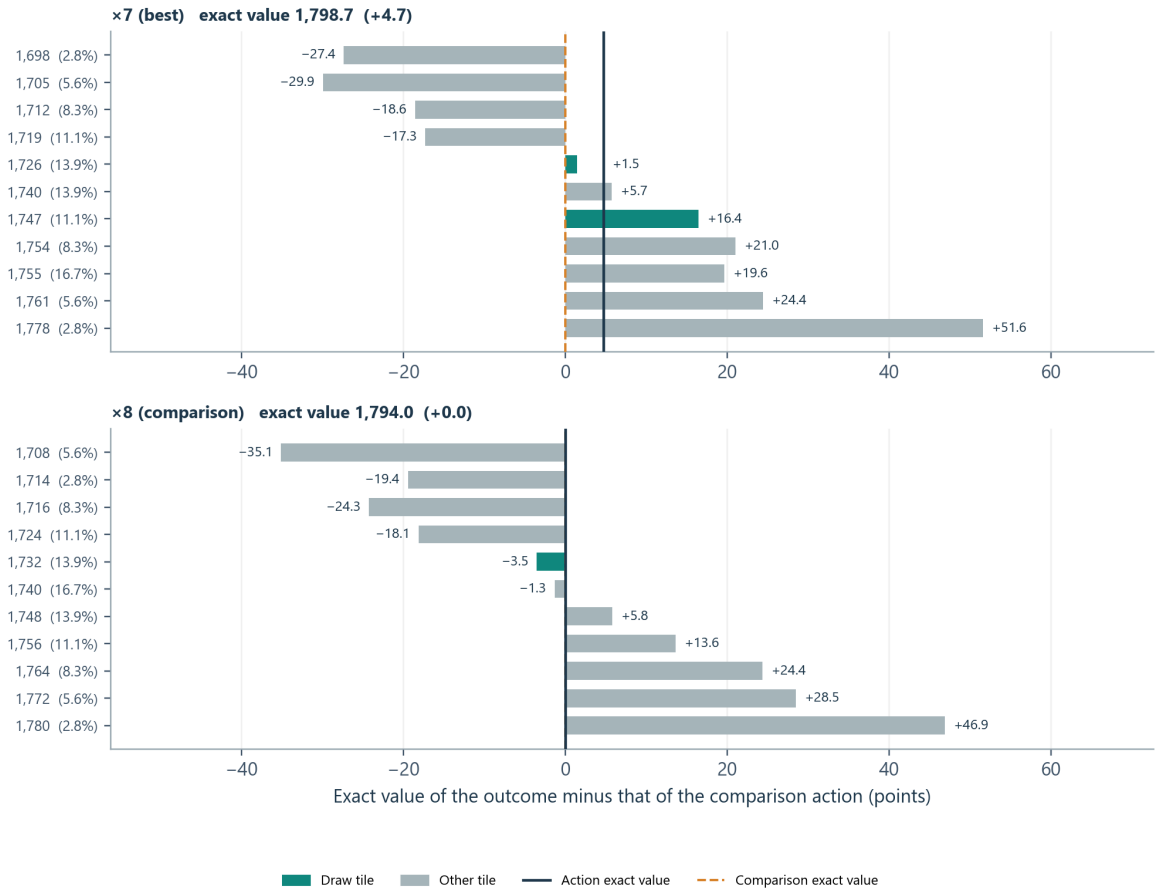


Figure 105: Exact outcome breakdown (reference: playing the ×8 first). Green marks draw tiles. On a sum of 7 the ×7 rides the jump at 1,733 to 1,755.

The difference comes from where the first roll lands. On the most common sum, 7 (16.7%), the ×7 reaches 1,733 and jumps to 1,755, and on sums of 6 (1,726) and 9 (1,747) it lands on draw tiles. On a sum of 7 the ×8 only reaches 1,740, and only a sum of 6 (1,732) lands on a draw tile. A better first landing means the second roll, with the ×8, starts from a better tile. Even when both cards will be used, the order is set by landing tiles, not by size. All three AIs picked the ×7.

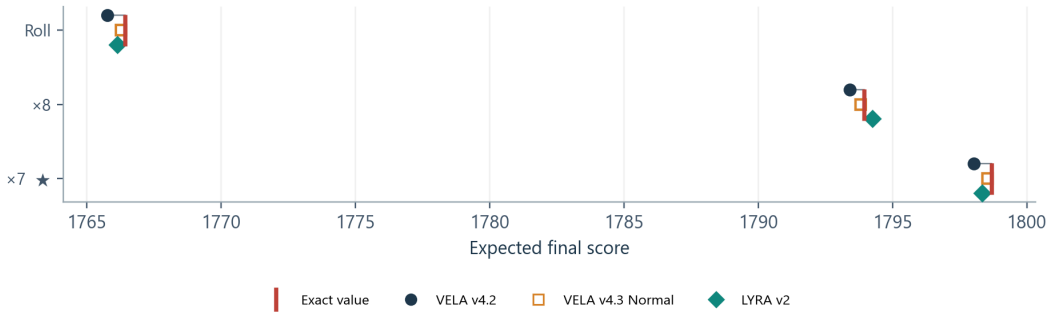


Figure 106: Exact value of each action (red vertical line) and the three AIs' values. ★ marks the best action.

PART 5

Where the AI goes wrong

Positions from the exact-value catalog where an AI lost more than 1 point, grouped by type. For each, the exact best line and the reason the AI missed it.

Most AI mistakes happen in close calls (Part 6), but occasionally an AI loses several points or more. In the eight cases below, the exact values settle who was right. Looking at how far each AI's value for each action sits above or below the exact value (its **error**) shows where each mistake came from.

The way each AI calculates decides the type of mistake. VELA v4.2 looks one move past the action and then uses its value table; LYRA v2 values the position one step ahead directly with its neural network. VELA v4.3 Normal and above calculate every action three steps ahead, and go up to six steps only when the top two actions are within 1.5 points.

Type	Cases
A draw two steps cannot see	14-1 (1,636), 14-2 (2,066): VELA v4.2 and LYRA v2 are both wrong; v4.3, which searches deeper, is right.
Error in the value table itself	14-3 (1,564), 14-4 (1,928): VELA is wrong at every depth or at one particular depth; LYRA is right.
Limits of a one-step calculation	14-5 (1,629), 14-6 (2,248): LYRA v2 and a one-step VELA make the same mistake.
New mistakes from deeper search	14-7 (1,632), 14-8 (2,061): v4.2 is right; v4.3 goes wrong at its three-step evaluation and does not check deeper.

14-1. A draw that only the third card makes (1,636)

VELA v4.2 and LYRA v2 both wrong

The draw should come before the jump

Position **1,636** · Rolls used **99** (1 left)
Deck (4): +8 · ×5 · ×8 · Stage





Figure 107: Board. Left (green): the exact best line. Right (orange dashed): the line VELA v4.2 and LYRA v2 chose (+7 first). Numbers give the order of play; thin dotted lines are jumps.

First action	Exact value	Loss vs best	Chosen by
−1 · best	1,695.69	0	v4.3 Normal
−3	1,695.69	0	
+7	1,680.93	−14.8	v4.2, LYRA v2
+9	1,665.77	−29.9	
Roll	1,661.14	−34.6	

The best line is −1 to 1,635, +7 to 1,642 and −3 to 1,639, which draws, then +9 to 1,648, with the drawn card played before the last roll. The +7 that both AIs chose rides the jump at 1,643 to 1,659, a big step forward, but the +9 then goes to 1,668 and the line ends without a draw.

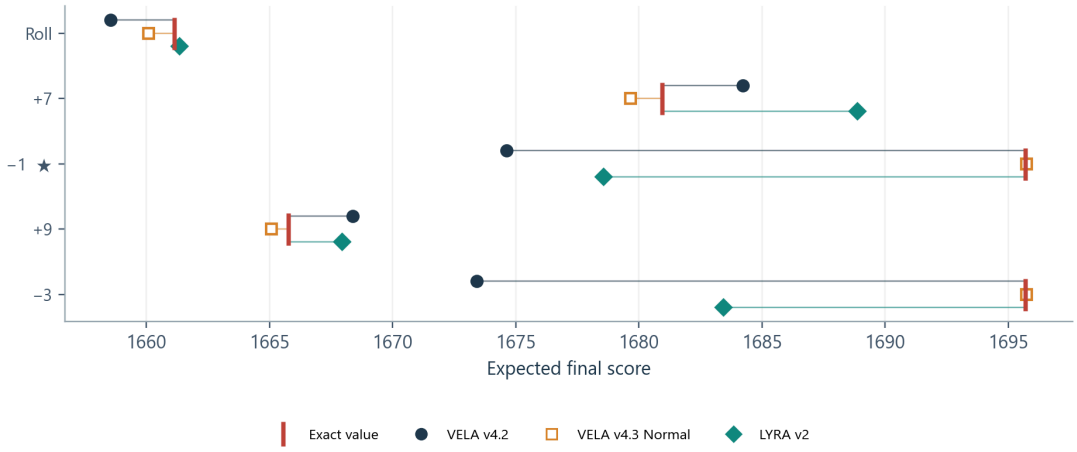


Figure 108: Exact value of each action (red vertical line) and the three AIs' values. ★ marks the best action. VELA v4.2 and LYRA v2 valued the −1 and the −3 far too low.

The draw comes from the third card (the −3). VELA v4.2's two-step calculation follows only the −1 and the +7 and then reads the position at 1,642 from its value table; LYRA v2 reads the position at 1,635 after the −1 with its network. Neither captures the full value of "one more draw with the free cards left", so they valued the −1 21.1 and 17.1 points too low. v4.3 Normal searched this position six steps deep, cut the −1's error to +0.1 points and chose the best action.

14-2. The order of three cards (2,066)

VELA v4.2 and LYRA v2 both wrong

+9 before +3

Position **2,066** · Rolls used **98** (2 left)
Deck (10): +1 · +4×2 · +5 · +6 · +8 · ×2 · ×5 · ×10 · Stage





Figure 109: Board. Green arrows: the best line and what follows; numbers give the order. Orange dashed: the line VELA v4.2 and LYRA v2 chose. Thin dotted lines are jumps.

First action	Exact value	Loss vs best	Chosen by
+9 · best	2,147.20	0	v4.3 Normal
−3	2,147.20	0	
Roll	2,143.55	−3.6	
+3	2,143.08	−4.1	v4.2, LYRA v2
×3	2,142.18	−5.0	

Two rolls left; the hand is +3, ×3, +9 and −3. The best line plays the +9 (or the −3) first. After the +9 to 2,075, the −3 rides the jump at 2,072 up to 2,083, and the +3 draws at 2,086. With the +3 first, as both AIs chose, the token goes to 2,069 and the −3 brings it back to the current tile (2,066), where it draws. Both lines draw one card, but without the jump the AI line starts its two remaining rolls 20 tiles behind (with the +9 still in hand). The loss is 4.1 points. VELA v4.2 also picks the +9 with a three-step search, and v4.3 Normal already gets it right at three steps. This is the same type of mistake as 14-1.

14-3. The value table undervalues a draw now (1,564)

VELA wrong, LYRA right

Draw now with the +8 instead of the stage card

Position **1,564** · Rolls used **99** (1 left)
Deck (4): +10 · +12 · ×10 · Stage





Figure 110: Board. Green arrows: the best line (+8 first) and what follows; numbers give the order. Orange dashed: VELA's line (stage card first). Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage. Dark labels mark tiles both lines pass.

First action	Exact value	Loss vs best	Chosen by
+8 · best	1,686.21	0	LYRA v2
Stage	1,682.20	−4.0	v4.2, v4.3 Normal
−3	1,662.35	−23.9	
+9	1,656.83	−29.4	
Roll	1,595.05	−91.2	

The +8 draws at once at 1,572. VELA v4.2 chose the stage card at both two and three steps, and so did v4.3 at every strength (a loss of 4.0 points). A one-step VELA and LYRA v2 chose the +8.

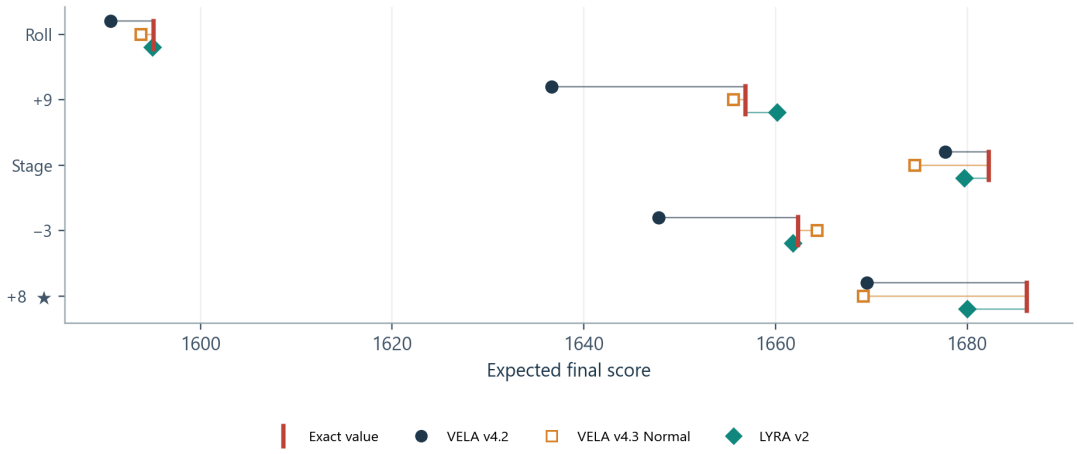


Figure 111: Exact value of each action and the three AIs' values. VELA valued the +8 far too low at every depth.

VELA's error is concentrated on one action, the +8. It valued the +8 16.7 points too low at two steps and 17.0 points too low with v4.3 Normal. A deeper search does not shrink the error, so the problem is not search depth: the value

table gives wrong values around this position. The neural network of LYRA v2 also undervalued the +8, but only by 6.2 points, and it chose the +8. The two AIs estimate value in different ways, so one can avoid an approximation error the other makes.

14-4. An error that appears only at depth 2 (1,928)

Only VELA v4.2 at depth 2 wrong

Stage card before −2

Position **1,928** · Rolls used **99** (1 left)

Deck (30): −3 · −2 · −1 · +1 · +2 · +3 · +4×2 · +5×2 · +6×2 · +7×2 · +8×2 · +9×2 · +10×2 · +11 · +12 · ×2 · ×3 · ×5 · ×7 · ×8 · ×10 · Stage×2





Figure 112: Board. Green arrows: the best line and what follows; numbers give the order. Orange dashed: the line VELA v4.2 chose at depth 2. Arrows that cross between boards are moves into another stage. Dark labels mark tiles both lines pass.

First action	Exact value	Loss vs best	Chosen by
Stage · best	2,000.03	0	v4.3 Normal, LYRA v2
−2	1,997.72	−2.3	v4.2
×5	1,969.82	−30.2	
Roll	1,947.56	−52.5	

The best line plays the stage card to 1,961, then the −2 to 1,959, and makes the last roll with the ×5. Playing the −2 first goes to 1,926, and the stage card then reaches 1,961. The two lines differ only in whether the last ×5 is rolled from 1,959 or from 1,961, and that difference is worth 2.3 points. VELA v4.2 is right at depths 1 and 3 and chooses the −2 only at depth 2, where it values the −2 2.2 points too high and the stage card 2.2 points too low, which reverses the order. A deeper search is not always more accurate: the value-table error in the positions where a given depth stops goes straight into the decision.

Why step back to 1,959 with the −2 at all? Because of where the last roll lands. The game ends after the last roll, so a card drawn then cannot be used unless a bonus roll comes up. On the last roll a jump matters more than a draw tile. From 1,959 the ×5 rides the jump at 1,994 to 2,022 on a sum of 7 (16.7%). From 1,961 no sum reaches a jump; only the chance of landing on a draw tile is higher (36%). The ×5's average landing tile is 1,998.7 from 1,959 and 1,996.0 from 1,961. The −2 would be worth nothing left over after the last roll, so spending it to step back two tiles costs nothing.

14-5. Limits of a one-step calculation with one card in the deck (1,629)

LYRA v2 and a one-step VELA both wrong

−3 before +8

Position **1,629** · Rolls used **99** (1 left)
Deck (1): +9





Figure 113: Board. Green arrows: the best line and what follows; numbers give the order. Orange dashed: the line LYRA v2 chose. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage.

First action	Exact value	Loss vs best	Chosen by
−3 · best	1,684.68	0	v4.2, v4.3 Normal
+8	1,672.57	−12.1	LYRA v2
+3	1,670.97	−13.7	
Roll	1,644.33	−40.4	

Only a +9 is left in the deck, so the next card drawn is known to be a +9. The best line plays the −3 to 1,626, then the +8, which draws the +9 at 1,634; that +9 then rides the jump at 1,643 to 1,659, and the +3 draws once more at 1,662. The +8 that LYRA v2 chose rides the jump at 1,637 to 1,651, but the +9 drawn next with the +3 cannot reach a jump and the line ends at 1,663. The loss is 12.1 points.

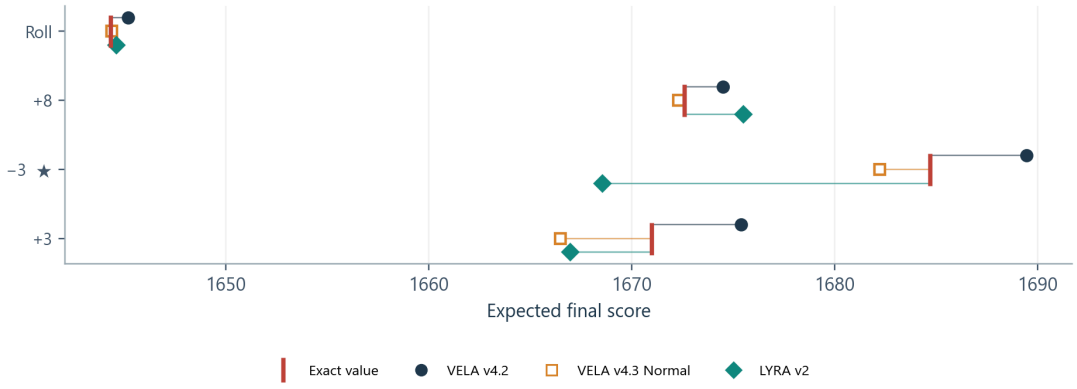


Figure 114: Exact value of each action and the three AIs' values. LYRA v2 valued the −3 far too low.

The −3 draws nothing on its own, so its value depends on how well the position one step later (1,626, hand +8 and +3, deck +9) is evaluated. LYRA v2 valued that position 16.1 points too low, and a one-step VELA chose the same action. VELA v4.2, which searches two steps, follows the −3 through to the +8 and sees the +9 being drawn, so it gets it

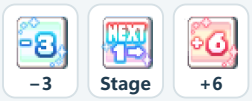
right. In rare positions such as a nearly empty deck, leaving the position one step ahead to an approximate value is a weak point.

14-6. The order of the stage card and the −3 (2,248)

LYRA v2 and a one-step VELA both wrong

Stage card before −3

Position **2,248** · Rolls used **99** (1 left)
Deck (8): +1 · +5 · +8 · +10 · +12 · ×2 · ×5 · ×8



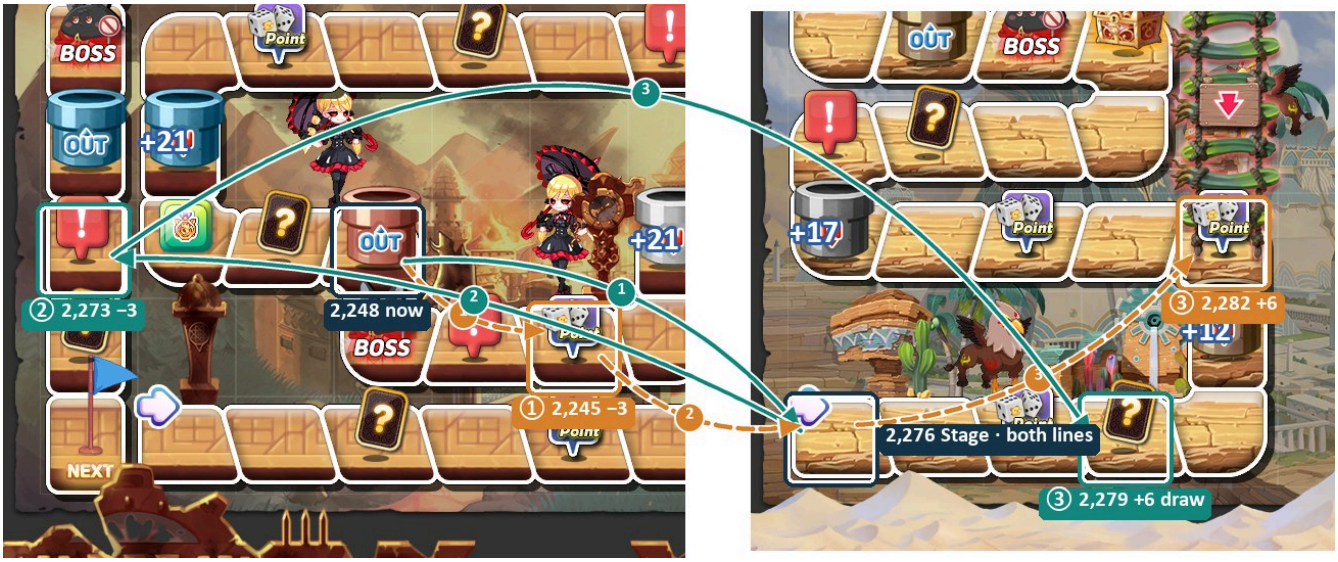


Figure 115: Board. Green arrows: the best line and what follows; numbers give the order. Orange dashed: the line LYRA v2 chose. Arrows that cross between boards are moves into another stage. Dark labels mark tiles both lines pass.

First action	Exact value	Loss vs best	Chosen by
Stage · best	2,301.34	0	v4.2, v4.3 Normal
−3	2,291.76	−9.6	LYRA v2
+6	2,288.88	−12.5	
Roll	2,266.03	−35.3	

The best line plays the stage card to 2,276, then the −3 to 2,273 and the +6 to 2,279, which draws. With the −3 first, as LYRA v2 chose, the token goes to 2,245 and the stage card still reaches 2,276, but the +6 then lands on 2,282 and draws nothing. The loss is 9.6 points. Neither line draws on the first move, so a one-step calculation leaves the difference to its evaluation of the next position. LYRA v2 valued the −3 6.4 points too high and the stage card 5.7 points too low.

14-7. A wrong three-step value stops the deeper search (1,632)

New mistake in VELA v4.3 (v4.2 right)

+7 before the stage card

Position **1,632** · Rolls used **99** (1 left)
Deck (4): −3 · +5 · ×8 · ×10

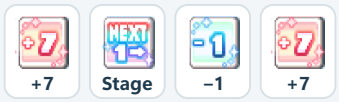




Figure 116: Board. Green arrows: the best line and what follows; numbers give the order. Orange dashed: the line v4.3 chose. Thin dotted lines are jumps. Arrows that cross between boards are moves into another stage. Dark labels mark tiles both lines pass.

First action	Exact value	Loss vs best	Chosen by
+7 · best	1,771.71	0	v4.2, LYRA v2
Stage	1,767.59	−4.1	v4.3 Normal
−1	1,729.74	−42.0	
Roll	1,663.56	−108.2	

The hand is +7, a stage card, −1 and +7. The best line draws at once with a +7 at 1,639, then plays the stage card and the other +7. v4.3 played the stage card first at every strength (a loss of 4.1 points).

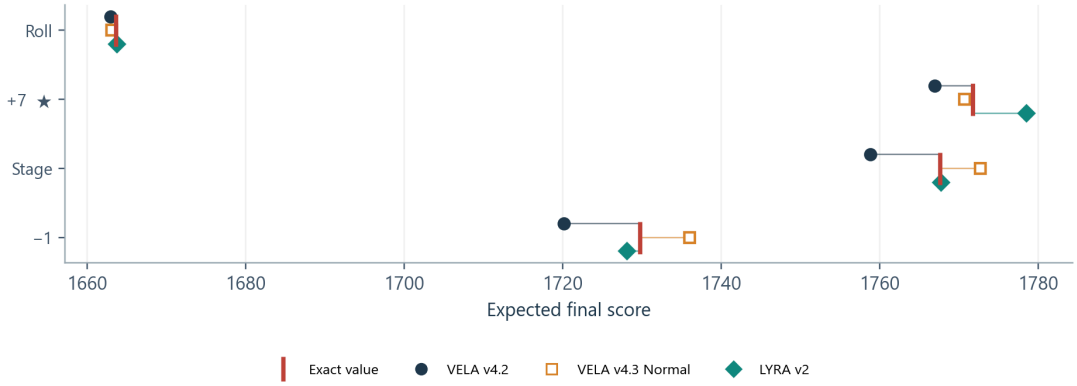


Figure 117: Exact value of each action and the three AIs' values. v4.3 Normal valued the stage card too high.

v4.3's three-step calculation valued the stage card 5.1 points too high, which put the stage card more than 1.5 points ahead of the +7. Beyond 1.5 points the decision is not treated as close and the search does not go deeper, so the three-step error becomes the decision. VELA v4.2's two-step calculation and LYRA v2 both chose the +7. Deeper search improves decisions on average (Part 6), but it occasionally creates positions like this one.

14-8. The −2 reaches the same tile with one card fewer (2,061)

New mistake in VELA v4.3 (v4.2 right)

Two stage cards instead of the −2

Position **2,061** · Rolls used **99** (1 left)

Deck (12): −3 · +4 · +7 · +8 · +9 · +10 · +12 · ×2 · ×3 · ×5 · ×7 · ×8





Figure 118: Board. Green arrows: the best line (two stage cards) and what follows; numbers give the order. Orange dashed: the line v4.3 chose (−2 first). Arrows that cross between boards are moves into another stage. Dark labels mark tiles both lines pass.

First action	Exact value	Loss vs best	Chosen by
Stage · best	2,123.55	0	v4.2, LYRA v2
−2	2,121.54	−2.0	v4.3 Normal
+1	2,120.55	−3.0	
Roll	2,086.13	−37.4	

The hand is a stage card, +1, −2 and a stage card. Playing the −2 first steps back to 2,059, and the stage card then reaches 2,063, the same tile the stage card reaches when played straight away. Both lines make the last roll from the same tile (2,107); the one that played the −2 simply has one card fewer. If the last roll earns a bonus roll, the remaining cards can be played then, so one card is worth little but not nothing. The loss is 2.0 points.

v4.3's three-step calculation valued the −2 4.1 points too high, and for the same reason as in 14-7 it did not search deeper. VELA v4.2 makes the same mistake with a three-step search. The error lies in the three-step evaluation of this position itself.

PART 6

How reliable the recommendations are

How often the AI is right, how much it loses when it is wrong, and what happens when a player overrides it.

6.1 What reliability means here

Reliability splits into two questions: is this one recommendation the best move, and how many points does following the recommendations score on average? The first can be checked directly only in the endgame, where exact values can be computed. The second is measured by playing thousands of games and averaging the scores.

The decisions in a typical game look like this (decisions with cards in hand, 2,000 VELA v4.2 games).

161 decisions per game

Decisions where the hand offers a choice. About 4 of them come with 0–2 rolls left.

About 11 close calls

Decisions where the AI rates the best and second-best actions less than 1 point apart. Either choice costs little.

Most gaps are large

The second-best action is rated 9.4 points below the best (median). Only 19% of decisions have a gap under 3 points.

Measured against exact endgame values, VELA v4.3 Normal chose a non-optimal action about 3 times per 1,000 decisions and lost more than 1 point 1.4 times. VELA v4.2 did so 12 and 5.1 times, and LYRA v2 26 and 7.4 times.

6.2 Comparison with exact endgame values

A catalog holds the exact value of every action in 4,312 positions reached in real games with 0–2 rolls left. We compared the action each AI chose with the exact optimal action. VELA v4.3 was run with the production search code; at Fast it chose the same action as v4.2 in every position.

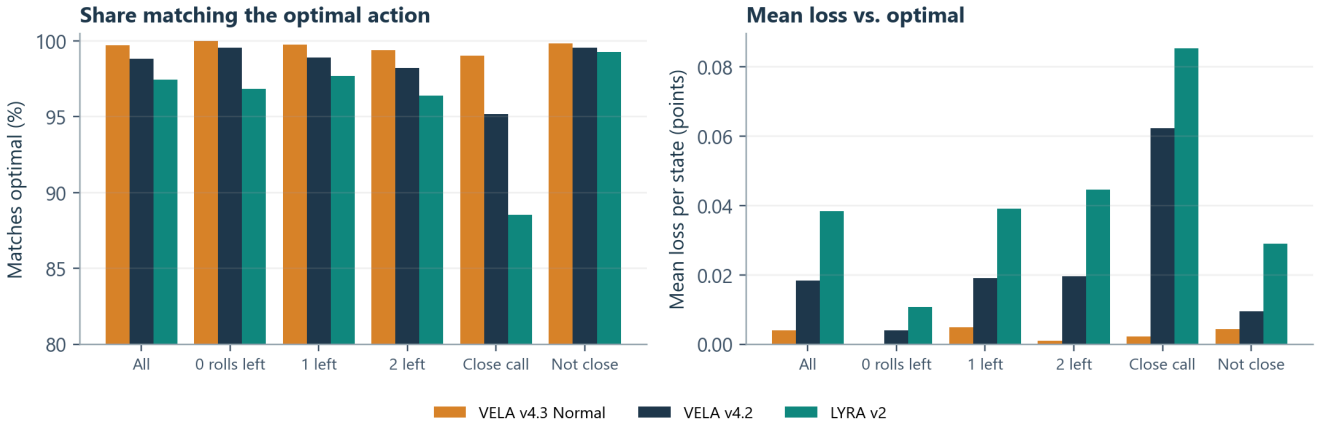


Figure 119: Share of positions where the AI matched the optimal action (left) and mean loss per position (right). The loss is the exact value of the optimal action minus the exact value of the action chosen. A close call is a position where the top two actions are less than 1 point apart in exact value.

Measure	v4.3 Normal	v4.3 High/Max	VELA v4.2	LYRA v2
Matches the optimal action	99.7%	99.8%	98.8%	97.4%
Mean loss per position	0.004	0.003	0.018	0.038
Positions losing more than 1 point	0.14%	0.12%	0.51%	0.74%
Largest loss	4.1	4.1	14.8	14.8
Match rate in close calls	99.0%	99.6%	95.2%	88.5%
Error in the chosen action's value	0.26	0.25	0.74	0.73

Losses and errors are in points.

VELA v4.2 and LYRA v2 chose the optimal action in more than 97% of endgame positions, and 70–75% of their mistakes came in close calls (17% of positions). VELA v4.3 Normal searches deeper in close calls, which raises its close-call match rate from 95.2% to 99.0%. It corrected 44 of the 51 positions where v4.2 went wrong and newly erred in 6 positions where v4.2 was right (losing up to 4.1 points). High and Max gave identical results on this catalog.

The catalog only includes positions that could be solved within a node limit, so it covers only about 22% of positions with two rolls left. In the endgame a six-step search nearly reaches the end of the game. There is no evidence of an equally large gain in the middle game; over whole games, Normal beats Fast by 1.4 points on average across 20,000 games (Part 1). Exact values cannot be computed before the endgame, so these match rates do not carry over to the game as a whole.

6.3 Why LYRA scores higher despite more endgame errors

In the endgame comparison LYRA v2 made more mistakes than VELA v4.2. Yet over the same 9,200 games, with the control-variate correction, LYRA v2 averaged 1,879.95 points and VELA v4.2 1,877.30, a lead of 2.65 points for LYRA. The two results measure different parts of the game. A game has only about 4 decisions with 0–2 rolls left, so the endgame gap of 0.020 points per position amounts to about 0.08 points per game. Nearly all of the 2.65-point lead comes from the remaining 157 or so decisions.

A comparison that mixed the two AIs by phase over the same games shows where the lead comes from.

Phase played by VELA v4.2 (LYRA v2 plays the rest)	vs. LYRA v2 alone
First 20 rolls (rolls used 1–20)	−0.13 (no difference)
First 50 rolls (1–50)	−1.20
First 80 rolls (1–80)	−2.12
Whole game	−2.65
Last 20 rolls (81–100)	−0.40

Differences in points.

The two AIs are even over the first 20 rolls; the gap opens mostly in the middle of the game, between 21 and 80 rolls used. Handing only the last 20 rolls to VELA v4.2 still lowers the score by 0.40 points, so LYRA v2 is better over the last 20 rolls as a whole, and VELA v4.2 is slightly more accurate only in the very short stretch with 0–2 rolls left. VELA v4.2's value table comes from a compressed exact calculation, which makes it strong in the endgame, where its two-step search reaches the end of the game. LYRA v2 is a neural network trained on its own games, and it appears to capture long-range links in the middle game (position, hand and deck) better than VELA v4.2. This report did not test which factor causes the difference.

VELA v4.3 Normal scores 1.4 points above v4.2 over 20,000 games, which would narrow the gap to LYRA v2 to a little over 1 point (not measured on the same games). Since the game is scored by the final tile, the average score is the measure to use when choosing an AI; the endgame match rate covers only a few decisions per game.

6.4 Overriding the AI

We measured what it costs to replace recommendations that look odd with another action. VELA v4.2 played 10,000 games per rule on the same dice sequences, with some decisions replaced by the rules below, and the average scores (control-variate corrected) were compared. A rule was applied only where the AI's recommendation differed from it. Following every recommendation averaged 1,877.8 points.

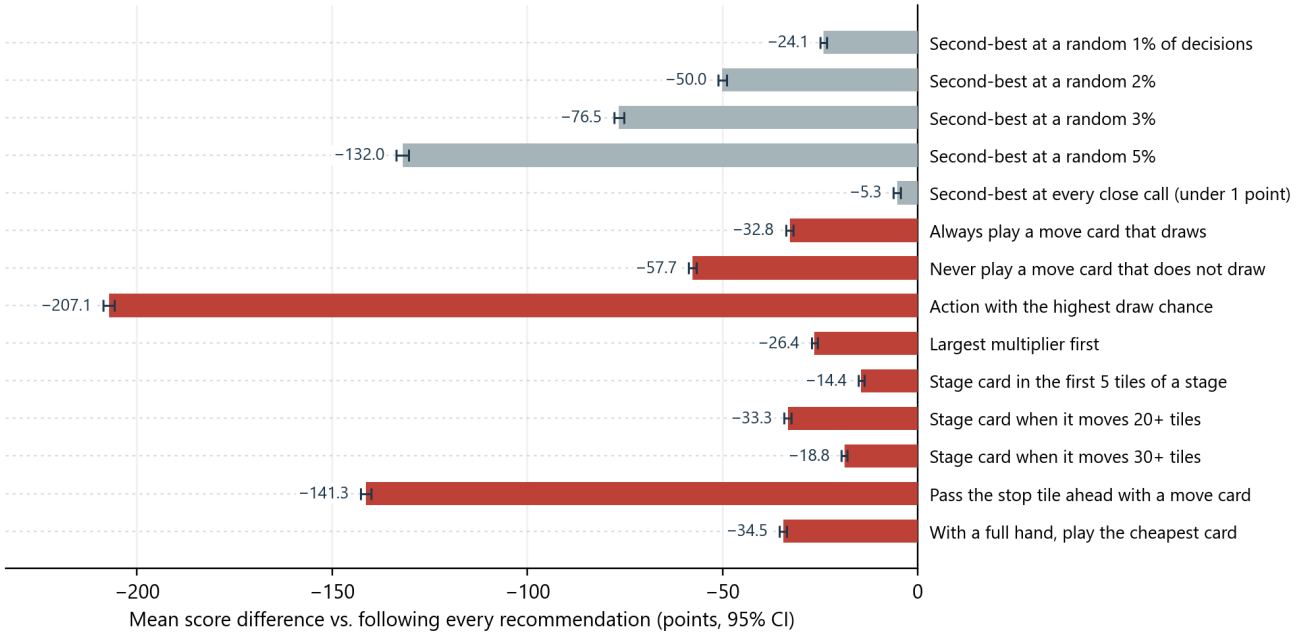


Figure 120: Average score loss per rule when the AI's recommendation is replaced by another action (95% interval). Games with the same dice sequence are compared. Gray: random and close-call baselines; red: rules a player might use.

Rule	Loss per game	Overrides per game	Loss per override
Second-best at a random 1% of decisions	−24.1	1.6	−15.1
Second-best at a random 2% of decisions	−50.0	3.2	−15.8
Second-best at a random 3% of decisions	−76.5	4.7	−16.4
Second-best at a random 5% of decisions	−132.0	7.5	−17.6
Second-best at every close call (under 1 point)	−5.3	12.9	−0.4
Always play a move card that draws	−32.8	8.1	−4.0
Never play a move card that does not draw	−57.7	3.3	−17.2
Action with the highest draw chance	−207.1	26.1	−7.9
Largest multiplier first	−26.4	2.7	−9.9
Stage card in the first 5 tiles of a stage	−14.4	1.7	−8.4
Stage card when it moves 20+ tiles	−33.3	3.0	−11.2
Stage card when it moves 30+ tiles	−18.8	2.2	−8.5
Pass the stop tile ahead with a move card	−141.3	4.2	−33.3
With a full hand, play the cheapest card	−34.5	1.9	−18.1

The player rules were defined as follows. "A move card that does not draw" applies to move cards only. "First 5 tiles of a stage" does not apply when a move card can draw now. "Pass the stop tile" applies when the next stop tile is within 11 tiles and the AI's action does not clear it on every dice result; the action becomes the move card with the highest AI value among those that clear it. "Full hand" applies only when the AI rolls, and plays the card with the lowest holding value from Section 1.4.

Every rule lowered the average score. Taking the second-best action at random only 1.6 times per game costs 24.1 points; three times, 50.0. In most decisions the second-best action is several points below the best. Taking the second-best only at close calls costs just 0.4 points per override, but telling which decisions are close calls is hard without the AI's values.

Among the player rules, the costliest are "action with the highest draw chance" (−207.1 points) and "pass the stop tile ahead with a move card" (−141.3 points). The first fires about 26 times per game and throws away both saving move cards to roll (Part 3, Section 1) and covering dozens of tiles with a multiplier (Section 8): judged by this action's draw chance alone, a move card that can draw always comes first. The stop-tile rule is the opposite of the AI's fourth

principle and loses 33.3 points each time it fires, because it spends a card to move a few tiles further. With a stop tile within 11 tiles it fires even when only a sum of 12 (1/36) would be blocked, so it replaces rolls that are almost never stopped.

The stage-card rules cost relatively little per override (8.4–11.2 points). The 30-tile threshold costs less than the 20-tile one: a stage card gains value the longer it is held (Section 1.4), so the stricter the threshold, the smaller the loss. "With a full hand, play the cheapest card" costs 18.1 points per override. As Part 3, Section 3 showed, the AI rolls with a full hand when a card in hand can draw right after the roll, and overriding that judgment costs points. "Always play a move card that draws" and "largest multiplier first" cost 4.0 and 9.9 points per override.

The AI is sometimes wrong too, but its mistakes are rare and mostly close calls, so they cost little (an average of 0.004–0.038 points per endgame position). Decisions a player changes because they "look wrong" include ones with large gaps, and in the random-override runs each change cost 15.1–17.6 points on average. The two AIs' averages (about 1,878.6 points for VELA v4.3 Normal and about 1,880 for LYRA v2) assume every recommendation is followed; changing some decisions breaks that average. These runs had VELA v4.2 playing on, and the decisions a player actually changes may be distributed differently from these rules. The control-variate correction used VELA v4.2's one-step search values (V1); its mean was −2.6 points (±2.0), not distinguishable from 0.

The AI's recommendation is not optimal either. "Rare and small" above was checked only in the endgame (0–2 rolls left), where exact values exist; earlier in the game there are no exact values, and the AIs can only be compared with each other. That comparison already shows sizable early- and mid-game errors: in Section 6.3, VELA v4.2 lost 2.65 points per game to LYRA v2, mostly in decisions at rolls used 21–80. Both AIs compute only a few steps ahead and leave the rest to an approximate value (VELA's value table, LYRA's network), so LYRA v2 can be wrong in the same way on judgments whose effect shows only dozens of rolls later. Action paths the AIs rarely choose in actual play, such as a long-term plan that collects and spends cards in an entirely different way, are seldom checked against real outcomes and may be badly misvalued; such a path could turn out better. These runs establish only that the rules above lose points. How far optimal play lies above the two AIs was not measured.

6.5 Evidence in the case positions

In all 29 cases of Part 3, VELA v4.2, v4.3 (Normal and High) and LYRA v2 chose the same action. Two checks back that choice.

First, starting from the chosen and the comparison action and playing on with the same dice until four rolls were used, 1,000 times each, the chosen action's value was above 0 within the 95% interval in 28 cases. The remaining one (11-2, three cards in the deck) is a close call with a value gap of 1.1 points.

Second, both actions were played on to the end of the game. With VELA v4.2 playing on, the chosen action came out ahead in all 29 cases (2,048 pairs; 16,384 for 6-2). Played to the end, 11-2 also favored the +10, by +4.3 points [+2.6, +5.9]. With v4.3 Normal playing on, 28 cases came out ahead; only 6-2 (the −1 riding a jump behind) was not distinguishable from 0, at +0.6 points [−0.6, +1.8]. The size of each gap was broadly similar to the gap after four rolls (see each evidence table).

The control-variate correction in these full-game comparisons used the AI's search values: one-step values (V1) with VELA v4.2 playing on and two-step values (V2) with v4.3 Normal. On the same data, the median standard deviation of the paired gap was 199.7 points raw, 41.5 with the value-table correction and 35.8 with the V1 correction (32.3 with V2 for v4.3 Normal), which needs 25–40% fewer pairs than the value-table correction for the same precision.

PART 7

High-scoring games

The score distribution of one million VELA v4.2 games, and how the games that scored 2,600 or more used their resources.

7.1 One million games

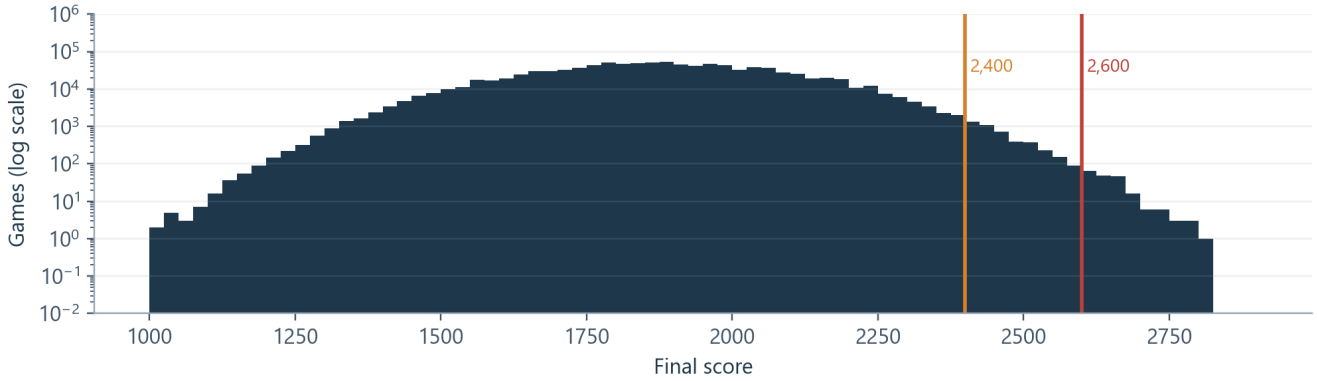


Figure 121: Final scores of one million VELA v4.2 games (log scale).

The mean is 1,877.0 points and the standard deviation (how widely scores spread from game to game) is 201.2. Half of the games end below 1,874, the top 1% reach 2,343 or more and the top 0.1% 2,501 or more. The best game scored 2,813 points, and no game reached the end of the board at 2,898.

Final score	Games	Share
2,000 or more	271,463	27.1463%
2,400 or more	4,558	0.4558%
2,600 or more	196	0.0196%
2,700 or more	19	0.0019%
2,800 or more	1	0.0001%

7.2 Resources in high-scoring games

We replayed all 196 games that scored 2,600 or more and 130 games from lower score bands to collect their action logs. Final scores, action counts and action sequences all matched the original records.

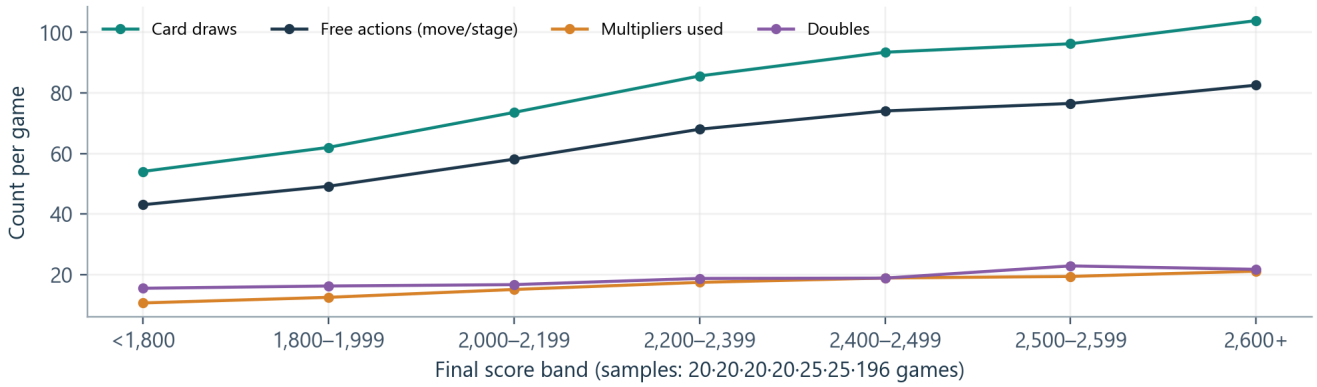


Figure 122: Actions per game by final score band. Higher bands drew more cards, took more free actions, played more multipliers and rolled more doubles.

Higher-scoring games drew more cards (about 104 per game at 2,600 or more, with about three deck refills), took more free actions and played more multipliers. They also rolled more doubles: about 22 per game at 2,600 or more, against about 15 below 1,800. The expected number over 100 paid rolls is about 16.7 (one in six), and each double gives a bonus roll that uses no roll, so games with more doubles move further. Doubles are pure dice luck that no play can increase. This comparison groups games by their final score after the fact. Good dice raise doubles, draws and the score together, so it does not show that drawing more cards causes higher scores.

What was left on first passing tile 2,000

Final score group	Tile reached	Rolls left	Cards in hand
2,600 or more (196 games)	2,020.0	21.0	4.2
2,400–2,599 (50 games)	2,019.2	16.0	4.0
Below 2,400 (sample that reached 2,000)	2,020.3	5.1	2.6

Games that finished at 2,600 or more still had 21 rolls left on average when they first passed tile 2,000. Games at 2,400–2,599 had 16, and lower-scoring games 5. The higher the final score, the earlier the game reached 2,000 and the more rolls it had left. This too is an observed difference, and it includes the good dice luck up to that point.

7.3 The top game: 2,813 points

Over 209 actions this game drew 106 cards (with three deck refills), played 22 multipliers and took 84 free actions. Near the end it crossed the stop run starting at 2,660 with two stage cards. By exact values its final multiplier order (×3 first) was 0.0006 points below the best (×2 first). Even the top game did not play every move optimally, and copying its line would not reproduce its score.

PART A

Glossary and data

The terms used in the report and the conditions behind each calculation.

Glossary

Term	Meaning
Full hand	Five cards in hand. Landing on a draw tile gives nothing (a played card leaves the hand first, so a card play can still draw).
Deck, refill	The cards not yet drawn (30 at the start). Drawing the last card refills the deck to 30.
Stop tile (boss tile)	A tile a plain roll cannot pass. Multiplier, move and stage cards go past it.
Bonus roll	A free next roll earned by a double on a paid roll; it does not use up a roll.
Draw possible	The action draws, or one card left in hand can draw from the landing tile.
AI value	The probability-weighted average of the AI's values for every outcome of an action; an estimate of the expected final score.
Holding value	The difference in AI value with and without a card in hand, in the same position.
Outcome breakdown	An action's value split by dice result and card drawn. The parts add back up to the value.
Full-game comparison	Two games played to the end with the same seed (the number that fixes the random sequence); the score gap is measured over many seeds. In Section 6.4, games with some decisions overridden by a rule are compared with games that follow every recommendation.
95% interval	The uncertainty range of a measured mean gap. If it includes 0, the sample cannot tell which action is better.
Exact value	The expected final score over every remaining possibility. Only computable in the endgame.
Rolls used	The number of rolls spent so far, as shown on the game screen. Rolls left = 100 minus rolls used.
Search depth	How many steps ahead the AI calculates: two for VELA v4.2, one for LYRA v2. VELA v4.3 has four search strengths (Fast = v4.2, Normal, High, Max); Normal and above search three steps and up to six in close calls.

Data and conditions

Data	Details
Case positions	Part 3: chosen from 320,000 positions reached in 2,000 VELA v4.2 games by per-strategy conditions, then checked with counterfactuals that change only the key factor and with agreement among the three AIs. Positions with a bonus roll pending were excluded. Parts 4 and 5: positions from the exact endgame catalog.
Outcome breakdowns	The values of the child positions in VELA v4.2's two-step evaluation. Recombined outcome values match the action value to within 0.001 points.
Holding values	Computed with the VELA v4.2 value model. The phase chart uses 7,000 random positions with cards in hand from 2,000 VELA v4.2 games.
Base probabilities	Computed from the rules over all 2,659 of the 2,898 tiles where the token can stand.
Choice patterns	Decision logs of 10,000 VELA v4.2 games and 10,000 LYRA v2 games. The two sets are different games.
Split after four rolls	For each case, 1,000 pairs starting from the chosen and the comparison action are played on by VELA v4.2 with the same dice, and VELA v4.2's values (with and without the hand) are compared once four rolls have been used.
Part 3 full-game comparisons	For each case, both actions were played on to the end with the same seeds: 2,048 pairs with VELA v4.2 playing on (control variate: one-step search values, V1) and 512 with v4.3 Normal (two-step search values, V2); cases whose 95% interval included 0 were extended to 16,384 and 2,048 pairs. Run on a 384-thread Linux machine, with scores on the same seeds matching the local results.
VELA v4.3	The production search code, run with the same model file. At Fast it chose the same action as v4.2 in all 4,312 catalog positions and gave the same scores as v4.2 in 12 sample full-game pairs.
Override experiment	VELA v4.2 played 10,000 games per rule on the same seeds, and control-variate-corrected scores were compared seed by seed. The correction used VELA v4.2's one-step search values (V1). The second-best action is the one with the second-highest VELA v4.2 value.
Exact values	4,312 positions from the exact endgame catalog (0–2 rolls left). The exact values and best lines in Parts 4 and 5 were recomputed with the production exact solver and match the catalog (differences of 10^{-6} or less).
One million games	Final scores of one million VELA v4.2 games. All 196 games at 2,600 or more and 130 lower-scoring games were replayed and matched the original records.